

Convergence of coronary artery disease genes onto endothelial cell programs

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Linking variants from genome-wide association studies (GWAS) to underlying mechanisms of disease remains a challenge^{1–3}. For some diseases, a successful strategy has been to look for cases in which multiple GWAS loci contain genes that act in the same biological pathway^{1–6}. However, our knowledge of which genes act in which pathways is incomplete, particularly for cell-type-specific pathways or understudied genes. Here we introduce a method to connect GWAS variants to functions. This method links variants to genes using epigenomics data, links genes to pathways *de novo* using Perturb-seq and integrates these data to identify convergence of GWAS loci onto pathways. We apply this approach to study the role of endothelial cells in genetic risk for coronary artery disease (CAD), and discover 43 CAD GWAS signals that converge on the cerebral cavernous malformation (CCM) signalling pathway. Two regulators of this pathway, *CCM2* and *TLNDR1*, are each linked to a CAD risk variant, regulate other CAD risk genes and affect atheroprotective processes in endothelial cells. These results suggest a model whereby CAD risk is driven in part by the convergence of causal genes onto a particular transcriptional pathway in endothelial cells. They highlight shared genes between common and rare vascular diseases (CAD and CCM), and identify *TLNDR1* as a new, previously uncharacterized member of the CCM signalling pathway. This approach will be widely useful for linking variants to functions for other common polygenic diseases.

Genetic variants that influence complex traits are thought to regulate genes that work together in biological pathways. Identifying convergence on particular pathways can help discover genes and cellular functions that causally influence disease risk^{1–6}. However, it is often challenging to identify such convergence. This is because complex traits involve contributions from multiple cell types, most risk variants are noncoding and can regulate multiple nearby genes, and it remains unclear which genes work together in which pathways in which cell types^{7–9}.

GWAS for CAD have discovered more than 300 independent signals^{10–12} (Supplementary Table 1). Overall, 75% of these signals are not associated with circulating lipids (Supplementary Table 1), which

indicates the presence of undiscovered disease mechanisms that may function through cells in the coronary artery, where atherosclerosis that causes CAD develops. Endothelial cells (ECs) are one of the most important of these arterial cells, controlling cholesterol uptake and efflux, smooth muscle cell responses, blood clotting and inflammatory immune cell recruitment^{13,14}. ECs are also highly enriched for CAD heritability¹⁵. At a few individual CAD GWAS loci, noncoding risk variants have been shown to regulate the expression of key EC genes such as endothelial nitric oxide synthase (*NOS3*) and endothelin 1 (*EDN1*), among others¹⁶. It remains unclear, however, which other genes in CAD GWAS loci might work together in which EC pathways to modulate disease risk.

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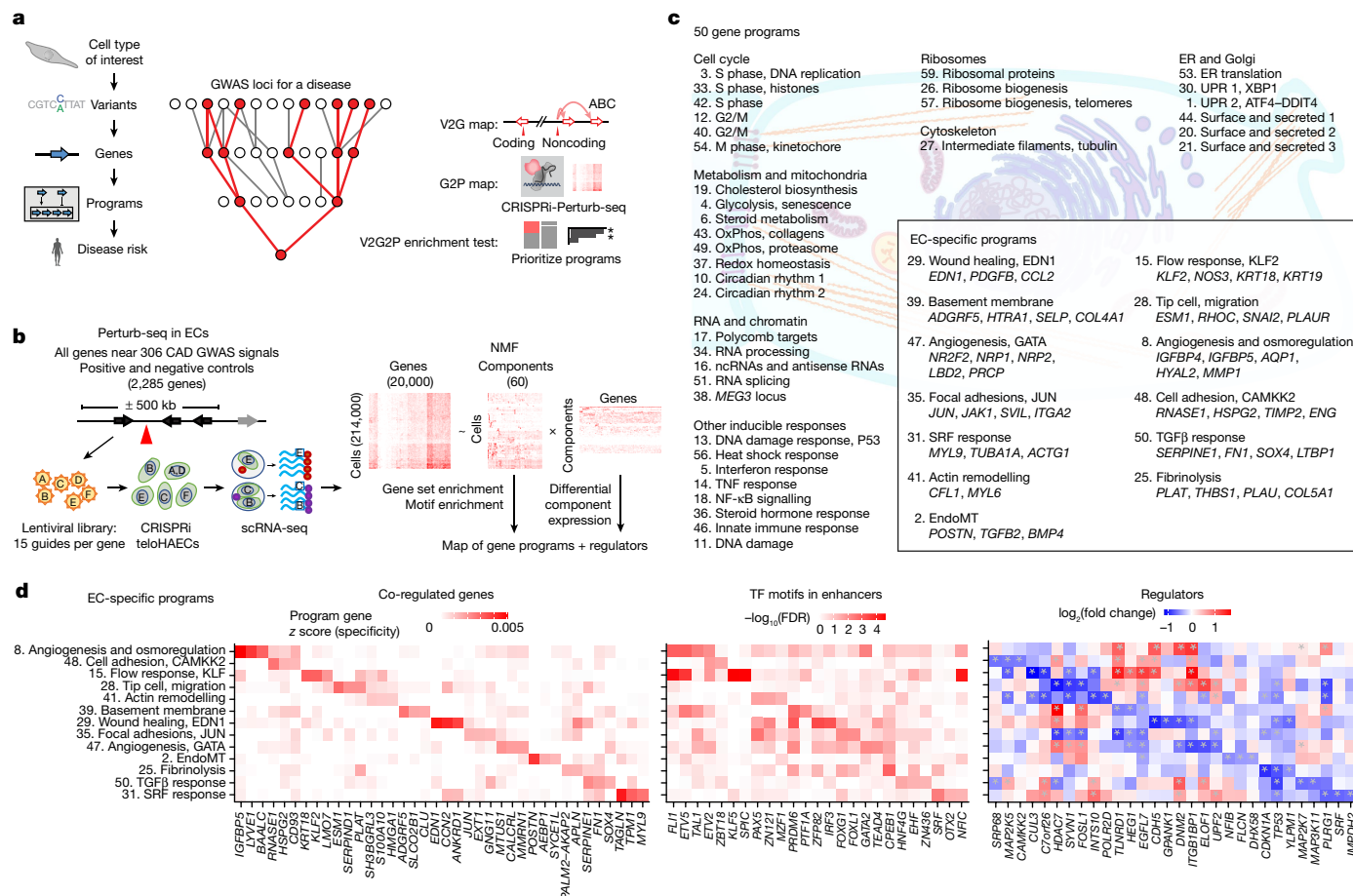


Fig. 1 | Building a map of gene programs in ECs using Perturb-seq. a, Overview of the V2G2P approach. Red nodes outline an example of the predicted convergence of multiple risk variants onto a small number of cell-type-specific programs related to disease. **b**, Diagram of the Perturb-seq G2P approach used to map genes to programs. **c**, The 50 gene programs were identified de novo from Perturb-seq and cNMF. Selected co-regulated program genes are shown in italics for the EC-specific programs. EndoMT, endothelial-to-mesenchymal

To address these challenges, we developed a new approach that links GWAS variants to genes in a systematic and unbiased manner and identifies their convergence onto specific disease-associated transcriptional programs. The five steps of this variant-to-gene-to-program (V2G2P) approach (Fig. 1a and Supplementary Note 1) and their application to EC functions in CAD are summarized below:

1. Identify a cell type and cellular model relevant to disease genetics through enrichment of disease risk variants in enhancers in that cell type. Here we focused on human arterial ECs using telomerase-immortalized human aortic ECs (teloHAECs) as a model.
2. Build a map of variant-to-gene (V2G) links in that cell type to link disease-associated variants to potential target genes. Here we considered evidence from variants in EC enhancers, coding regions and splice sites.
3. Build a map of gene-to-program (G2P) links in that cell type using Perturb-seq^{17–20} to systematically knockdown all possible candidate disease genes and identify sets of genes that act together in biological pathways. Here we knocked down all expressed genes within ±500 kb of 306 CAD GWAS signals, read out the effects of each perturbation with single cell RNA sequencing (scRNA-seq) and used unsupervised machine learning to define gene ‘programs’, unbiased by previous knowledge of gene sets or pathways.
4. Identify disease-associated programs by developing a statistical test to determine whether the genes with links to risk variants are

transition; ER, endoplasmic reticulum; OxPhos, oxidative phosphorylation; UPR, unfolded protein response. **d**, Heatmaps show for each EC-specific program the top three program co-regulated genes (ranked by specificity to the program; Methods), the top three TF motifs in enhancers (ranked by enrichment FDR, see Extended Data Fig. 3m for sequence logos) and the top three regulators (ranked by fold change in component expression after perturbation; see also Extended Data Fig. 3e).

enriched in (that is, converge on) particular programs. Here we found that many CAD GWAS loci converge on five gene programs identified de novo with Perturb-seq, which seem to correspond to branches of the CCM signal transduction pathway.

5. Study the genes in disease-associated programs. Here we nominate 41 genes that may influence CAD risk through effects in ECs and dissect two in detail. Specifically, we show that knockdown of *TLNDR1* or *CCM2* mimics the effects of atheroprotective laminar blood flow, and that the poorly characterized gene *TLNDR1* is a newly identified regulator in the CCM pathway.

In summary, the V2G2P approach defines cellular programs de novo using Perturb-seq and intersects these programs with enhancer-to-gene maps from the same cell type. It provides an interpretable, systematic and unbiased framework for tracing the path from variant to gene to disease program simultaneously for all GWAS loci for a given disease and cell type.

A V2G map in ECs

To implement this V2G2P approach, we collected GWAS signals for CAD^{10,12} and defined a set of nearby genes for each GWAS signal to include the 2 closest genes on either side and all genes within ±500 kb. We focused on the 228 non-lipid GWAS signals that were not associated

with circulating lipid levels (see Methods ‘Defining variants in CAD GWAS signals’) because lipid-associated signals are likely to act in hepatocytes or other non-EC types. This analysis produced 1,942 total candidate genes, with a median of 8 nearby genes per GWAS signal (Supplementary Table 1).

We selected teloHAECs as a well-established arterial EC model²¹ and collected bulk RNA-seq, ATAC-seq and H3K27ac ChIP-seq data in resting conditions and several stimulated conditions (addition of IL-1 β , TNF and VEGFA) to identify expressed genes and candidate enhancers (Supplementary Table 2). Variants in teloHAEC enhancers were 11-fold to 13-fold enriched for CAD heritability by stratified linkage disequilibrium score regression (S-LDSC; Extended Data Fig. 1a). Moreover, genes near CAD GWAS loci that were expressed in teloHAECs were also expressed in primary coronary artery ECs in vivo (Extended Data Fig. 1b–d). These results support the choice of this cellular model.

To link risk variants to genes (V2G), we identified genes predicted to be regulated by EC enhancers containing CAD variants using the activity-by-contact (ABC) model^{9,22}. We also considered genes containing coding variants (see Methods ‘Linking variants to genes’). We identified 254 out of 1,942 nearby genes with a link to a CAD risk variant (genes with a V2G link or simply V2G genes) at 125 out of 228 non-lipid GWAS signals (range of 1–5 genes per signal; Supplementary Tables 3 and 26).

A G2P map in ECs

To link genes to programs, we applied CRISPR interference (CRISPRi)–Perturb-seq to identify de novo sets of genes that act in the same transcriptional pathways. Although Perturb-seq is emerging as a powerful tool to study gene pathways^{17–20,23,24}, new design and analysis approaches are needed to discover pathways enriched for genetic risk for common diseases. Accordingly, we developed an approach whereby we systematically knocked down all expressed genes near all CAD GWAS signals and **applied an unsupervised matrix factorization approach to identify sets of co-regulated genes. We then linked upstream perturbed genes with the downstream genes they regulate to define gene programs in a systematic fashion**, unbiased by previous knowledge of annotated pathways or gene sets (Fig. 1b).

We engineered teloHAECs to express dCas9-KRAB (CRISPRi; Extended Data Fig. 1e,f) and transduced these cells with a guide RNA (gRNA) library targeting all 1,661 expressed genes nearby CAD GWAS signals and 624 control genes (15 guides per promoter), plus 1,000 control gRNAs, for a total of 37,637 guides (Supplementary Tables 4–6). After 5 days of doxycycline induction of CRISPRi, we collected 20 lanes of 10 $\times$ 3’ scRNA-seq data (Methods). In total, we obtained data for 214,449 cells expressing a single gRNA at an average depth of 929,000 total transcript unique molecular identifiers (UMIs) per targeted promoter (Extended Data Figs. 1g–l and 2a,b and Supplementary Table 7). Target genes were effectively knocked down, and knockdown of common essential genes decreased fitness. Also, 10.7% of perturbations of expressed targets significantly affected the transcriptome (Extended Data Fig. 1m–q and Supplementary Tables 8–10).

We applied an unsupervised approach to this Perturb-seq data to discover gene programs, independent of previous knowledge of annotated pathways or gene sets (Fig. 1b, right). First, we used consensus non-negative matrix factorization (cNMF)²⁵ to model the gene-by-cell matrix as a linear combination of latent components representing co-regulated gene sets that co-vary in the population of cells (Methods, Extended Data Fig. 2c–g and Supplementary Table 11). From each cNMF component, we defined a ‘program’ as a set of genes comprising both co-regulated genes (the 300 marker genes for which expression was most specific to that component) and regulators (the 0–35 genes for which perturbations significantly affected the expression of each component relative to negative control gRNAs, false discovery rate (FDR) < 0.05; Extended Data Fig. 2h). This analysis established

a G2P map that included 18,606 links from 7,692 unique genes to 50 programs (Fig. 1c, Extended Data Fig. 3a–c and Supplementary Tables 12–14).

After defining these 50 programs using an unsupervised approach, we annotated each program based on their regulators and co-regulated genes. This process included manual curation, analysis of transcription factor (TF) motifs in their promoters and predicted enhancers, and gene set enrichment (see Methods ‘Definition and annotation of gene expression programs’). We identified programs representing a range of cellular functions: from ubiquitously expressed (housekeeping) processes to a wide variety of inducible responses (for example, unfolded protein response, DNA damage, heat shock and inflammation) despite the absence of stimuli for these responses in our culture system (Fig. 1c). **We annotated 13 programs as EC-specific because they included genes that were on average more highly expressed in ECs than in other cell types** (Fig. 1c, Extended Data Fig. 3c, Supplementary Table 13 and Methods ‘Defining EC-specific programs’). These EC-specific programs included distinct combinations of genes enriched for roles in angiogenesis, extracellular matrix remodelling, barrier function and the endothelial-to-mesenchymal transition, and the promoters of their co-regulated genes were enriched for different TF motifs (Fig. 1d and Extended Data Fig. 3d–f). An analysis of regulators (perturbations) identified cases in which programs were co-ordinately or oppositely regulated by the same perturbations (for an example, see Extended Data Fig. 3g). Ten genes were regulators of five or more EC-specific programs, including genes known to have important functions in ECs such as *EGFL7* and *ITGB1BP1* (also known as *ICAP1*)^{26,27} (Extended Data Fig. 3h–l).

this is a DE level analysis

Taken together, this G2P map represents a wide range of cellular pathways, links upstream regulators to coherent sets of downstream genes and provides a resource for understanding the functions and potential disease-relevance of genes in ECs.

CAD GWAS signals converge on five programs

We next applied a simple statistical test (V2G2P enrichment) to determine, in an unbiased fashion, **whether GWAS variants for a trait converge onto particular gene programs**. Specifically, we tested whether genes for each program (genes with a G2P link; Fig. 2a) were more highly enriched in genes likely to be affected by CAD risk variants (genes with a V2G link; Fig. 2a) than expected by chance (see Methods ‘Identifying CAD-associated programs through V2G2P analysis’).

We identified significant V2G2P enrichment for 5 programs, each including 12–18 genes linked to CAD variants (compared with 4.5 expected by chance; 2.6-fold to 4-fold enrichment, FDR < 0.05; Fig. 2b and Supplementary Table 13). Together, these 5 programs included 41 unique V2G2P genes (genes linked to CAD variants and part of at least 1 of the 5 V2G2P programs), including genes near 43 out of 228 non-lipid GWAS signals (Fig. 2c and Supplementary Tables 1 and 15).

The five V2G2P programs corresponded to distinct sets of genes related to extracellular matrix organization, cell migration and angiogenesis (Fig. 2b and Extended Data Fig. 4). Program 8 included genes involved in negative regulation of angiogenesis (*IGFBP4* and *IGFBP5*) and osmotic balance (*SLC12A2* and *AQP1*). Program 48 included genes involved in cell adhesion and migration such as *FSLT1* and *TIMP2*, and was regulated by *MEK5* (*MAP2K5*), *ERK5* (also known as *MAPK7*) and calcium/calmodulin-dependent (*CAMKK2*) signalling. Program 39 expressed genes involved in the basement membrane (*COL4A1* and *COL4A2*) and platelet recruitment (*VWF* and *SELP*). Program 35 expressed genes involved in focal adhesions (*ITGA2*) and the JAK–STAT signalling pathway. Program 47 expressed genes involved in angiogenesis, including *NR2F2*, *NRP1* and *NRP2*, and two genes specifically associated with a stalk cell phenotype (*VWF* and *EHD4*).

Several independent lines of evidence supported the associations of these 5 programs and 41 genes with CAD. (1) All five V2G2P programs

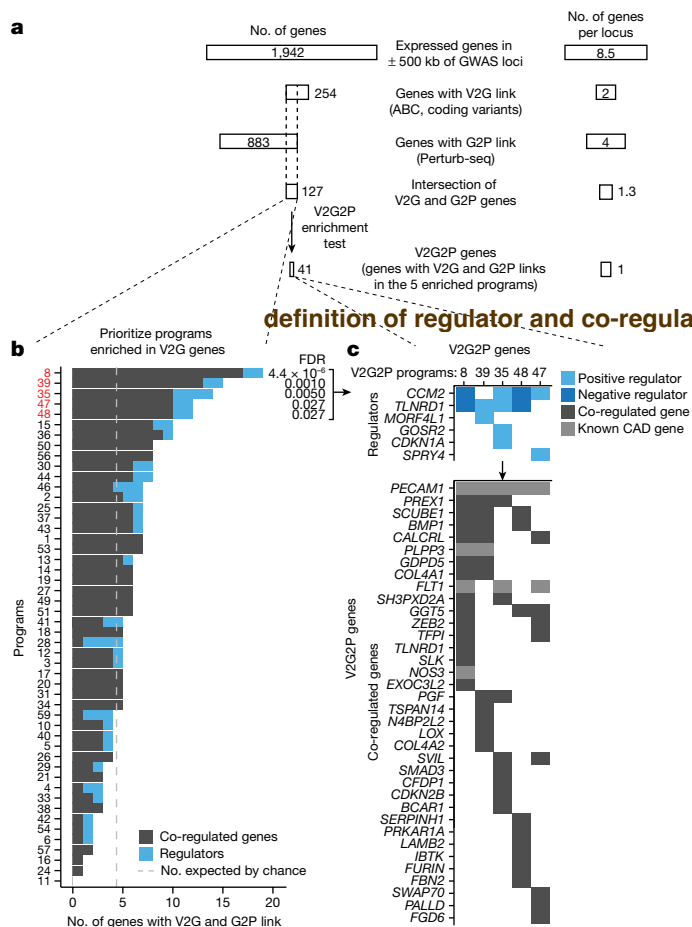


Fig. 2 | CAD genes converge on five programs in ECs. a, Counts of CAD genes (left) or genes per locus (right) at each step of the V2G2P analysis. **b**, The 50 programs are ordered (y axis) by the number of program genes linked to CAD variants (x axis). We define the 5 programs with FDR < 0.05 as V2G2P programs. Grey dashed line indicates the number of genes linked to CAD variants expected by chance. **c**, Relationships among the 41 V2G2P genes for CAD and the 5 V2G2P programs. Top, six V2G2P genes were regulators of one or more V2G2P programs (FDR < 0.05). Positive and negative regulators indicate genes for which loss of function leads to decreased or increased program expression, respectively. Bottom, 36 V2G2P genes were co-regulated genes in one or more V2G2P programs. Known CAD gene: one of the eight gold standard CAD genes known to function in ECs (Supplementary Table 16).

were EC-specific programs that included at least one out of the eight gold standard genes for which variant-to-gene-to-disease effects in ECs have previously been characterized (known EC CAD genes in Fig. 2c and Supplementary Tables 15 and 16). Program 8 included four such genes: *NOS3*, *PLPP3*, *FLT1* and *PECAM1*. (2) All five V2G2P programs were significantly enriched for CAD heritability based on MAGMA⁴ analysis, and two were significantly enriched for CAD heritability by S-LDSC²⁸ (FDR < 0.05; Methods and Extended Data Fig. 5a,b). (3) The 41 V2G2P genes were highly ranked using an independent gene prioritization method, polygenic priority score (PoPS)⁵, compared with other nearby genes at the same GWAS signals (rank-sum test $P = 2.5 \times 10^{-53}$; Extended Data Fig. 5c,d and Supplementary Table 15). (4) Nine out of the 41 V2G2P genes have previously been found to affect atherosclerosis and/or vascular barrier integrity in studies of mouse models in a way that is consistent with their acting in ECs (Supplementary Table 15).

In summary, the V2G2P approach identified the convergence of CAD GWAS signals onto 5 EC-specific gene programs that are enriched in CAD heritability and include 41 unique genes linked to CAD risk variants.

Benchmarking and comparison of methods

We compared our V2G2P-prioritized genes to those from seven previous studies that used various approaches to prioritize genes in CAD GWAS loci (Supplementary Note 2 and Supplementary Table 17). Overall, 31 out of the 41 V2G2P genes were not prioritized in the two EC-specific studies, and 17 were not prioritized by any of these 7 studies (Supplementary Note 2 and Supplementary Table 17). These 17 genes included the 2 strongest regulators of the 5 V2G2P programs, *TLNDR1* and *CCM2*, which we explore in detail below.

We also benchmarked the ability of the V2G2P strategy to identify the eight gold standard EC CAD genes (Supplementary Note 2). Compared with other studies that nominated CAD genes in an EC-specific fashion (based on expression quantitative trait locus (eQTL) co-localization²⁹ or variant-targeted CRISPR screens³⁰), V2G2P achieved higher recall for the gold standard genes (50% compared with 12.5% for the others) while also achieving high precision (80%). V2G2P also performed well compared with studies that nominated CAD genes without specificity for ECs and compared with methods that prioritize gene sets or programs from GWAS data (Supplementary Note 2 and Extended Data Fig. 5g).

Both V2G data from the ABC model and G2P data from Perturb-seq were essential for the ability of the V2G2P analysis to identify disease-associated genes and programs (Fig. 2a, Supplementary Note 3 and Extended Data Figs. 5–7), consistent with recent observations that combining locus-specific variant-to-gene links with genome-wide enrichments for gene pathways can improve the specificity of disease gene identification^{5,31,32}. In particular, at most GWAS signals, neither V2G nor G2P information alone was sufficient to identify probable disease genes: 119 GWAS signals had 2 or more genes with a V2G link (up to 5), and 195 GWAS signals had 2 or more genes with a G2P link (up to 25), including links to all 50 programs. These observations are consistent with the expectation that noncoding variants often regulate multiple nearby genes^{9,33}, and that, by chance, a given GWAS signal might have several nearby genes involved in various cellular pathways. Combining these two layers of information in the V2G2P enrichment test provided more specificity. That is, for the 43 signals with V2G2P links to these programs, only 6 had more than 1 linked gene (up to 2; Extended Data Fig. 6h). We performed other internal benchmarking studies to confirm the value of each component of the V2G2P approach (including cell-type-specific versus cell-type-non-specific ABC data, and Perturb-seq versus scRNA-seq without perturbations; Supplementary Note 3).

In summary, V2G2P identified known CAD genes more accurately than other cell-type-specific gene prioritization studies, and identified 17 new genes not nominated by any previous study of CAD loci. V2G2P required both cell-type-specific V2G data and systematic Perturb-seq data for its ability to identify disease-relevant programs and genes.

Linking the CCM pathway to CAD risk

A key feature of the V2G2P approach is that it provides mechanistic hypotheses that link variants to genes to pathways at all prioritized loci. This aspect can accelerate further functional studies to understand the molecular mechanisms that drive effects on disease risk. We used this information to propose mechanisms for previously uncharacterized GWAS loci (Supplementary Note 4) and explored in detail two specific genes—*CCM2* and *TLNDR1*—that were the strongest regulators of the five CAD-associated programs (Fig. 2c). **These investigations revealed that both *CCM2* (a known member of the CCM complex) and *TLNDR1* (a previously poorly studied gene with no known function in ECs) act together in the CCM signalling pathway to regulate many other CAD genes.**

We first examined *CCM2*, which was prioritized in our V2G2P analysis, because it harbours a missense coding variant associated

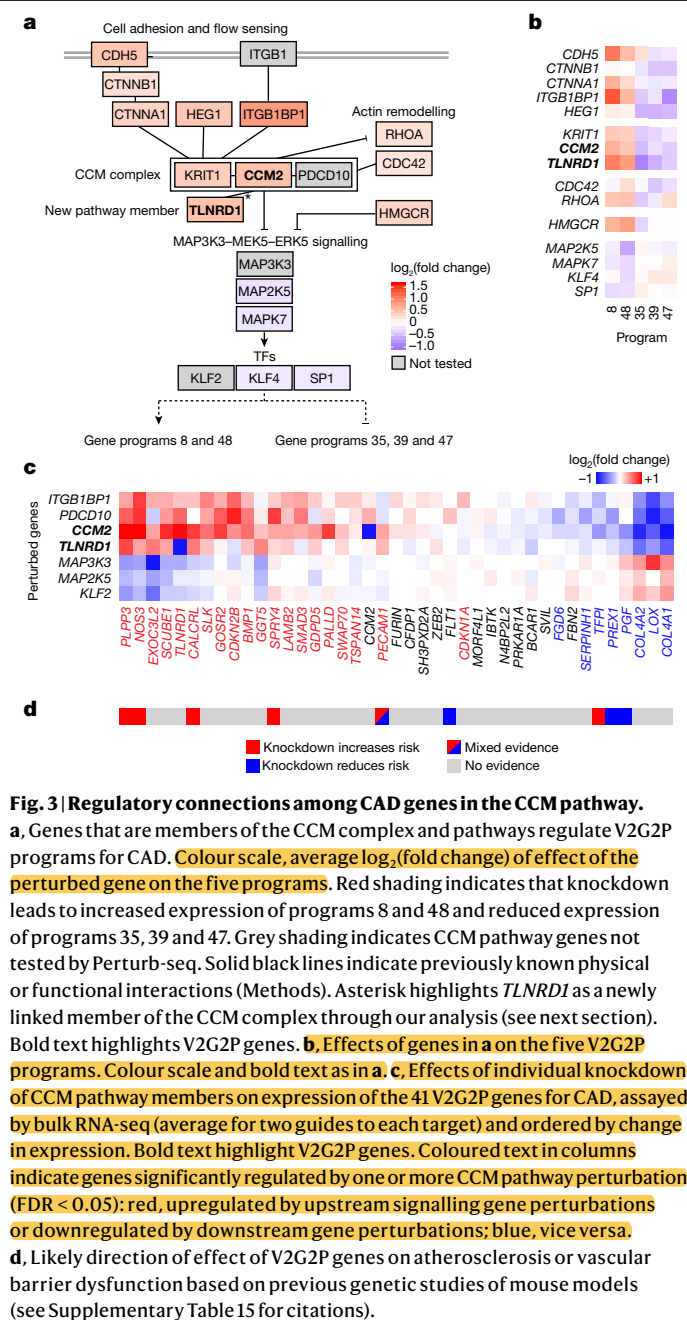


Fig. 3 | Regulatory connections among CAD genes in the CCM pathway. **a**, Genes that are members of the CCM complex and pathways regulate V2G2P programs for CAD. Colour scale, average log₂(fold change) of effect of the perturbed gene on the five programs. Red shading indicates that knockdown leads to increased expression of programs 8 and 48 and reduced expression of programs 35, 39 and 47. Grey shading indicates CCM pathway genes not tested by Perturb-seq. Solid black lines indicate previously known physical or functional interactions (Methods). Asterisk highlights *TLNRD1* as a newly linked member of the CCM complex through our analysis (see next section). Bold text highlights V2G2P genes. **b**, Effects of genes in **a** on the five V2G2P programs. Colour scale and bold text as in **a**. **c**, Effects of individual knockdown of CCM pathway members on expression of the 41 V2G2P genes for CAD, assayed by bulk RNA-seq (average for two guides to each target) and ordered by change in expression. Bold text highlight V2G2P genes. Coloured text in columns indicate genes significantly regulated by one or more CCM pathway perturbation (FDR < 0.05): red, upregulated by upstream signalling gene perturbations or downregulated by downstream gene perturbations; blue, vice versa. **d**, Likely direction of effect of V2G2P genes on atherosclerosis or vascular barrier dysfunction based on previous genetic studies of mouse models (see Supplementary Table 15 for citations).

with a decreased risk of CAD^{10,12} (rs2107732, V74I; odds ratio = 0.92, $P = 1.53 \times 10^{-8}$). Moreover, knockdown of *CCM2* in Perturb-seq significantly regulated four out of the five CAD-associated programs (Fig. 2c and Extended Data Fig. 8a,b).

CCM2 encodes one of three known components of the CCM complex. Rare loss-of-function mutations in proteins forming the CCM complex can cause rare monogenic vascular malformations through effects on microvascular ECs, including activation of MAP3K3–MEK5–ERK5 signalling to KLF2 and KLF4 (refs. 34,35). However, no mechanistic link between CCM signalling and genetic risk for CAD has previously been described.

Notably, examination of the Perturb-seq data revealed that *CCM2* and other known members of the CCM signalling pathway regulate the CAD-associated programs in a consistent pattern (Fig. 3a,b). Knockdown of *CCM2*, another member of the CCM complex (*KRIT1*) and other genes in the pathway that act upstream of the CCM complex (*CDH5*,

ITGB1BP1, *CTNNA1* and *HEG1*) showed directionally concordant effects on the V2G2P programs (upregulating programs 8 and 48 and downregulating programs 35, 39 and 47; Fig. 3a,b and Extended Data Fig. 8c). Knockdown of downstream genes that are repressed by the CCM complex—including *MEK5*, *ERK5* and *KLF4*—affected the expression of the five CAD-associated programs in the opposite direction (Fig. 3a,b).

To validate observations from the Perturb-seq screen and to further characterize the role of the CCM signalling pathway on gene expression, we individually knocked down *CCM2* and five other genes in the CCM pathway (*ITGB1BP1*, *CCM2*, *PDCD10*, *MAP3K3* (also known as *MEKK3*), *MAP2K5* and *KLF2*) and measured effects using bulk RNA-seq. Overall, 28 out of the 41 V2G2P genes were significantly differentially expressed after CCM pathway perturbation (FDR < 0.05; Fig. 3c, Extended Data Fig. 8d and Supplementary Tables 15 and 18).

Notably, the directionality of the effects on downstream CAD V2G2P genes indicated that inhibition of CCM signalling probably has a protective effect on CAD—opposite of its direction of effect on risk for monogenic CCM disease³⁶. In particular, eight of the V2G2P genes that are regulated by the CCM pathway have previously been studied in mice, and show effects on atherosclerosis and or vascular permeability in ways that are consistent with functions in ECs (Supplementary Table 15). The direction of effect on disease phenotypes and response to *CCM2* knockdown were similar (Fig. 3d). Specifically, of the five genes previously shown to maintain vascular barrier function or be protective for atherosclerosis, four (*NOS3*, *PLPP3*, *CALCL* and *SPRY4*) were upregulated in response to *CCM2* knockdown. Conversely, both of the genes previously shown to promote atherosclerosis or vascular dysfunction (*PGF* and *PREX1*) were downregulated. An additional gene (*PECAMI*) had mixed directions of effect on disease depending on the genetic model (Fig. 3d and Supplementary Table 15). Thus, downregulation of the CCM complex leads to changes in gene expression that may be protective in CAD.

Together, these data show that many of the V2G2P CAD genes can be placed in a transcriptional pathway downstream of *CCM2*, which implicates the CCM complex in genetic risk for CAD beyond its known role in rare monogenic CCM disease.

V2G2P for *TLNRD1*

We next examined *TLNRD1*, the V2G2P gene with the strongest combined effect on the five V2G2P programs (Figs. 2a and 4a and Extended Data Fig. 8e–h). *TLNRD1* (talin rod domain containing 1) is a poorly studied gene that has previously been found to interact with F-actin³⁷ and to affect cell migration in a cancer cell line³⁸, but has not been linked to CAD or any function in ECs. Notably, the transcriptional effect of knocking down *TLNRD1* was almost identical to that of knocking down *CCM2* (Fig. 4b). *TLNRD1* regulated the five CAD V2G2Ps in the same direction as *CCM2* and other upstream CCM signalling components (Fig. 3a,b), and had a similar effect on the expression of the 41 V2G2P genes (Fig. 3c). These observations suggested that *TLNRD1* could be a newly identified regulator in the CCM signalling pathway and that its downregulation may be protective in CAD.

We experimentally validated the predicted V2G link for *TLNRD1* by testing whether the protective CAD allele would downregulate *TLNRD1* expression in ECs. *TLNRD1* is located in the 15q25.1 CAD risk locus (lead variant $P = 2.63 \times 10^{-10}$; rank 159 of 241), where our V2G2P analysis identified a noncoding variant in a predicted EC-specific enhancer (Extended Data Fig. 8e; rs1879454; hg19 chromosome 15: 81377717: C (major, risk allele)→A (minor, protective allele); minor allele frequency = 0.16). We used CRISPRi–FlowFISH²² to perturb this and other enhancers near *TLNRD1*. The chromatin-accessible element containing rs1879454 indeed regulated *TLNRD1* expression (estimated –21% effect, FDR-corrected two-sided Student's *t*-test, $P = 0.001$; Fig. 4c,d and Extended Data Fig. 8i–k). rs1879454 also seemed to affect the regulatory activity of this element. The protective A allele was predicted to disrupt a GATA motif, and, in cells heterozygous for this variant, the A allele was

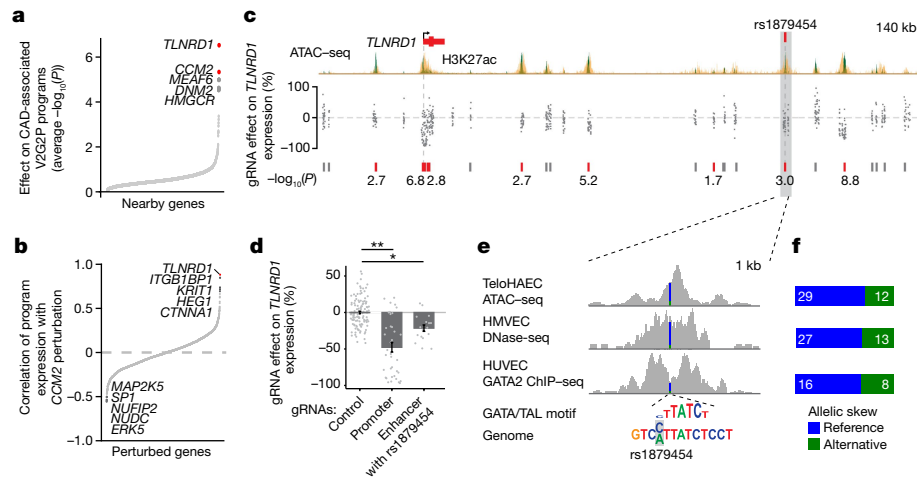


Fig. 4 | Linking CAD risk variants at 15q25.1 to *TLNRD1*. **a**, The 1,503 perturbed nearby genes to CAD GWAS loci, ordered by effect on the 5 V2G2P programs for CAD (average $-\log_{10}(P)$ value), two-sided statistical test from the model-based analysis of single cell transcriptomics (MAST)⁵⁰ package). The top five genes are labelled, with V2G2P genes in red. **b**, The 2,284 perturbed genes ordered by their similarity with *CCM2* perturbation (correlation in $\log_2$ effects on program expression). The top five genes are labelled, with V2G2P genes in red. **c**, Top, ATAC-seq and H3K27ac ChIP-seq tracks in teloHAECs. Middle, each dot is the effect of a gRNA on *TLNRD1* expression averaged across four CRISPRi-FlowFISH replicates. Bottom, boxes represent elements tested in the screen. Labels for

red boxes represent $-\log_{10}(\text{FDR})$ for significant elements (heteroscedastic two-sided t -test; Methods). **d**, FlowFISH quantitation for guides targeting the indicated elements. Bar and whiskers indicate the mean $\pm$ s.e.m. Dots as in **c**. N for guides: negative controls (117), *TLNRD1* promoter (37), enhancer containing rs1879454 (17). *FDR = 0.001, **FDR = 2×10^{-7} . **e**, Top, zoom-in on the enhancer containing rs1879454. Coloured bars indicate read coverage of the reference (blue) and alternative (green) alleles. Bottom, position-weight matrix for a composite GATA/TAL motif and the genome sequence with reference and alternative alleles highlighted in grey. **f**, Quantitation for allele-specific counts at rs1879454 from **e** (Methods).

associated with a twofold decrease in allele-specific GATA2 ChIP-seq signals in human umbilical vein ECs (HUVECs; binomial $P = 0.0758$). Moreover, we observed a 2.4-fold decrease in allele-specific ATAC-seq signal in teloHAECs ($P = 0.0058$) and a 1.9-fold decrease in allele-specific DNase-seq signal in human microvascular ECs (HMVECs; $P = 0.0192$) (Fig. 4e,f).

These data show that the effects of *TLNRD1* knockdown are highly similar to those of *CCM2* knockdown. They also link a protective noncoding CAD allele to decreased *TLNRD1* expression in ECs, which suggests that these genes function together and that a decrease of either one may protect against CAD.

TLNRD1 interacts with the CCM complex

Given the strong similarity in the transcriptional effects of *TLNRD1* and *CCM2*, we sought to gain further insight into the molecular role of this previously poorly characterized gene in the CCM signalling pathway.

We first considered whether the two proteins might physically interact. *CCM2* physically interacts with other proteins in the CCM complex and downstream pathways³⁶, and a recent genome-wide yeast-2-hybrid screen³⁹ provided preliminary evidence of a direct interaction between *CCM2* and *TLNRD1*. We used AlphaFold2.3 Multimer to model potential interactions between the three core CCM proteins and *TLNRD1*. *TLNRD1* was predicted to directly bind the carboxy-terminal helix of *CCM2* (C-helix, residues 417–443; Fig. 5a, right inset) as part of a consistent high-confidence arrangement that also recapitulated the known *CCM2*–*KRT1* binding site in the PTB domain of *CCM2* (ref. 40) (Fig. 5a, left inset), as well as published interactions with *PDCD10* (ref. 41) (Extended Data Fig. 9a–c). The predicted *CCM2*–*TLNRD1* interaction depended on the C-helix of *CCM2*, which bound the *TLNRD1* nine-helix bundle (Fig. 5a). Testing the *TLNRD1*–*CCM2* interaction in human cells showed that *TLNRD1* immunoprecipitated with *CCM2* pull-down and vice versa, and this interaction was lost after deletion of the C-helix of *CCM2* (Fig. 5b, Extended Data Fig. 9d–f and Supplementary Fig. 1).

A key molecular function of the CCM complex is to repress downstream signalling through *MAP3K3* (refs. 34,36,42) (Fig. 3a). To test whether the transcriptional effects of *TLNRD1* knockdown were also related to *MAP3K3* signalling, we knocked down these genes alone or in combination. Individual knockdown of *TLNRD1* or *CCM2* upregulated *KLF2*, *KLF4*, *NOS3* and other probable atheroprotective genes and downregulated probable atherogenic genes. *MAP3K3* knockdown had the opposite effect, and knockdown of both *MAP3K3* and *TLNRD1* or both *MAP3K3* and *CCM2* partially rescued the transcriptional effect of knockdown of each individual gene (Fig. 5c, Extended Data Fig. 10a–d and Supplementary Table 19).

To determine whether the relationship between *TLNRD1* and *CCM2* might extend beyond human ECs in vitro, we tested *tlrnd1* function in zebrafish—a model system in which *ccm2* has specific effects in heart and vascular development^{34,43,44}. Targeting of *tlrnd1* or *ccm2* using CRISPR led to highly similar effects on cardiac and vascular development, including atrial chamber enlargement, pericardial oedema, atrioventricular valve defects and thin ventricular walls (Fig. 5d, Extended Data Fig. 11b,c,g and Supplementary Table 20). Embryos with *tlrnd1* or *ccm2* knocked down using CRISPR also had vascular defects, including posterior cardinal vein dilation and increased vascular permeability to red dextran particles (Extended Data Fig. 11f,h). *tlrnd1* was expressed in the heart and vasculature (Extended Data Fig. 11a), and *tlrnd1* knockdown led to increased *klf2b* expression, similar to the effect of human *TLNRD1* knockdown on *KLF2* expression in teloHAECs (Extended Data Fig. 11i,j). Finally, a 100 μM dose of a *tlrnd1* or *ccm2* morpholino had similar effects as CRISPR-mediated perturbations, and a 50 μM dose of either morpholino had no effect. However, 50 μM of both morpholinos had similar effects as 100 μM of either morpholino alone (Extended Data Fig. 11b,d,e), a result consistent with both proteins functioning in the same pathway.

Together, these data indicate that *TLNRD1* is a previously unrecognized, evolutionarily conserved member of the CCM signalling pathway. They also provide an example of molecular convergence in which V2G2P analysis identified two new CAD genes that not only regulate the same transcriptional pathway but also physically interact.

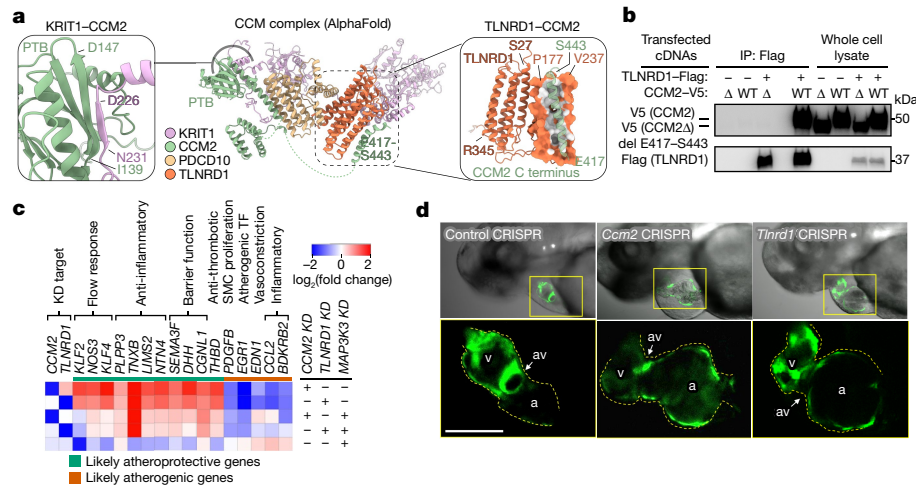


Fig. 5 | TLNRD1 interacts with CCM2 and phenocopies CCM2 in ECs and zebrafish. **a**, AlphaFold2.3 Multimer model of TLNRD1, CCM2, PDCD10 and KRIT1. Right, predicted interaction between TLNRD1 (residues P177–V237) and the C-terminal helix of CCM2 (residues E417–S443). Left, recapitulation of the known interaction between the phosphotyrosine binding (PTB) domain of CCM2 and KRIT1 (ref. 40). Dashed green lines indicate flexible loops. Amino acid positions indicate boundaries of predicted α -helix and β -sheet features. **b**, Flag-tagged TLNRD1 and/or V5-tagged full length (WT) CCM2 or its C-terminal truncation (Δ) were expressed in HEK293T cells. Extracts were co-immunoprecipitated (IP) with mouse anti-Flag and blotted with rabbit

anti-V5 (top) or anti-TLNRD1 (bottom). For gel source data, see Supplementary Fig. 1a. Similar results were seen in two separate experiments. **c**, Heatmap of top genes regulated by both *TLNRD1* and *CCM2* that affect CAD-relevant EC functions (Methods) in cells with the indicated knockdown (KD). **d**, *ccm2* and *tlnrD1* CRISPR knockdown induce atrial enlargement and atrioventricular valve (av) dilation in zebrafish embryos. Top, representative merged light microscopy and fluorescence (cardiac myosin light chain 2–GFP in cardiomyocytes) confocal microscopy images 50 h post-fertilization (anterior to the left). Bottom, $\times 3$ zoomed-in fluorescent-only image of the heart (yellow boxes, above). $N = 5$ embryos were analysed per group. a, atrium; v, ventricle. Scale bar, 100 μ m.

CAD-relevant phenotypes of CCM2 and TLNRD1

We further experimentally characterized how *TLNRD1* and *CCM2* might affect EC phenotypes relevant to CAD. Because atherosclerosis predominantly develops in regions of disrupted or turbulent blood flow¹³, we tested how the effects of *TLNRD1* or *CCM2* knockdown compared with the effects of laminar flow. First, we noted that all of the genes most strongly upregulated by both *TLNRD1* and *CCM2* knockdowns, for which effects on relevant EC functions have previously been assessed, had functions consistent with an atheroprotective effect (including *NOS3*, in which damaging mutations have recently been shown to be a non-lipid driver of CAD⁴⁵). By contrast, genes downregulated by *TLNRD1* and *CCM2* knockdown had functions consistent with an atherogenic effect (Fig. 6a, top two rows, and Supplementary Table 29). These observations suggest molecular mechanisms by which reduction of *TLNRD1* and *CCM2* could decrease CAD risk. Notably, the transcriptional effects of *TLNRD1* or *CCM2* knockdown in static culture were similar to the effects of laminar shear stress (flow, 12 dynes cm^{-2}) on control cells (Pearson $R = 0.40$, $P = 1.5 \times 10^{-54}$ for *CCM2*; $R = 0.52$, $P = 1.6 \times 10^{-94}$ for *TLNRD1*; Fig. 6a, Extended Data Fig. 10e,f and Supplementary Table 19).

Consistent with the similar transcriptional effects of flow and either *TLNRD1* or *CCM2* knockdown, *CCM2* or *TLNRD1* knockdown in static culture increased the number and parallel arrangement of actin stress fibres (Fig. 6b–e), a characteristic of flow response in unperturbed ECs¹³. This finding was consistent with previous studies of *CCM2* knockdown in HUVECs⁴⁶. Conversely, *TLNRD1* or *CCM2* knockdown cells showed reduced alignment to flow relative to control cells (Extended Data Fig. 10i–k). They also exhibited a weaker transcriptional response to flow, perhaps because the flow-response program was already partially active (Extended Data Fig. 10g,h). Endothelial dysfunction that is thought to contribute to CAD is also characterized by decreased barrier function (a leaky endothelium that allows inflammatory cells into the arterial wall¹³). CRISPRi knockdown of either *TLNRD1* or *CCM2* in teloHAECs reproducibly increased EC barrier function, as measured by trans-endothelial electrical resistance (TEER) (Fig. 6f,g).

Together, these observations demonstrate that *TLNRD1* and *CCM2* similarly regulate arterial EC phenotypes relevant to CAD. They indicate that downregulation of *TLNRD1* or *CCM2* by common variants may be atheroprotective by conferring a flow-like response and improving barrier function in ECs not exposed to laminar flow, which are most prone to atherogenesis¹³.

V2G2P generalizes to other traits

To test whether the V2G2P approach would generalize to other complex traits, we applied our ABC maps and Perturb-seq data from ECs to study two other vascular traits: mean arterial blood pressure and pulse pressure (Methods). The V2G2P enrichment test identified programs significantly associated with each of these two traits, which were distinct from those we identified for CAD (Extended Data Fig. 12a and Supplementary Table 21). For example, for pulse pressure, the V2G2P test identified significant enrichment for program 50 (TGF β response, FDR = 0.0046) and program 29 (EDN1, wound healing, FDR = 0.0316), and identified genes known to regulate vascular tone and stiffness such as *FHL2*, *SMAD3* and *TGFB1* (Extended Data Fig. 12a). These observations confirm that V2G2P does not simply identify generic EC programs but identifies different cell-type-specific programs relevant to each vascular trait.

To test whether our approach is generalizable to other cell types, we applied V2G2P to study seven traits related to red blood cells using ABC maps⁹ and a recent genome-scale Perturb-seq dataset from K562 erythroleukaemia cells¹⁹ (Extended Data Fig. 12b and Supplementary Table 22). The results showed that different traits had significant enrichment for different, relevant programs. For example, genes linked to variants associated with mean corpuscular haemoglobin were most significantly enriched in K562 program 13, which included many haemoglobin genes and known regulators (including *GFI1B* and *CBFA2T3*). By contrast, variants associated with platelet count showed most significant enrichment in K562 program 4. Genes in this program showed high promoter enrichment of motifs for the known megakaryocyte

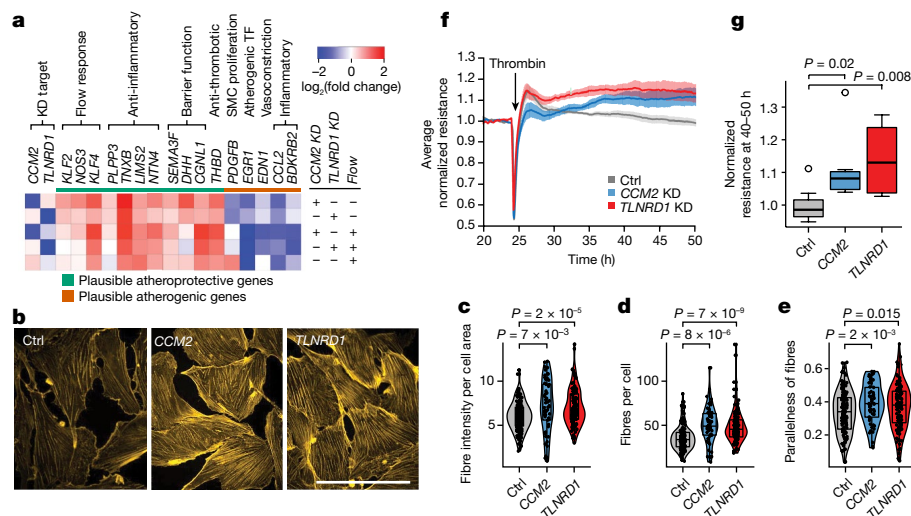


Fig. 6 | *CCM2* and *TLNRD1* knockdown mimic the atheroprotective effects of laminar flow in ECs. **a, Heatmap of genes strongly regulated by both *TLNRD1* and *CCM2* that affect CAD-relevant EC functions (as per Fig. 5c), in CRISPRi telHAECs with the indicated treatments versus control (Ctrl) cells in static culture. **b**, Representative maximum projection images of phalloidin-stained CRISPRi telHAECs with control, *TLNRD1* or *CCM2* guides. Scale bar, 100 μm. **c**, Quantitation of actin fibre (phalloidin stain) intensity per cell area (Methods). $N = 145$ (control), 47 (*CCM2*) and 117 (*TLNRD1*). Boxplots: centre line, median; box limits, upper and lower quartiles. P values: two-sided t -test. **d**, As in **c**, but showing the number of actin fibres per cell. **e**, As in **c**, but showing the parallel arrangement of actin fibres. A score of 0 indicates randomly oriented fibres,**

and a score of 1 indicates that all fibres in a cell are parallel to each other. **f**, TEER measurements for CRISPRi telHAECs with the indicated guides (two guides per target), each normalized to average resistance over the 4 h before thrombin was added to disrupt cell junctions. $N = 8$ (control), 7 (*CCM2* KD) and 6 (*TLNRD1* KD). Ranges indicate s.e.m. **g**, Boxplot of normalized TEER signals from **f** averaged for hours 45 to 50 (20–25 h after thrombin treatment). Quantification as in **c**. Boxplot whiskers indicate $1.5 \times$ the interquartile range and points indicate outliers. These observations suggest that the effects of *CCM2* KD on barrier function may differ between human arterial ECs (where atherosclerosis develops) and human dermal microvascular ECs, where *CCM2* KD has been found to decrease resistance^{51,52}.

regulators *SP1* and *SP3*, and included genes known to be involved in megakaryocyte differentiation and platelet count such as *VASP* and *TPM4* (Extended Data Fig. 12b and Supplementary Table 22).

In summary, variants associated with different traits map onto different programs corresponding to relevant biological pathways, even within a single cell type, and the V2G2P approach can be successfully applied to a second cellular model. Thus, the V2G2P analysis pipeline is likely to be generally applicable to provide insights into complex traits and cell types beyond CAD.

Discussion

GWAS have identified hundreds of loci for many common complex diseases such as CAD. An emerging paradigm for understanding their function is that risk variants should map onto genes that act together in biological pathways^{1–6}. Yet, such pathway-level convergence has been difficult to identify for many diseases because existing approaches can be limited to studying one gene or pathway at a time, underpowered and/or biased towards rediscovering known genes and pathways.

Our study introduced a new method to address this challenge, in which we built unbiased maps of genome function using epigenomic data and Perturb-seq and then combined these maps to identify convergence of risk variants onto pathways. The method is systematic in that it measures the full transcriptomic effects of all genes in all relevant GWAS loci, facilitating the discovery of new disease-associated pathways and new functions for uncharacterized genes (Supplementary Note 5). Applying this method to ECs revealed that 43 out of 306 CAD GWAS signals indeed converge onto 5 transcriptional programs, all related to CCM signalling. Two newly prioritized CAD genes, *CCM2* and *TLNRD1*, strongly regulated these programs (which included dozens of other CAD genes), showed highly similar transcriptional and cellular phenotypes, and physically interacted with one another. This strong signature of molecular convergence identified the poorly characterized gene *TLNRD1* as a new member of the CCM signalling pathway,

implicated the CCM pathway in genetic risk for CAD and demonstrated that Perturb-seq identifies transcriptional programs that help interpret disease risk variants.

Our data suggest a model whereby protective alleles affecting *CCM2* and *TLNRD1* downregulate the activity of the CCM complex in ECs, increasing the expression of atheroprotective genes (including two V2G2P-prioritized gold standard genes, *NOS3* and *PLPP3* (refs. 47,48)) and downregulating atherogenic genes (Figs. 3, 5c and 6a and Supplementary Table 15). Of note, downregulation of *CCM2* or *TLNRD1* in cells in static culture mimicked the effect of laminar flow, promoting atheroprotective gene regulation, actin stress fibre formation and endothelial barrier function (Fig. 6). Given that atherosclerosis in vivo predominantly develops in areas of disrupted or turbulent blood flow¹³, these data suggest that reduction of *CCM2* or *TLNRD1* function might confer a resistant phenotype on ECs precisely in those regions of disrupted flow that are most prone to atherogenesis.

The convergence of 43 CAD risk loci onto CCM signalling related to flow responses in ECs (a number of loci comparable to the 78 CAD loci that are associated by GWAS with circulating lipid levels) suggests that this is a key mechanism that controls risk for CAD in the human population. We anticipate that future application of the V2G2P approach in other atherosclerosis-relevant cell types, including smooth muscle cells and monocytes, will be a powerful approach to provide a more comprehensive analysis of genetic risk for CAD.

The new link we found between CCM signalling and common, polygenic CAD is in the opposite direction of its known role in rare, monogenic CCM disease. That is, whereas complete loss of function of CCM complex proteins causes cavernous malformations in the brain and spinal cord, our results indicate that common variant alleles that quantitatively downregulate CCM complex function in arterial ECs reduce CAD risk. Further studies in vivo will be required to understand how CCM signalling has such different effects in these two diseases. Additionally, our finding that *TLNRD1* is a new regulator of CCM signalling suggests that future studies could evaluate mutations in *TLNRD1* as

an alternative possible cause of disease in the 47% of sporadic CCM cases with multiple lesions that lack pathogenic mutations in the three previously known members of the CCM complex⁴⁹.

In summary, our approach establishes a new, generalizable path to systematically link risk variants to disease genes and to convergent transcriptional programs, providing a rich foundation for further studies to dissect new disease mechanisms. By applying Perturb-seq and the V2G2P approach across many cell types and states relevant to various complex diseases, it should be possible to nominate causal disease genes for a large fraction of GWAS loci and map how they converge onto particular cellular pathways. Such a project is becoming increasingly feasible and would provide a foundation for systematic efforts to leverage human genetic data to discover disease mechanisms.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41586-024-07022-x>.

- Uffelmann, E. et al. Genome-wide association studies. *Nat. Rev. Methods Primers* **1**, 59 (2021).
- Visscher, P. M. et al. 10 years of GWAS discovery: biology, function, and translation. *Am. J. Hum. Genet.* **101**, 5–22 (2017).
- Claussnitzer, M. et al. A brief history of human disease genetics. *Nature* **577**, 179–189 (2020).
- de Leeuw, C. A., Mooij, J. M., Heskes, T. & Posthuma, D. MAGMA: generalized gene-set analysis of GWAS data. *PLoS Comput. Biol.* **11**, e1004219 (2015).
- Weeks, E. M. et al. Leveraging polygenic enrichments of gene features to predict genes underlying complex traits and diseases. *Nat. Genet.* **55**, 1267–1276 (2023).
- Pers, T. H. et al. Biological interpretation of genome-wide association studies using predicted gene functions. *Nat. Commun.* **6**, 5890 (2015).
- Westra, H.-J. & Franke, L. From genome to function by studying eQTLs. *Biochim. Biophys. Acta* **1842**, 1896–1902 (2014).
- Maurano, M. T. et al. Systematic localization of common disease-associated variation in regulatory DNA. *Science* **337**, 1190–1195 (2012).
- Nasser, J. et al. Genome-wide enhancer maps link risk variants to disease genes. *Nature* **593**, 238–243 (2021).
- van der Harst, P. & Verweij, N. Identification of 64 novel genetic loci provides an expanded view on the genetic architecture of coronary artery disease. *Circ. Res.* **122**, 433–443 (2018).
- Tcheandjieu, C. et al. Large-scale genome-wide association study of coronary artery disease in genetically diverse populations. *Nat. Med.* **28**, 1679–1692 (2022).
- Aragam, K. G. et al. Discovery and systematic characterization of risk variants and genes for coronary artery disease in over a million participants. *Nat. Genet.* **54**, 1803–1815 (2022).
- Gimbrone, M. A. Jr & Garcia-Cardena, G. Endothelial cell dysfunction and the pathobiology of atherosclerosis. *Circ. Res.* **118**, 620–636 (2016).
- Gupta, R. M. et al. A genetic variant associated with five vascular diseases is a distal regulator of endothelin-1 gene expression. *Cell* **170**, 522–533.e15 (2017).
- Turner, A. W. et al. Single-nucleus chromatin accessibility profiling highlights regulatory mechanisms of coronary artery disease risk. *Nat. Genet.* **54**, 804–816 (2022).
- Pepin, M. E. & Gupta, R. The role of endothelial cells in atherosclerosis: insights from genetic association studies. *Am. J. Pathol.* <https://doi.org/10.1016/j.ajpath.2023.09.012> (2023).
- Dixit, A. et al. Perturb-seq: dissecting molecular circuits with scalable single-cell RNA profiling of pooled genetic screens. *Cell* **167**, 1853–1866.e17 (2016).
- Adamson, B. et al. A multiplexed single-cell CRISPR screening platform enables systematic dissection of the unfolded protein response. *Cell* **167**, 1867–1882.e21 (2016).
- Replogle, J. M. et al. Mapping information-rich genotype–phenotype landscapes with genome-scale Perturb-seq. *Cell* **185**, 2559–75.e38 (2022).
- Datlinger, P. et al. Pooled CRISPR screening with single-cell transcriptome readout. *Nat. Methods* **14**, 297–301 (2017).
- Bouis, D., Hospers, G. A., Meijer, C., Molema, G. & Mulder, N. H. Endothelium in vitro: a review of human vascular endothelial cell lines for blood vessel-related research. *Angiogenesis* **4**, 91–102 (2001).
- Fulco, C. P., Nasser, J., Jones, T. R. & Munson, G. Activity-by-contact model of enhancer–promoter regulation from thousands of CRISPR perturbations. *Nat. Genet.* **51**, 1664–1669 (2019).
- Norman, T. M. et al. Exploring genetic interaction manifolds constructed from rich single-cell phenotypes. *Science* **365**, 786–793 (2019).
- Morris, J. A. et al. Discovery of target genes and pathways at GWAS loci by pooled single-cell CRISPR screens. *Science* **380**, eadh7699 (2023).
- Kotliar, D. et al. Identifying gene expression programs of cell-type identity and cellular activity with single-cell RNA-seq. *eLife* **8**, e43803 (2019).
- Nichol, D. & Stuhlmann, H. EGFL7: a unique angiogenic signaling factor in vascular development and disease. *Blood* **119**, 1345–1352 (2012).
- Brütsch, R. et al. Integrin cytoplasmic domain-associated protein-1 attenuates sprouting angiogenesis. *Circ. Res.* **107**, 592–601 (2010).
- Finucane, H. K. et al. Partitioning heritability by functional annotation using genome-wide association summary statistics. *Nat. Genet.* **47**, 1228–1235 (2015).
- Stolze, L. K. et al. Systems genetics in human endothelial cells identifies non-coding variants modifying enhancers, expression, and complex disease traits. *Am. J. Hum. Genet.* **106**, 748–763 (2020).
- Wünnemann, F. et al. Multimodal CRISPR perturbations of GWAS loci associated with coronary artery disease in vascular endothelial cells. *PLoS Genet.* **19**, e1010680 (2023).
- Stacey, D. et al. ProGeM: a framework for the prioritization of candidate causal genes at molecular quantitative trait loci. *Nucleic Acids Res.* **47**, e3 (2019).
- Jagadeesh, K. A. et al. Identifying disease-critical cell types and cellular processes by integrating single-cell RNA-sequencing and human genetics. *Nat. Genet.* **54**, 1479–1492 (2022).
- GTEX Consortium. The GTEx Consortium atlas of genetic regulatory effects across human tissues. *Science* **369**, 1318–1330 (2020).
- Snellings, D. A. et al. Cerebral cavernous malformation: from mechanism to therapy. *Circ. Res.* **129**, 195–215 (2021).
- Zhou, Z. et al. The cerebral cavernous malformation pathway controls cardiac development via regulation of endocardial MEK3 signaling and KLF expression. *Dev. Cell* **32**, 168–180 (2015).
- Riolo, G., Ricci, C. & Battistini, S. Molecular genetic features of cerebral cavernous malformations (CCM) patients: an overall view from genes to endothelial cells. *Cells* **10**, 704 (2021).
- Gingras, A. R. et al. Central region of talin has a unique fold that binds vinculin and actin. *J. Biol. Chem.* **285**, 29577–29587 (2010).
- Cowell, A. R. et al. Talin rod domain-containing protein 1 (TLNDR1) is a novel actin-bundling protein which promotes filopodia formation. *J. Cell Biol.* **220**, e202005214 (2021).
- Luck, K. et al. A reference map of the human binary protein interactome. *Nature* **580**, 402–408 (2020).
- Fisher, O. S. et al. Structural basis for the disruption of the cerebral cavernous malformations 2 (CCM2) interaction with Krev interaction trapped 1 (KRIT1) by disease-associated mutations. *J. Biol. Chem.* **290**, 2842–2853 (2015).
- Draheim, K. M. et al. CCM2–CCM3 interaction stabilizes their protein expression and permits endothelial network formation. *J. Cell Biol.* **208**, 987–1001 (2015).
- Zhou, Z. et al. Cerebral cavernous malformations arise from endothelial gain of MEK3–KLF2/4 signalling. *Nature* **532**, 122–126 (2016).
- Renz, M. et al. Regulation of β 1 integrin–Klf2-mediated angiogenesis by CCM proteins. *Dev. Cell* **32**, 181–190 (2015).
- Donat, S. et al. Heg1 and Ccm1/2 proteins control endocardial mechanosensitivity during zebrafish valvulogenesis. *eLife* **7**, e28939 (2018).
- Khera, A. V. et al. Gene sequencing identifies perturbation in nitric oxide signaling as a nonlipid molecular subtype of coronary artery disease. *Circ. Genom. Precis. Med.* **15**, e003598 (2022).
- Macek Jilkova, Z. et al. CCM proteins control endothelial β 1 integrin dependent response to shear stress. *Biol. Open* **3**, 1228–1235 (2014).
- Knowles, J. W. et al. Enhanced atherosclerosis and kidney dysfunction in $eNOS^{-/-}Apoe^{-/-}$ mice are ameliorated by enalapril treatment. *J. Clin. Invest.* **105**, 451–458 (2000).
- Mueller, P. A. et al. Coronary artery disease risk-associated *Plpp3* gene and its product lipid phosphate phosphatase 3 regulate experimental atherosclerosis. *Arterioscler. Thromb. Vasc. Biol.* **39**, 2261–2272 (2019).
- Denier, C. et al. Genotype–phenotype correlations in cerebral cavernous malformations patients. *Ann. Neurol.* **60**, 550–556 (2006).
- Finak, G. et al. MAST: a flexible statistical framework for assessing transcriptional changes and characterizing heterogeneity in single-cell RNA sequencing data. *Genome Biol.* **16**, 278 (2015).
- Whitehead, K. J. et al. The cerebral cavernous malformation signaling pathway promotes vascular integrity via Rho GTPases. *Nat. Med.* **15**, 177–184 (2009).
- Zheng, X. et al. CCM3 signaling through sterile 20-like kinases plays an essential role during zebrafish cardiovascular development and cerebral cavernous malformations. *J. Clin. Invest.* **120**, 2795–2804 (2010).

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Methods

Cell culture and creation of CRISPRi teloHAECs

TeloHAECs were purchased from the American Type Culture Collection (ATCC, CRL-4052) and grown in Lifeline VEGF endothelial cell medium (LL-0005) with 1× penicillin–streptomycin. Cells were plated at a density of $0.5\text{--}1.0 \times 10^6$ cells per 10 cm plate and split before reaching 4×10^6 per plate (after 3–4 days). To create the teloHAEC CRISPRi line, cells were transduced with lentiviral vectors containing (1) doxycycline-inducible (tetracycline operator controlled) dCas9-KRAB-BFP (CRISPRi machinery, which targets epigenetic repressors to efficiently silence enhancers or promoters^{53–55}; Addgene, 85449) and (2) rtTA (tetracycline activator) with a hygromycin marker (Addgene, 66810). After hygromycin selection ($250 \mu\text{g ml}^{-1}$ for 4 days), cells were treated with $1 \mu\text{g ml}^{-1}$ doxycycline (a stable tetracycline analogue) for 3 days before FACS for the top 15% of BFP-positive cells. After a period in culture without doxycycline, cells were treated again with doxycycline and re-sorted (Extended Data Fig. 1e). Diagnostic FACS performed immediately before the Perturb-seq screen showed no leaky BFP expression in the absence of doxycycline, and 93% BFP-positive cells in the presence of doxycycline (Extended Data Fig. 1f). CRISPRi teloHAECs were passaged for routine maintenance in the absence of doxycycline. Eahy926 cells (a HUVEC and A549 hybrid line) and HEK293T cells were purchased from the ATCC (CRL-2922 and CRL-3216, respectively) and grown in DMEM with 10% FBS. To study responses to CAD-associated cytokines, cells were untreated (control) or treated with 10 ng ml^{-1} recombinant human IL-1 β (Millipore IL038), 10 ng ml^{-1} recombinant TNF (Millipore GF023) or with normal medium lacking VEGF (for teloHAECs) or supplemented with VEGF (1× concentration from LifeLine VEGF medium for Eahy926 cells) for 24 h. All cell lines were mycoplasma-free. In addition to authentication by the provider, we further authenticated each line by analysis of microscopic morphology (for example, teloHAECs displayed the characteristic EC cobblestone morphology and showed localization of VE-cadherin to endothelial cell junctions), functionality (high transfectability and protein expression for HEK293T), mapping of ATAC-seq, ChIP-seq and RNA-seq reads to the human genome and RNA-seq profiles and responses (for example, for Eahy926 and teloHAEC, expression of EC-specific genes and observation of previously observed responses to stimuli, such as IL-1 β).

Bulk RNA-seq

Total RNA was collected from teloHAECs (parental or CRISPRi lines) using a Qiagen RNeasy kit (74016, Qiagen), DNase treated (TURBO DNase, Invitrogen, AM2238, 15 min at 37 °C) and purified on MyOne Silane beads. For flow response and MAP3K3 knockdown studies, DNase treatment was performed on the spin column between two buffer RW1 washes for 20 min at room temperature with Purelink DNase (Invitrogen, 12185010) using 10 μl in 80 μl of 1× buffer. mRNA was purified from 400 ng to 1 μg of total RNA using the NEBNext Poly(A) mRNA Magnetic Isolation module (NEB), processed for RNA-seq library generation using a NEBNext Ultra II RNA Library kit for Illumina (NEB) and sequenced to a depth of 10–30 million reads per library. Reads were mapped to the human hg19 genome build, and counts per gene tables assembled as previously described^{9,22} or, for flow response and MAP3K3 knockdown studies, using kallisto⁵⁶. Differential expression calls were made using Limma Voom⁵⁷ (for parental teloHAECs and Eahy926) or edgeR⁵⁸ (for all other libraries). Bulk RNA-seq data are available from the Gene Expression Omnibus (GEO) database with accession numbers GSE210522 (cytokine treatment of parental lines and single guide knockdown) GSE232400 (flow response and MAP3K3 double knockdown studies).

ATAC-seq, H3K27ac ChIP-seq and identification of teloHAEC enhancers

For ATAC-seq, 1 well of a 12-well plate (around 200,000 cells) was directly lysed using a custom TN5 buffer (33 mM Tris acetate pH 7.8,

66 mM potassium acetate, 10 mM magnesium acetate, 16% dimethylformamide and 0.1% NP40). Next, 47.5 μl of lysed cells was added to 2.5 μl Tn5 tagmentation enzyme (Illumina) and incubated at 37 °C for 1 h, and the reaction was stopped by adding 20 μl buffer RLT (Qiagen). Products were purified by adding 1.8 volumes Ampure XP beads (Beckman-Coulter) and magnetic separation of beads, followed by two 80% ethanol washes, brief drying of pellets and resuspension in 23 μl water. Barcoded ATAC-seq libraries were then generated as previously described^{9,22} and sequenced to a depth of 10–20 million reads per library. ChIP for H3K27ac was performed as previously described^{9,22} using anti-H3K27ac antibody (39685, Active Motif) at 1:200 dilution. ChIP-seq libraries were prepared using a KAPA Hyper Prep kit (KAPA Biosystems). ATAC-seq libraries were prepared in biological triplicate, and ChIP-seq libraries in biological duplicate. For both types of libraries, reads were mapped to the human genome (hg19 build) using Bowtie2, and peaks were identified using MACS2, essentially as previously described^{9,22}. Raw and processed data are available from the GEO database with accession numbers GSE210489 (ATAC-seq) and GSE210491 (ChIP-seq). Enhancers and their predicted target genes were identified by applying the ABC model to these data, using ATAC-seq and H3K27ac ChIP-seq as the measures of enhancer activity, and using a cross-cell type average of Hi-C maps as the measure of 3D enhancer–promoter contact frequency (<https://github.com/broadinstitute/ABC-Enhancer-Gene-Prediction>)^{7,8}. An ABC fractional score threshold of ≥ 0.015 was used⁹.

Selection of genes for the Perturb-seq library

We constructed a library of promoter-targeted CRISPRi guides to all potential causal CAD genes (Fig. 1b). First, we identified all coding genes within a 1 Mb window surrounding the lead SNPs from CAD loci identified in either or both of two previous studies^{10,12} that were expressed in teloHAECs (1+ transcript per million (TPM) from bulk RNA-seq). If fewer than 2 expressed genes were found within 500 kb upstream or downstream of the lead SNP, the window was expanded to include the closest 2 genes to each side (for a total of 1,661 genes). Noncoding genes were generally excluded, unless there was strong evidence for regulatory functions, particularly in ECs. Selected genes with TPM < 1 were included, particularly if they were known to be important for CAD in tissues where they were more highly expressed (for example, PCSK9) or were regulated by IL-1 β in bulk RNA-seq data in teloHAECs (FDR < 0.05, fold change > 1.3). As negative controls, we included guides targeting 48 coding genes expressed in other cell types but not detectably expressed in ECs, and the 132 expressed coding genes within 1 Mb of 16 randomly selected lead SNPs associated with inflammatory bowel disease, Crohn's disease or ulcerative colitis⁵⁹, and which did not overlap with CAD loci. As positive controls, and to aid in connecting candidate CAD genes to known pathways in ECs, we targeted the promoters of an additional 284 genes with known roles in a wide range of CAD-relevant EC functions such as barrier formation, TGF β signalling and inflammation, and major classes of expressed TFs and common essential genes. We also targeted an additional 160 promoters of expressed genes predicted to be regulated by EC enhancers containing fine-mapped variants associated with other disease phenotypes expected to be modulated by ECs (migraine, blood clotting in leg, systolic blood pressure, diastolic blood pressure and mean arterial blood pressure, from the UK Biobank; Supplementary Table 23). This process produced a total of 2,285 genes, some of which were members of more than one category.

Guide library production and validation

sgRNA guides were designed to target promoters of the chosen CAD and control genes (15 guides spanning from –150 to +100 relative to the transcription start site (TSS)) using our established pipeline^{9,22} (<https://github.com/EngreitzLab/CRISPRDesigner>). We included 400 non-targeting guides (which do not have close matches to any region in the human genome) and 600 safe-targeting guides (targeting non-genic

regions lacking enhancer marks)⁵³. Because teloHAECs are puromycin resistant, we adapted the CROP-opti vector²⁰ (Addgene, 106280) for blasticidin resistance (CROP-opti Blast) by digesting the vector with BsiWI and MluI, PCR-amplifying the blasticidin-resistance gene from lenti-dCas-VP64_Blast (Addgene, 61425) with added homology arms and performing Gibson assembly (Gibson master mix, New England Biolabs). To create CROP-opti-BC-Blast, we added HyPR-Seq barcodes between the WPRE element and the U6 promoter of CROP-opti-Blast, as previously described⁶⁰. A pool of oligonucleotides encoding the guide sequences, plus extensions with homology to the U6 promoter and downstream scaffold (TATCTGTGGAAAGGACGAAACACCG and GTTAAAGAGCTATGCTGGAACAGCATAG) was synthesized by Agilent Technologies and cloned into Crop-Opti-BC-Blast by Gibson assembly and bacterial electroporation as previously described⁵³ at an average coverage of 202 transformants per guide. Note that because the vector was prepared from a single clone, diversity of the HyPR-seq barcodes (which were not required for Perturb-seq) was not preserved. The library was sequenced and shown to include all 37,637 designed guides with relatively equal coverage of each (the difference in count frequency between the top and bottom 10th percentiles of guides was 2.8). A lentiviral library was produced using a standard 3-plasmid protocol⁵³ at a scale to yield 10 ml of virus stored in aliquots at -80 °C, with each aliquot thawed only once.

Experimental procedure of Perturb-seq

To transduce this library into CRISPRi teloHAECs, cells were resuspended in medium containing 10 µg ml⁻¹ polybrene at a density of 1 × 10⁶ cells per ml, mixed with virus and plated 4 ml per well to 6-well plates, centrifuged at 2,000 r.p.m. for 2 h at 30 °C and incubated at 37 °C for 2 h before addition of another 4 ml medium without polybrene. The next day, cells were collected and plated onto 15 cm plates and treated with 15 µg ml⁻¹ blasticidin for 4 days. The effective viral titre was determined using this same protocol, and a volume of virus was chosen that gave a final measured 15.7% infection rate (such that most successfully transduced cells have only 1 gRNA). For the Perturb-seq study, 127.5 million CRISPRi teloHAECs were transduced and selected for blasticidin resistance for a coverage of approximately 360 cells per guide (as back-calculated from the yield at the first post-blasticidin split, using the 36.7 h doubling time observed in routine culture) to 461 cells per guide (as estimated from the initial number of cells and infection rate). After blasticidin selection, cells were treated with 2 µg ml⁻¹ doxycycline for 5 days (plating 18 × 10⁶ cells at each split to maintain complexity of the library). We reasoned that because atherosclerotic plaques develop slowly, the longer-term transcriptional effects of causal CAD gene disruption would provide the greatest insights into disease mechanisms. Thus, although we found that knockdown of guide-targeted genes is near maximal after 2 days of doxycycline treatment (inducing the CRISPRi machinery), we treated guide-containing cells with 2 µg ml⁻¹ doxycycline for 5 days to measure the longer-term consequences of each perturbation. We also used this same 5-day doxycycline treatment protocol for downstream validation studies (for example, bulk RNA-seq of sgRNA clones).

The presence of gRNAs in cells enables multiplets (droplets containing two or more cells) to be unambiguously identified as droplets containing more than one guide. This allowed us to load about tenfold more cells per 10X Genomics lane than the maximum number recommended in the manufacturer's protocol. In brief, cells were collected, resuspended in PBS with 1% BSA, counted and loaded at 150,000 cells per lane on a 10X Genomics Chromium Controller using a 3' scRNA-seq V3 kit (20 lanes, for a total of 3 million cells). Cells were isolated in 2 batches, with 6 lanes for the first batch and 14 lanes, across 2 cassettes, for the 2nd batch, 6 h later. scRNA-seq libraries were generated using the 10X Genomics protocol and given lane-specific indexes. From the initial amplified cDNA, we used a two-stage PCR protocol to generate 'dialout' libraries for each lane. Because the CROP-seq vector expresses a Pol II

polyadenylated transcript that ends immediately downstream of the guide sequence, the dialout libraries identify the gRNA sequences associated with each droplet²⁰. PCR1 oligonucleotides for the guide dialout PCR were as follows: CTACACGACGCTCTTCCGATCT and GTGACTG GAGTTCAGACGTGTGCTCTTCCGATCTGTGGAAAGGACGAAACACC. PCR2 oligonucleotides were AATGATACGGCGACCGACCGAGATCT ACACCTTTTCCCTACACGACGCTC and CAAGCAGAAGACGGCATACG AGAT-8 bp index sequence-GTGACTGGAGTTCAG.

Assignment of gRNAs to cells

To obtain complete information about guide assignments, dialout libraries were sequenced to approximately 40-fold saturation. Guides were identified from read 1 sequences using Bowtie2 to align dialout reads to a 'genome' composed of all 37,637 guide sequences, requiring no-mismatches. Aligning read 1 and read 2 sequences linked gRNA sequences with cell barcodes (CBCs, unique to each bead per droplet) and UMIs. To avoid low-frequency PCR chimeras, we required that each CBC-UMI-guide combination be duplicated at least 4 times (Extended Data Fig. 1g). We then identified the guides associated with each CBC and the number of different UMIs for each CBC-guide combination. We selected four UMIs for any single guide as the threshold to call a cell as containing a guide (Extended Data Fig. 1h). We defined singlets (one cell and one guide per CBC) as having ≥4 UMIs for the most frequent guide and ≥4 less than this for the second most frequent guide (choosing these thresholds to give a good balance between power to detect transcriptional effects and accuracy in measuring the magnitude of these effects, as described under 'Selection of singlet thresholds', below). Doublets and higher multimers were cells with ≥4 UMIs for the top guide, and one or more additional guides with more than one quarter this number of UMIs. The counts of identified singlets, doublets and higher multiplets are shown in Extended Data Fig. 1i.

scRNA-seq data pre-processing and subsetting to singlets

scRNA-seq libraries were sequenced on two Illumina NovaSeq S4 flow cells, producing 20,245,734,673 total reads across all 20 libraries. The FASTQ files were processed on the 10X Cloud to run CellRanger count with the hg38 reference genome. We used the 'filtered' features (that is, CBCs corresponding to droplets that contain a cell) and combined the outputs from all 20 10X lanes into a single genes × cell matrix. This analysis identified 822,156 cell-containing droplets (see Supplementary Table 7 for other CellRanger output statistics). To measure the effects of individual guides on individual cells, we selected only those CBCs identified in the dialout analysis as corresponding to singlet cells. This identified 214,449 singlets (droplets containing one cell and one guide), defined as 4+ UMIs for the top guide and ≥4-fold fewer UMIs for any other guide (Extended Data Fig. 1i). This gave an average of 5.7 cells per guide and 85.5 cells per target promoter. The average sequencing depth was 10,870 transcriptome-mapped UMIs per singlet cell, and 929,000 transcript UMIs, across all 15 guides for each target promoter. Raw and processed data and supplementary files for downstream analyses are available from the GEO database with accession number GSE210681.

Estimation of fitness effects

To estimate the fitness effects of guides, we compared the relative frequency of all 15 guides to a given target in the original library to the frequency of the same guides in singlet cells, and estimated significance by Benjamini-Hochberg adjusted binomial tests. Essential genes were defined as those that scored as fitness reducing in five out of seven tested lines as previously described⁶¹.

Differential gene expression analysis and knockdown efficacy

To measure the differential effects of guides to specific target promoters on individual genes, we used edgeR⁵⁸, with settings for scRNA-seq from a previously published study⁶², comparing all singlet cells with guides

to each target to all singlet cells with any of the 1,000 non-targeting and safe-targeting guides. Genes with fewer than 10 UMI counts across all singlet cells were excluded from the analysis. To control for possible batch effects, we included the 10X lane number as a covariate. For average knockdown efficacy for each perturbation (across all 15 guides), we used the $\log_2$ fold change and P values reported by edgeR. To measure the knockdown efficacy of individual guides, we performed binomial tests on the number of transcripts for the guide's target in singlet cells with that guide (hits), and all transcripts in singlets with that guide (tests) versus a background frequency of (transcripts to the target in other singlet cells)/(all transcripts in other singlet cells). Note that with an average of 5.7 cells per guide, assigning significance for knockdown effects of individual guides was only possible for genes with high expression in unperturbed cells (for example, TPM > 100). To identify perturbations with a significant effect on the transcriptome, we used the edgeR results for the 48 negative control promoters (for genes not detectably expressed in teloHAECs) to estimate the number of differentially expressed (DE) genes found by chance at thresholds of nominal P value < 0.01 and fold change > 1.15. Perturbations with a significant effect on the transcriptome (across all 15 guides to each target) were defined as having more DE genes, by these same thresholds, than the 48 non-expressed controls (using binomial tests with a background rate equal to the average DE gene count for controls over all genes tested, and multiple hypothesis correction using the Benjamini–Hochberg method).

Selection of singlet thresholds

Expression of the CROP-seq guide mRNA in teloHAECs is lower than in some other cell lines, such as K562 and HEK293T cells¹², which results in the absence of a clear gap between noise (low UMI–CBC–guide combinations that are probably PCR chimeras) and higher UMI-count true guide reads (Extended Data Fig. 1h). We predicted that reducing stringency for singlet calls could potentially reduce power to detect perturbation effects on transcription (owing to increased noise from mis-calling some true doublets as singlets) or could increase power (by increasing the total number of called singlets analysed). To test which of these was true, we measured the correlation between differential expression calls for cells with guides to a given target in the full Perturb-seq library versus a smaller pilot library tested in resting teloHAECs, reasoning that parameters that improved the correlation between these separate studies would also increase the power of the full-scale library to detect real transcriptional effects. Information about guides and raw and processed data for the '200 gene' pilot library can be found in the GEO database with accession number GSE212396. For the pilot library, we chose singlets with the very stringent threshold of 6 UMIs for the top guide and more than 5-fold less for the next most frequent guide ('6<5×'). For the full Perturb-seq dataset, we chose 4 UMIs for the top guide and equal to or more than 4-fold less than the next most frequent guide ('4<=4×', our final applied standard, yielding 214,449 singlets) or the relaxed thresholds '3<=3×' (284,466 singlets) and '2<=2×' (389,792 singlets). We identified 37 gene targets that were shared between libraries and showed a FDR < 0.1 effect on the transcriptome in the full Perturb-seq 4<=4× dataset (measured as described above). We then ran EdgeR^{58,62} for differential expression testing (cells with guides to each of these 37 targets versus cells with control guides) for each library and singlet definition (pilot 6<5×, or full library 4<=4×, 3<=3× and 2<=2×). Then, for all genes called as DE in either the pilot library or the full library (raw P value < 0.01), we measured the correlation in $\log_2$ (fold changes) between the pilot and full scale data, repeating this analysis for each singlet definition.

Last, we measured the difference in correlation coefficients (R) between the relaxed threshold comparisons (pilot versus full library 3<=3×, and pilot versus full library 2<=2×) and the base comparison (pilot versus full library 4<=4×). The median correlation between

pilot and full-scale studies significantly improved with the relaxed singlet thresholds (Supplementary Fig. 2a, with significance assessed by two-sided t -test). This result indicated that the increased number of called singlets with the relaxed thresholds increased the power to detect real transcriptional effects, despite an expected increase in doublets mis-assigned as singlets. Plotting the change in R for each target for the 2<=2× singlet definition ($(R$ for pilot versus full library 2<=2×) – R for pilot versus full library 4<=4×, y axis) against the R value for the base correlation (between the pilot and the 4<=4× full library singlet definition, x axis), we found that in all 13 cases in which R started high (>0.15, probably real correlations between strong transcriptional effects), R increased (Supplementary Fig. 2b). R also increased in all but one case in which it started out negative (correcting anti-correlations probably driven by noise). Weak positive base correlations were adjusted up or down, potentially improving true correlations and correcting spurious ones. As such, relaxed singlet thresholds might improve power to detect reproducible transcriptional changes more than is indicated by simple mean differences in R values. Conversely, lower stringencies reduced the apparent knockdown effect on these target genes themselves (Supplementary Fig. 2c, median $\log_2$ (fold changes) of –0.53 for 4<=4×, –0.41 for 3<=3× and –0.42 for 2<=2×), which is probably due to the fact that a mis-called singlet that was actually 2 cells with different guides would show half-magnitude transcriptional effects of each guide. Reduced singlet thresholds also decreased median $\log_2$ (fold changes) for target genes across all targets in the full-scale library (Supplementary Fig. 2d; –0.368 for the 4<=4× singlet definition, and –0.327 for the 2<=2× singlet definition). Based on these observations, we chose the thresholds of 4 UMIs for the top guide and $\leq 1/4$ this for the next (4<=4×) to provide a good balance between overall power and accurate detection of the magnitude of effects.

Data processing before defining gene programs

To remove noncoding RNA from the analysis, we removed genes with names starting with 'LINC' and gene names with patterns starting with two letters and six digits. We retained cells with a minimum of 200 unique detected genes and a minimum of 200 UMIs. We retained genes detected in a minimum of ten cells.

cNMF

To identify sets of genes that are co-expressed across single cells in a dataset, we used NMF. NMF decomposes an input cell × gene UMI count matrix (X) into a cell × component matrix (W) and a component × gene matrix (H), such that $X = W \cdot H + E$, where E is the error term. The cell × component matrix W represents the contribution of each component to the cell's transcriptional profile, and the component × gene matrix H encodes information about gene expression programs. The number of components (K) is a hyperparameter defined before performing matrix factorization (see below). To account for the fact that the NMF algorithm is a stochastic algorithm that depends on the initial seed, we used a previously described cNMF method²⁵. The cNMF method, after normalizing the expression of each gene to unit standard deviation, has the following steps: factorizes the normalized matrix multiple times (here, 100 repeats); clusters the components from the repeat runs based on their pairwise Euclidean distances; removes the components that show low similarity to any other component (here, threshold on Euclidean distance = 0.2); defines 'consensus components' as the median of each of the component clusters; and recomputes the cell × component matrix W using these consensus components. A technical note about applying the cNMF pipeline as previously described²⁵, we found that including all genes, as opposed to the 2,000 most variable genes, was important for finding certain programs observed only infrequently in the dataset (data not shown). This is because genes for which expression changes in only a small fraction of cells (for example, cells with a particular perturbation) would not end up being included in the 2,000 most variable genes.

Choosing the number of components for cNMF analysis

To choose the free parameter K (number of components; Extended Data Fig. 2d,e), we defined a set of benchmarking statistics and compared the results of cNMF run for $K = [3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 17, 19, 21, 23, 25, 27, 29, 30, 35, 40, 45, 50, 55, 60, 100]$. We ultimately chose $K = 60$ for all downstream analyses.

We examined the following benchmarking statistics:

1. Number of unique gene ontology (GO) terms enriched in program co-regulated genes (Extended Data Fig. 2d, see below).
2. Number of unique enriched TF motifs in the promoters or enhancers of program co-regulated genes (Extended Data Fig. 2e, see below).
3. Number of perturbations significantly regulating any component (Extended Data Fig. 2f).
4. Error of cNMF (difference between the original normalized data and reconstructed data, calculated by taking the sum of squares of the element-wise difference of the data, Extended Data Fig. 2g).
5. Stability of cNMF (a measure of consistency of the components output from repeated runs, represented by the silhouette score²⁵; Extended Data Fig. 2g).

We chose $K = 60$ for further analysis as the number of components that gave a low cNMF error value while near-maximizing each other metric.

Excluding components associated with batch effects

We examined whether some components identified by cNMF were likely to represent batch effects. To do so, we calculated the Pearson correlation between each of the 20 batches (that is, 10X lanes) and the expression of each component across all cells. Based on the distribution of batch $\times$ program Pearson correlation (Supplementary Tables 11 and 13), we assigned 10 components with Pearson correlation > 0.15 as likely to represent batch effects (Extended Data Fig. 2c). We chose the threshold of 0.15 because of the following reasons: (1) most components showed very low correlation with batch; (2) above this threshold, sample correlation with batch greatly increases (indicating particularly strong association with batch for these 10 components); (3) 8 out of 10 of these components were associated with mitochondrial or ribosomal genes—sets of genes commonly observed as batch effects in 10X scRNA-seq; (4) these components showed absolute $R > 0.15$ across multiple 10X lanes (average of 9); and (5) importantly, none of these components showed enrichment by the V2G2P test (described below), and so would not have been identified as significant for CAD risk. We used the remaining 50 components for further analysis. This approach (including batch as a covariate in the differential expression test) has theoretical advantages, in particular reducing bias when groups (here, perturbed genes) are not evenly distributed across batches⁶³.

Defining co-regulated genes for each program

We defined co-regulated genes for each cNMF component as the 300 marker genes with the highest z score regression coefficient as defined by cNMF²⁵. Essentially, cNMF uses a linear regression model to identify coefficients indicating the number of standard deviations that each gene's expression would change with the increased usage of a given component. Consequently, marker genes of a component are those with the highest marker gene regression coefficients (or specificity scores) for that component, and we selected the top 300 of these marker genes as the set of co-regulated genes for each gene expression program (as defined below).

Defining regulators for each program

We tested whether gRNAs targeting a given gene led to a significant change in expression of each component from the cNMF model. We used MAST⁵⁰ to compare the expression of each component in cells carrying gRNAs targeting a given gene versus cells carrying

control gRNAs (1,000 safe-targeting and negative control guides), including the 10X lane as a covariate to account for batch effects. We removed the guides present in fewer than three singlet cells and the perturbations with fewer than two guides. We used the Benjamini–Hochberg method to account for multiple hypothesis testing on the MAST P values (Extended Data Fig. 2h) and assigned regulators of a program as those genes for which perturbation affected component expression with $FDR < 0.05$ accounting for 140,760 total tests (60 programs $\times$ 2,346 perturbations, which includes the 2,285 targeted TSSs and targeted enhancers that were not further analysed in this study; Supplementary Table 6).

To confirm that these FDR values were well calibrated, we also conducted a simulation-based test. For each perturbed gene, we sampled from the control cells (all singlet cells with non-targeting or safe-targeting guides) the same number of cells and compared these sampled cells to the rest of the control cells using the same MAST⁵⁰ procedure. We identified 0 significant regulators in this approach (Extended Data Fig. 1q), which indicated that our $FDR < 0.05$ threshold is a conservative estimate. We also performed the same procedure to estimate the background rate for perturbations called by EdgeR as having a significant effect on the transcriptome^{58,64,65}.

Definition and annotation of gene expression programs

We defined a gene expression program as the set of genes comprising both the 300 co-regulated genes and the significant regulators for each cNMF component. We annotated programs based on features of their co-regulated genes and regulators by manual curation of genes with known biological functions, by enrichment of TF motifs in the promoters and predicted enhancers of co-regulated genes and by GO term enrichment (see below). The manual labels we assigned to each program (for example, program 8, angiogenesis and osmoregulation in Fig. 1c) represent our attempt to annotate the program based on functions of known genes. However, we note that each program includes many genes and regulators that have not been studied in combination before and may represent new, specific gene–pathway relationships that we do not currently have the vocabulary to describe.

Identifying motifs enriched in promoters and enhancers

To identify TFs that might regulate program co-regulated genes, we calculated enrichment of human TF motifs in the sequences of the promoter and enhancers of the top 300 genes ranked by component specificity score (Extended Data Fig. 2e and Supplementary Table 24).

We obtained promoter sequences by taking 500 bp surrounding the TSS as previously annotated²² and enhancer regions from the ABC model at an ABC score threshold of 0.015 in the teloHAEC control condition. For a gene that had multiple enhancers, we counted motif instances across all of its enhancers. To match motifs to sequences, we used HOCOMOCO (v.11) human full scan motifs (https://hocomoco11.autosome.ru/downloads_v11), and Find Individual Motif Occurrences (FIMO) (https://meme-suite.org/meme/meme_5.3.2/tools/fimo), with the default settings, and P value thresholds of 10^{-6} for enhancers or 10^{-4} for promoters.

For a given motif and a given program, we counted the number of occurrences of a motif in the promoter sequences of either the top 300 program co-regulated genes or all expressed genes in teloHAECs and compared these two vectors of motif counts using a two-sided t -test. We computed enrichment by dividing the program gene's average motif match count by the rest of the expressed gene's average motif match count. We tested all pairs of matched motifs (570 for promoter and 590 for enhancer) $\times$ 60 programs, and used the Benjamini–Hochberg method to account for multiple hypothesis testing on the t -test P values.

Determining enrichment of annotated gene sets in components

To determine whether the gene expression programs align with annotated and publicly available pathways, we tested whether the

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co-regulated genes in each component were enriched in gene sets from the Molecular Signatures Database (MSigDB). To do so, we used the clusterProfiler R package⁶⁶ and MSigDB gene sets⁶⁷ (here, the gene sets labelled as 'all' for all gene sets and 'c5' for GO terms only). We filtered the MSigDB gene sets to only those with more than 3 genes and fewer than 800 genes. We annotated each program with the gene sets that showed significant enrichment among the program genes (FDR < 0.05; Supplementary Table 25) and compared the number of gene sets showing significant enrichment as a function of the number of programs K (Extended Data Fig. 2d).

Defining EC-specific programs

To annotate programs as EC-specific, we analysed the degree to which program co-regulated genes were expressed in ECs versus other cell types. We took gene expression TPM data across all available cell types from FANTOM5 and calculated the expression z score of each gene across all cell types. To give each gene an EC specificity score, we calculated the average of all z scores for a gene across EC samples. We defined EC specificity scores for each program as the average of the 300 co-regulated genes' specificity scores and selected 0.19 (90% percentile) as the threshold to call programs as EC-specific (Extended Data Fig. 3c and Supplementary Table 13).

Variance explained by all cNMF components

To jointly quantify the fraction of variance explained by all 60 programs, we compared the residual variance in the dataset after subtracting the consensus matrix factorization to the total variance in the dataset: $V = 1 - \text{Var}(X - WH)/\text{Var}(X)$, where X is the (cell $\times$ gene) normalized data matrix input to cNMF, W is the (cell $\times$ program) usage matrix, H is the (program $\times$ gene) spectra or weight matrix, and matrix variance is defined by summing the column-level or gene-level variances: $\text{Var}(X) = \sum_j \text{Var}(X_j)$. Note that cNMF normalizes the input data so each $\text{Var}(X_j) = 1$.

Variance explained by individual gene programs

To rank gene programs by variance explained, we devised a method to quantify variance explained by NMF or cNMF components separately. For the k 'th program H_k , we consider the effective matrix decomposition given only this program; the effective usage matrix B_k in this case is given simply by orthogonal projection or ordinary least squares: $B_k = XH'_k/||H_k||^2$, where the prime indicates transposition. We then define the variance explained in terms of the residual fraction as above: $V_k = 1 - \text{Var}(X - B_kH_k)/\text{Var}(X)$. Our method may be generalized to any set of programs, but with more than one program, the effective usage matrix must be obtained by non-negative least squares (a single iteration of NMF).

Defining variants in CAD GWAS signals for V2G analysis

CAD lead GWAS variants were derived from previous studies^{10,12}. We excluded lead variants from ref. 10 if the variants were in strong linkage disequilibrium (LD; $r^2 \geq 0.7$) with a lead variant from ref. 12 or were ≤ 5 kb away from a lead variant from ref. 12. A LD expansion was performed to include variants that are both within a 1 Mb window of, and are in strong LD ($r^2 \geq 0.9$) with, any of these lead GWAS variants in the 1000 Genome European ancestry dataset (pLink --ld-window-kb 1000 --ld-window 99999 --ld-window-r2 0.9). For each lead variant, we also included variants prioritized through functionally informed fine-mapping (PIP ≥ 0.1) in either study^{10,12}. We defined a 'GWAS signal' as this collection of variants around, and including, each lead variant.

Identifying CAD variants associated with lipid levels

We classified CAD GWAS signals as lipid or non-lipid based on their association with lipid levels in other GWAS studies because the CAD GWAS signals also associated with lipids are presumed to act through non-ECs such as hepatocytes. For lead signals included in the

previous study¹², we defined a CAD GWAS signal to be associated with lipids if the lead variant was linked to 'LDL-direct', 'triglycerides', 'cholesterol', 'HDL-cholesterol', 'apolipoprotein A', 'apolipoprotein B', 'HDLc' or 'LDLc' in the phenome-wide association scan¹². For GWAS signals exclusively nominated in ref. 10, we used a different procedure, whereby we considered a signal to be associated with lipids if its lead variant was associated ($P < 5 \times 10^{-8}$) with HDLc, LDLc, triglycerides, apolipoprotein A or apolipoprotein B based on GWAS from the UK Biobank (H. Finucane and J. Ulirsch: <https://www.finucanelab.org/data>). We refer to the remaining GWAS signals not associated with lipid levels as 'non-lipid CAD GWAS signals', and focused on this subset of signals as cases in which CAD variants might plausibly act in ECs.

Linking variants to genes

We used a combination of V2G methods to identify a list of genes linked to CAD variants that could plausibly act in ECs. At each CAD GWAS signal, we considered as candidate genes at least two genes upstream or downstream of the lead GWAS SNP, and all the genes within ± 500 kb of the lead variant to be potentially regulated by the GWAS signal. We focused our analysis on protein-coding genes and excluded long noncoding RNAs ('LINC'), gene isoforms ('-AS'), microRNAs ('MIR'), small nuclear RNAs ('RNU') and genes of uncertain functions ('LOC'). To link CAD variants to genes, we intersected the variants with ABC enhancers⁹ in ECs to identify the top 2 genes most likely to be regulated by each variant (highest 2 ABC fractional scores over 0.015). Specifically, we used ABC data, for enhancers and predicted target genes, from teloHAEC and Eahy926 cells (control, or treated with IL-1 β , TNF or VEGF, this study) and from a previous ABC analysis⁹ of HUVEC ('endothelial_cell_of_umbilical_vein_Roadmap', 'endothelial_cell_of_umbilical_vein_VEGF_stim_12_hours-Zhang2013', and 'endothelial_cell_of_umbilical_vein_VEGF_stim_4_hours-Zhang2013' datasets). To account for cell-state-specific regulation that was not predicted by ABC, we also intersected candidate CAD variants at each signal with ATAC peaks and considered the two genes closest to variant-containing peaks as plausibly regulated. We also linked variants to genes if the variant was in a coding sequence or within 10 bp of a splice site annotated in the RefGene database (downloaded from UCSC Genome Browser on 24 June 2017)⁶⁸. We confirmed that these candidate CAD variants were significantly enriched for matching any or all of these criteria (Extended Data Fig. 5e). We identified 254 candidate CAD genes, defined as 'genes with V2G links', at 125 out of 228 non-lipid CAD GWAS signals (Supplementary Table 1).

Transcription profile comparisons between teloHAECs and human right coronary artery ECs

To confirm the validity of teloHAECs as a relevant model for ECs in human coronary artery (where atherosclerosis that leads to CAD develops), we compared scRNA-seq gene expression from control guide carrying teloHAECs from our Perturb-seq screen to scRNA-seq data from explanted human right coronary artery ECs (RCAECs)⁶⁹. We compared the gene expression at two levels: for all perturbed genes (2,285 genes) and for the 41 CAD associated genes. Among the perturbed genes in teloHAECs, 2,107 genes were expressed at TPM > 1 in healthy or disease RCAECs. We observed high correlation of gene expression in TPM between teloHAECs and RCAECs (Pearson correlation = 0.66, P value = 6.45×10^{-280} ; Extended Data Fig. 1b). We observed similar correlations of gene expression for the 41 CAD-associated genes (Pearson correlation = 0.63, P value = 9.29×10^{-6} ; Extended Data Fig. 1c). Furthermore, 40 out of 41 CAD associated genes were expressed at >1 TPM in RCAECs (Extended Data Fig. 1d).

Identifying CAD-associated programs using V2G2P analysis

We developed an approach to identify gene programs that are likely to affect CAD risk through functions in ECs. To do so, we tested whether the 254 genes with V2G links (between CAD variants and enhancers or

coding regions in ECs) were enriched in each Perturb-seq program. Specifically, we performed a one-tailed Fisher exact test separately for co-regulated genes and for regulators. For co-regulated genes, we constructed a contingency table for whether a gene is a co-regulated gene (out of 17,472 expressed genes) and whether a gene has a V2G link. For regulators, we constructed a contingency table for whether a gene is a regulator (out of all perturbed genes) and whether a gene has a V2G link. We then multiplied the *P* values from co-regulated gene and regulator Fisher exact tests together to get a final program enrichment *P* value. We use Benjamini–Hochberg method for multiple hypothesis correction across all 50 non-batch programs. For an example of this analysis, see Extended Data Fig. 6g. Five programs showed significant enrichment by this method (FDR < 0.05: programs 8, 35, 39, 47 and 48), referred to as ‘V2G2P programs for CAD’.

Defining CAD-associated V2G2P genes

We defined V2G2P genes for CAD as those 41 genes that were both a gene with a V2G link to a CAD variant and a member of one of the 5 CAD-associated programs (as a regulator and/or co-expressed gene). The 41 genes were linked to 43 GWAS signals due to cases in which independent GWAS signals are linked to the same gene.

Identifying enriched programs using MAGMA

We tested whether the co-regulated genes in each program were significantly enriched near variants associated with CAD using MAGMA⁴. To do so, we took the CAD summary statistics from a previous study¹² (https://data.mendeley.com/public-files/datasets/2zdd47c94h/files/5b4eb0d7-96e8-4c7e-b109-046107ebd480/file_downloaded) and used the MAGMA --annotate function to summarize CAD association *P* values for variants within a 50 kb window of all human genes, using the 1000 Genomes European reference data for base allele frequencies (https://ctg.cncr.nl/software/MAGMA/ref_data/g1000_eur.zip). We then ran MAGMA to test for enrichment of CAD heritability within 50 kb of the top 300 program genes and corrected for multiple testing (60 components) using the Benjamini–Hochberg method.

Identifying programs and cell types enriched for CAD heritability through stratified LD score regression

We used S-LDSC to estimate the enrichment of CAD heritability linked to program genes and to enhancers in teloHAECs. Although the original implementations of S-LDSC linked variants to genes based on genomic distance^{28,70}, we additionally required that variants either overlap exonic regions of the gene or overlap nearby candidate enhancers in ECs (as in refs. 32,71). In particular, for co-regulated genes in each program, we derived an annotation for S-LDSC by including exonic regions (exons from transcripts with Ensembl_canonical, appris_principal, appris_candidate or appris_candidate_longest tags, as indicated in the GENCODE v38lift37 annotations) as well as endothelial *cis*-regulatory elements derived from snATAC-seq⁷², from which we merged the 9 adult and 8 fetal sets of endothelial peaks into a single annotation, and for each gene set included all peaks within 50 kb of the gene starts and ends. For all peaks, we first converted coordinates from the GRCh38 to the GRCh37 reference assembly using UCSC LiftOver, discarding peaks that could not be converted. To estimate the enrichment of CAD heritability in teloHAEC enhancers, we required the variants to overlap enhancers predicted by ABC from ATAC-seq and H3K27ac ChIP-seq data in teloHAECs under control conditions or treated with IL-1 β , TNF or VEGF (ABC score > 0.015). For each set of variants (programs or teloHAEC enhancers), we ran S-LDSC using 1000 Genome EUR Phase 3 genotype data to estimate LD scores, baseline v.2.2 annotations as recommended by the LDSC developers⁷³ and HapMap 3 SNPs excluding the MHC region as regression SNPs. We ranked programs by their enrichments and reported the *P* values of these enrichments (Extended Data Fig. 5b). Full S-LDSC results for teloHAEC enhancers can be found in Supplementary Table 27.

PoPS

PoPS is a method to nominate probable causal genes in a GWAS locus, which prioritizes genes based on their being members of many gene sets enriched for heritability genome-wide⁵. We applied PoPS to previously published summary statistics¹² using the predefined set of gene sets as previously described⁵ (Extended Data Fig. 5c,d). For each GWAS signal, we calculated the PoPS rank among ‘nearby genes’ (2 to either side of the lead SNP, and all within ± 500 kb). Previously we have shown that genes with the highest PoP score in the locus are strongly enriched for likely causal genes, as identified by fine-mapped coding variants⁵, and that this enrichment increases when further focusing on genes that are both the closest gene and have the highest PoP score. In this analysis, we did not use any features from Perturb-seq and, as such, this method represents an entirely independent method that validates the high likelihood of causality of the set of CAD-associated V2G2P genes.

Defining gene expression programs for cells carrying control guides

To examine the gene programs in normal, unperturbed teloHAECs, we used the same analysis pipeline on the subset of cells carrying control guides (5,506 cells). We used cNMF to discover *K* = 60 components, and defined 60 ‘control programs’ based solely on the 300 co-regulated genes defining each component (because control guides did not target any genes, so there was no regulator information). Of the 60 programs, 4 programs correlated with batch (programs 2, 17, 22 and 41). We compared the program co-regulated genes between control cells and full library programs (Extended Data Fig. 6d). Control program 10 highly overlapped with full library programs 8 and 39. The four control programs that correlated with batch also had high overlap in co-regulated genes with the full library’s batch programs. We then utilized the V2G2P approach to prioritize these programs and found that none of the control programs was enriched for genes with V2G links (Extended Data Fig. 6e).

Identifying genes in the CCM pathway

Figure 3 shows a curated set of genes previously reported to interact physically or functionally with the CCM complex and/or downstream ERK5–MEK5 signalling^{46,51,74–78}, plus one additional gene (*TNLRD1*) that we identified here as a member of the CCM pathway. These genes were manually selected through an iterative process involving examining genes known to interact with the CCM complex and that were found to regulate the enriched programs in Perturb-seq.

Allelic imbalance analysis for a variant linked to *TNLRD1*

We calculated allelic imbalance in ATAC-seq and ChIP-seq signal for the rs1879454 variant, accounting for any mapping bias towards the reference allele using previously described methods⁷⁹. Specifically, we created two reference genome FASTA files that harboured the reference or alternative alleles at rs1879454, aligned ATAC-seq data to both genome files, selected reads that overlapped the variant coordinate and used PySuspenders⁷⁹ and PySAM (<https://github.com/pysam-developers/pysam>) to assign and count reads that uniquely aligned to one or the other allele. We applied this procedure to ATAC-seq data from teloHAECs and the ENCODE datasets ENCSR000EVW (GATA2 ChIP-seq on HUVECs) and ENCSR000EOB (DNase-seq and DGF on HMVEC-dLy-Neo).

CRISPRi–FlowFISH for *TNLRD1*

We used CRISPRi–FlowFISH to test the effects of 61 candidate enhancers on *TNLRD1* expression in teloHAECs, including the enhancer containing rs1879454. We designed gRNAs tiling across all accessible regions (here defined as the union of the peaks in the chromatin accessibility dataset called by MACS2 with a lenient *P* value cut-off of 0.1, and 150-bp regions on either side of the MACS2 summit) in the range chromosome 15: 81267614–81427246 in ATAC-seq data from teloHAECs. We excluded

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gRNAs with low specificity scores or low-complexity sequences as previously described²². We infected teloHAECs with the gRNA lentiviral library with 15 $\mu\text{g ml}^{-1}$ blasticidin selection for 3 days and activated CRISPRi with 2 $\mu\text{g ml}^{-1}$ doxycycline incubation for 5 days. We performed FlowFISH using ThermoFisher PrimeFlow (ThermoFisher 88-18005-210) as previously described²², using ThermoFisher probe sets VA1-3010837-PF for *TLNRD1* and VA4-13187-PF for *RPL13A*. We observed an approximately 2.6-fold signal for *TLNRD1* in cells with all probes applied (stained) versus cells without target gene probes applied (unstained) (Extended Data Fig. 8j). We analysed these data as previously described²². In brief, we counted gRNAs in each bin using Bowtie to map reads to a custom index, normalized gRNA counts in each bin by library size, then used a maximum-likelihood estimation approach to compute the effect size for each gRNA. We used the limited-memory Broyden–Fletcher–Goldfarb–Shanno algorithm (implemented in the R stats4 package) to estimate the most likely log-normal distribution that would have produced the observed guide counts, and the effect size for each gRNA is the mean of its log-normal fit divided by the average of the means from all negative control gRNAs. As previously described, we scaled the effect size of each gRNA in a screen linearly so that the strongest 20-guide window at the TSS of the target gene has an 85% effect²². This was to account for non-specific probe binding in the RNA FISH assay (this is based on our observation that promoter CRISPRi typically shows 80–90% knockdown by qPCR). We averaged the effect sizes of each gRNA across replicates and computed the effect size of an element as the average of all gRNAs targeting that element. We assessed significance using a two-sided *t*-test comparing the mean effect size of all gRNAs in a candidate element to all negative control guides. We computed the FDR for elements using the Benjamini–Hochberg procedure and used a FDR threshold of 0.05 to call significant regulatory effects.

Generation of single-guide CRISPRi teloHAEC derivatives

Paired oligonucleotides for individual guides (newly designed, as described for the Perturb-seq library, or with the best knockdown efficacy in Perturb-seq) were annealed and cloned into the BsmBI site of a CROP-Opti-Blast vector (plasmid available upon request), which were then used to generate lentivirus (as per ref. 53). CRISPRi teloHAECs were infected with each virus, in separate wells, and selected for using blasticidin (15 $\mu\text{g ml}^{-1}$ for 4 days), before 5 days of doxycycline induction and analysis by bulk RNA-seq, fluorescence imaging or physiological assays. The following guides (TargetGene_CloneIndex: ForwardSequence) were used: CCM2_C2: GGCAAGAAGGTGAGCGTGCG; CCM2_F6: GAGCCGCTA CATGCTCGACCC; CDH5_B8: GCCAGCTGGAACCTGAAG; CDH5_D5: GTTGACTGCCTGTCCGTCCA; ITGB1BP1_C7: GAAGCCCGCGGCACTC CCACG; ITGB1BP1_G8: GAAGTCCGCAACCCGGGGAT; KLF2_C9: GGAC CCGGGGAGAAAGGACG; KLF2_G10: GCCGCGGTATATAAGCCGGC; MAP2K5_A11: GCCGAGGCCGCGCGGACTGG; MAP2K5_B5: GTCTGC CCCACCCGGAGACAC; MAP3K3_A4: GTTCTGAGGTGGAGAACGG; MAP3K3_C3: GCCAATAACAAGAAGGAAGT, MEF2A_C10: GCGGCGC GAAGCGCTGGTGG; MEF2A_H10: GACTGAATTATCTCTCGGT; Negative_control_B6: GCAACGGTGTAACCGCGGATC; negative_control_D2: GTGGTTTCAACACCGGACCCA; negative_control_D8: GGTGGTTCC GTTGGCTGGCC; Negative_control_F4: GCTGGCGGAGCATTTGGGATA, NFAT5_D4: GGCTCGCTTCTCCGCGCG; NFAT5_D7: GGTCCCCGTCC CGCCGGGG; PDCD10_D11: GACCGAGCAGAAGAGGTCTA, PDCD10_G1: GCCGCTTTACGCCACTCGCGT; TLNRD1_B3: GTGGCTGCGCCGCCGC CCGCA; TLNRD1_D12: GCCTCCGGCAGCCCCCTGCGGG.

Ribonucleoprotein-based CRISPR–Cas9 genome editing

For some experiments, we used Synthego's ribonucleoprotein (RNP) technology as an orthologous method to knockdown target genes as previously described⁸⁰. In brief, teloHAECs were nucleofected with Synthego's Gene Knockout kit v2 for non-targeting negative control, *CCM2*, *TLNRD1* or *MAP3K3* using a Lonza 4D-Nucleofector system. For

each nucleofection reaction, we used 150,000 cells with 20 pmol of Cas9 and 50 pmol of sgRNA. The cells were then nucleofected (program CA-210) using SG cell line nucleofection solution (Lonza, V4XC-3024). The nucleofected cells were seeded in teloHAEC culture medium and collected 48 h later for RNA extraction for RT–qPCR analysis and/or RNA-seq to measure gene knockdown efficiency and perturbation effects. For *MAP3K3* knockdown in single-guide CRISPRi lines, cells were treated with 2 $\mu\text{g ml}^{-1}$ doxycycline for 72 h before nucleofection and for 48 h afterwards.

Identification of disease-relevant genes regulated by both *TLNRD1* and *CCM2*

For the heatmaps in Figs. 5c and 6a, we identified genes that were most-strongly regulated by both *TLNRD1* and *CCM2* CRISPRi knockdowns from bulk RNA-seq data as follows. We included genes that were members of the eight gold standard, known EC-acting CAD genes that were significantly regulated by both *TLNRD1* and *CCM2* knockdowns (both $P < 0.05$). We included genes that were in the top 40 *TLNRD1* and *CCM2* upregulated or downregulated genes (by highest P value across both knockdowns), had an average fold change in *TLNRD1* and *CCM2* knockdowns of >2 (either positive or negative) and had previous evidence for functions predicted to increase or decrease CAD-relevant EC phenotypes (Supplementary Table 29).

Computational prediction of the *TLNRD1* and *CCM2* protein structure

AlphaFold2.3 Multimer (v.3)⁸¹ was run using sequences for KRIT1 (UniProt identifier O00522), CCM2 (UniProt identifier Q9BSQ5, with and without deletion of residues 417–444), PDCD10 (UniProt identifier Q9BUL8) and TLNRD1 (UniProt identifier Q9HIK6). Models were visualized using UCSF ChimeraX (v.1.61). Predicted alignment error was extracted using AlphaPickle⁸² and plotted using combinations of AlphaPickle, Matplotlib (v.3.7.0) and Seaborn.

Co-IP of *CCM2* and *TLNRD1*

HEK293 cells were transfected with V5-tagged CCM2 full length, V5-tagged CCM2 C-terminal truncation, Flag-tagged TLNRD1 and/or Flag-tagged AKT1, as indicated in Fig. 5b and Extended Data Fig. 9d–f, using FuGENE (E2311, Promega) or PEI MAX (Polysciences). Two days after transfection, cell lysates were extracted using IP lysis buffer (87787, Thermo Scientific) supplemented with 1× Halt Protease Inhibitor cocktail (1862209, Thermo Scientific). The protein concentration was determined using a Pierce BCA Assay (ThermoFisher), and an equal mass of protein was used for each sample. Immunoprecipitation was carried out using magnetic beads (88805, Thermo Scientific) conjugated with 5 μg of either rabbit anti-V5 (13202, Cell Signaling Technology) or mouse anti-Flag (F1804, Millipore Sigma) antibody. Cell lysates were incubated with the antibody-conjugated beads for 20–30 min at room temperature. Beads were then washed three times with IP lysis buffer (1861603, Thermo Scientific), and precipitants were eluted using 2×LDS sample buffer (NP0007, Thermo Fisher Scientific). Precipitants and input lysates were separated by 10% SDS–PAGE and transblotted to nitrocellulose. For the anti-Flag IP, blots were immunoblotted with 1:1,000 rabbit anti-V5 (13202, Cell Signaling Technology), followed by 1:5,000 anti-rabbit HRP secondary (7074, Cell Signaling), then stripped (21059, Thermo Scientific) and re-probed with 1:1,000 rabbit anti-TLNRD1 (HPA071766, Sigma). For the anti-V5 IP, blots were immunoblotted with 1:1,000 primary mouse anti-Flag (F1804, Millipore Sigma) and 1:5,000 secondary anti-mouse HRP (7076, Cell Signaling), and stripped and re-probed with 1:1,000 mouse anti-V5 (ab27671, Abcam). The AKT1–Flag vector is Addgene, 9021. CCM2–V5 (ccsbBroad304_04281) and TLNRD1–V5 (ccsbBroad304_03872) vectors were obtained from the Broad Institute Gene Perturbation Platform⁸³. For TLNRD1–Flag, cDNA sequences were amplified from the TLNRD1–V5 vector using primers that incorporated an in-frame Flag tag and cloned

into the pcDNA3.1 backbone. The CCM2 C-terminal truncation was created in the CCM2-V5 vector by site-directed mutagenesis using a QuickChange II Site-Directed Mutagenesis kit (Agilent 200523-5) and the oligonucleotides F-CACCCTCAGAGGGGTCAGCATGCCCAAC and R-GTTGGGCATGCTGACCCCTCTGAGGGTG, which resulted in a deletion of amino acids 419–442 at the C terminus of CCM2, in frame with the V5 tag.

TEER measurements

For TEER measurements, we used an ECIS Z-Theta instrument from Applied BioPhysics in a 96-well plate system (Applied BioPhysics; 96W10idf). CRISPRi teloHAECs expressing individual guides to *TLNRD1*, *CCM2* or non-targeting guides (2 guides each) were treated for 5 days with 2 $\mu\text{g ml}^{-1}$ doxycycline. A gold-electrode-containing 96-well ECIS plate was incubated at 37 °C and 5% CO₂ with culture medium for 30 min to equilibrate before coating with 2.5 mg ml⁻¹ fibronectin in 0.1 M bicarbonate buffer at pH 8.0. Then, the coated wells were inoculated with 45,000 cells in 100 μl medium. An additional 100 μl of medium was added to each well before initiating the measurements at 4,000-Hz AC. At 25 h, after the cells formed a confluent layer, the culture medium was replaced with 200 μl of fresh culture medium with 1 U ml⁻¹ thrombin to disrupt cell–cell junctions, and measurements continued until 50 h to observe cell junction recovery after thrombin treatment.

Measurement of EC responses to laminar flow

200,000 CRISPRi teloHAECs with individual control, *CCM2* or *TLNRD1* guides were seeded on flow chamber slides (80176, Ibidi) that had been pre-coated with 0.2% gelatin. After 24 h, cells were cultured under laminar flow (12 dynes cm⁻²) for 48 h (10902, Ibidi pump system). Static culture controls were seeded at the same density. Cells were treated with 2 $\mu\text{g ml}^{-1}$ doxycycline for 2 days before seeding, and throughout, for a total of 5 days. RNA was collected with 300 μl of Trizol and extracted with 60 μl of chloroform. After addition of 1 volume 70% ethanol, RNA was loaded onto a Qiagen RNeasy spin column, washed with 350 μl of buffer RW1 and treated for 20 min at room temperature with 10 μl Purelink DNase (Invitrogen, 12185010) in 80 μl of 1 $\times$ buffer. Subsequent RNA purification steps were as per the Qiagen RNeasy protocol.

Fluorescence imaging and quantification of teloHAECs

For quantification of actin fibre characteristics, CRISPRi teloHAECs expressing individual gRNAs (targeting *CCM2*, *TLNRD1* or negative control) were treated with 2 $\mu\text{g ml}^{-1}$ doxycycline for 5 days. Cells were fixed in situ with by addition of paraformaldehyde to 3.2% for 30 min at 37 °C, washed with PBS, permeabilized by addition of PBS with 0.1% Triton X-100 for 15 min at room temperature, washed with PBS and stained with PerkinElmer Cell Painting dyes (Phenovue Fluor 568–phalloidin, Phenovue Fluor 488–concanavalin A, Phenovue Hoechst 33342 nuclear stain and Phenovue 512 nucleic acid stain) according to the manufacturer's instructions. Cells were imaged in four channels as previously described⁸⁴ on a Perkin Elmer Opera Phenix Imaging System-106513, confocal $\times 63$ magnification with 1 $\times$ binning. The stacks of images for the Phalloidin and Hoechst channels were converted to single images using maximum projection, output ranges were standardized and images were exported. Cell boundaries were drawn by hand on a Phalloidin/Hoechst composite image in Fiji and saved as regions of interest (ROIs). Phalloidin channel images were loaded into Fiji, converted to 16-bit greyscale, and cell areas and dimensions for each ROI were extracted using the Measure function (reporting Area and Fit Ellipse). Actin fibres were detected and quantified using the LPX Fiji plugin as previously described⁸⁵, with lineExtract parameters: gwsiter=5, mdnmsLen=8, pickup=above (10.0), shaveLen=3, delLen=5, and line properties for each ROI measured using LineFeature. Parallelness (a_normAvgRad) ranges from 0 (for randomly oriented fibres) to 1 (all fibres parallel).

Zebrafish husbandry and transgenic lines

Adult wild-type AB, transgenic Tg(flk1:EGFP) (that express EGFP at the surface of blood vessels) and transgenic Tg(cmlc2:EGFP) (that express EGFP in heart muscle) zebrafish lines were maintained at 28.5 °C in circulating system water on a 14-h light/10-h dark cycle under standard conditions. Male and female embryos and larvae (≤ 5 days post-fertilization (d.p.f.)) were kept in the dark in an incubator at 28.5 °C for subsequent experiments. At the end point, embryos were euthanized by tricaine overdose (MS-222, Western Chemical) followed by freezing (for RNA isolation), fixing in paraformaldehyde (for histological analysis) or bleach treatment. All animal experiments were performed in accordance with relevant guidelines and regulations and with approval from the Mayo Clinic Institutional Animal Care and Use Committee.

tlnd1 and *ccm2* CRISPR knockdown in zebrafish

crRNAs for both *ccm2* and *tlnd1* were designed using the Alt-R Predesigned Cas9 crRNA Selection Tool using the Integrated DNA Technologies (IDT) database. All the crRNAs were selected based on published criteria⁸⁶. For *ccm2*, guides were designed to target two distinct exons shared by all transcripts (AA: TTGAACGAGACACGATACC, AF: ATGGAGCCACAACACCCACC). For *tlnd1*, guides either targeted the 5' untranslated region (AN.1: GGAAACACAAGGGACGTCTC, AF: GCTGAAAGTTACACCCAACG) or the single *tlnd1* exon (AN.2: CTGC CGCTAAGGATGTTGGT, DG: CAAGAGCAAATGCAGCTGG). For *ccm2* and *tlnd1*, RNPs were prepared as previously described^{86,87}. In brief, the crRNA (bearing the guide sequence) was annealed with an equal molar amount of tracrRNA (bearing the gRNA scaffold, IDT, 1072532) in duplex buffer (IDT, 11010301), to form gRNA, by heating at 95 °C for 5 min and subsequently cooling on ice. gRNA was assembled with an equal molar amount of Alt-R S.p. Cas9 Nuclease V3 (IDT, 1081058) to form the RNP complex (28.5 μM final concentration) by incubation at 37 °C for 5 min followed by storage at –20 °C, following the published protocol^{86,87}. RNP complexes prepared from the tracrRNA/scaffold only were used as a negative control. Next, 3 nl of each RNP complex (28.5 μM final concentration) was injected into the yolk of one-to-two cell stage embryos (wild-type, Tg;Flu:EGFP (for the permeability analysis) or Tg;cmlc2:EGFP (for visualization of the atrioventricular valve)).

tlnd1 and *ccm2* morpholino knockdown in zebrafish

Morpholinos to knockdown *tlnd1* and *ccm2* were designed and injected using standard protocols⁸⁸. The *ccm2* morpholino has been validated to cause cardiovascular phenotypes at the 100 μM dose⁷⁸. A custom morpholino for *tlnd1* (TTCCCCGAGCCACTAGCCATAG) was designed to target the TSS and was ordered from Gene Tools. The control oligonucleotide was a single sequence, CCTCTTACCTCAGTTA CAATTTATA, that is a validated negative control⁸⁸. Wild-type zebrafish embryos were injected with 3 nl of diluted morpholinos at multiple concentrations (50 μM , 100 μM , 200 μM or 300 μM of control, *tlnd1* or *ccm2* morpholinos) at the one cell stage using a pico-injector (Harvard Apparatus). For co-injection, *tlnd1* and *ccm2* morpholinos were mixed to give 50 μM of each, and 3 nl of the mixture was injected.

Zebrafish imaging and phenotyping

Embryos were observed for mortality and visible phenotypes at 2 d.p.f. and 3 d.p.f. using a light microscope. Images were captured at 2 and 3 d.p.f. on an EVOS microscope (Life Technology) and Zeiss Axio-observer Z1. The 3 d.p.f. embryos (knockdown or control) were scored as having a heart phenotype if they displayed visible atrial chamber enlargement, moderate to severe pericardial oedema and slow blood flow in the tail veins. Note that normal zebrafish undergo cardiac looping between approximately 2 d.p.f. and 3 d.p.f. (wherein the atrium and ventricle change from a linear posterior-to-anterior arrangement to a right-to-left asymmetric arrangement). Most of the *ccm2* or *tlnd1* knockdown embryos that scored positive by the criteria above

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also showed a looping defect, maintaining the posterior-to-anterior arrangement of atrium and ventricle at 3 d.p.f. However, because looping is a time-dependent phenomenon that normally occurs near the 3 d.p.f. time when we examined the embryos for heart phenotypes, we did not include this as a scoring criterion. For the additional phenotypic analyses described below (confocal imaging, haematoxylin and eosin (H&E) staining, tail vein morphology, blood flow and vascular permeability), we selected *ccm2* or *tlrnl1* knockdown embryos that scored as positive for heart phenotype at 2 d.p.f. High-resolution images for the vascular permeability and cardiac chamber analyses were acquired using a confocal microscope LSM 800 (Zeiss).

Histological staining of zebrafish embryos for atrial and ventricular thickness

H&E staining was performed by staff at the Mayo Clinic Comprehensive Cancer Center Histology core laboratory. In brief, 3 d.p.f. zebrafish larvae were fixed in 4% paraformaldehyde overnight at 4 °C. To obtain paraffin sections, fixed larvae were dehydrated stepwise in ethanol/1× PBS dilutions (5, 25, 50, 75 and 100% ethanol). Transverse sections at a thickness of 5 µm using a microtome (MICROME) were produced from the anterior beginning of the otic vesicle and included posterior structures until the cloacal vent. The sectioned region therefore spanned from the glomerulus up to the cloaca and included the complete pronephros. Sections were stained with Gills 1, eosin Y and Harris haematoxylin (Richard Allan Scientific) according to the manufacturer's protocol.

FITC-dextran 2,000 kDa and Texas Red-dextran 70 kDa injections and imaging for tail vein morphology and vascular permeability

Microangiography was performed as previously described^{89,90}. In brief, at 3 d.p.f., CRISPR-Cas9-injected embryos were anaesthetized in 0.015% tricaine methanesulfonate (Western Chemical) and microangiography was performed by inserting a glass microneedle (World Precision Instruments) through the pericardium directly into the ventricle. For assessment of vascular morphology, 2,000 kDa FITC-dextran (Sigma, FD2000S-100MG) was diluted to 2 mg ml⁻¹ in zebrafish embryo medium⁹¹, and a total of 4.5 nl was injected. For measurement of vascular permeability, Texas Red-dextran with a molecular weight of 70 kDa was solubilized in embryo medium at a 2 mg ml⁻¹ concentration and a total of 4.5 nl was injected. Images were acquired after 30 min using a Zeiss LSM 880 confocal microscope, standard FITC and dsRed filter sets and ×10 objective at room temperature. For quantification of permeability, the Raw .czi images were preprocessed using Zeiss software (ZEN2) to generate a maximum intensity projection image. The maximum intensity projection images of controls as well as CRISPR mutants were then processed using the Matlab programming platform, as previously described⁹⁰. Movies for the blood flow in the heart and tail veins were taken by capturing 60 s bright field-time-lapse images at 60 frames per s using an EVOS microscope at ×20 magnification, as previously described⁹².

RT-qPCR assays in zebrafish

klf2b, *ccm2* and *tlrnl1* expression was measured by RT-qPCR on RNA isolated from 100 µM *tlrnl1* morpholino embryos or CRISPR *tlrnl1*, *ccm2* or control embryos at 3 d.p.f. The following primers were used: *klf2b*, F: GAAGAGACACCTGTGAGGGC and R: GGACACCGATTCTGAGGACC; *ccm2*, F: GGCGGATCAGATGAGGGAAC and R: CAGACAGCAATACG GACCGA; and *tlrnl1*, F: ACACGCGAGAGTACCTGTTG and R: TCATCC CGCGACAAATCCAA.

In situ hybridization for *tlrnl1* expression in zebrafish. In situ hybridization was performed using previously validated methods⁹³. In brief, a 437 bp fragment of *tlrnl1* was amplified from gDNA using the PCR primers F: CATTAACGGAATGGCAGGCG and R: TGCCCGGATAAAGGCAAAGT, subcloned and verified by sequencing. Anti-sense in situ hybridization

probes were generated using a M13 reverse primer with SpeI-linearized plasmid, whereas sense (negative control) probes were generated using an M13 forward primer with NotI-linearized plasmid. In situ hybridization of embryos was conducted at 24 and 72 h post-fertilization using these anti-sense or sense (control) probes against *tlrnl1*.

Applying the V2G2P approach to additional GWAS traits and cell types

We tested whether the V2G2P method was generally applicable to other traits beyond CAD in EC and to other cell types.

We first examined whether the same Perturb-seq dataset in ECs could be applied to interpret variants for other vascular traits related to EC functions beyond CAD. We applied V2G2P to two additional GWAS traits (pulse pressure (PP) and mean arterial pressure (MAP)) from the UK Biobank⁹⁴, with finemapping information from H. Finucane and J. Ulirsch (<https://www.finucanelab.org/data>). We performed V2G analysis by mapping variants associated with these traits onto the same EC enhancer map we used for CAD, and identified genes linked to PP or MAP variants in ECs. We then performed V2G2P analysis by testing for enrichment of the PP or MAP V2G gene sets in the 50 EC programs we identified from Perturb-seq. Note that we performed the V2G2P enrichment test using only the 300 co-regulated genes in each program because not all the genes at GWAS loci for these blood pressure traits were targeted for perturbation in our EC Perturb-seq screen.

We next examined whether the entire analysis framework could be applied to another cell type: K562 erythroid cells, which are a relevant model for red blood cell and platelet traits. Here we examined seven GWAS traits for red blood cell and platelet measures: mean corpuscular haemoglobin, mean corpuscular volume, platelet count, red blood cell count, mean corpuscular haemoglobin concentration, haemoglobin A1c and haemoglobin. We also assessed four traits for which K562 cells are not likely to be an appropriate model: PP, MAP, systolic blood pressure (SBP) and diastolic blood pressure (DBP). Data were from the UK Biobank⁹⁴, with finemapping by H. Finucane and J. Ulirsch (<https://www.finucanelab.org/data>).

We constructed V2G maps for each trait using ABC data in K562 cells (K562-Roadmap⁹⁵) to identify variant-containing enhancers, and identified the set of V2G genes for each trait (genes with links to variants associated, by GWAS, with each trait). We then constructed a G2P map by applying cNMF to the genome-scale Perturb-seq data previously collected in K562 cells¹⁹. We tested *K* values over a broad range, and selected *K* = 90 as the number of components that minimized cNMF error and maximized other ranking metrics (see 'Choosing the number of components for cNMF analysis' above). Finally, we performed the V2G2P enrichment test (considering both the 300 co-regulated genes for each program and the regulators of each program, identified as the perturbations significantly affecting expression of each program, from Perturb-seq). Of the 90 programs, we found, 32 programs were prioritized for at least one of 6 GWAS traits (Extended Data Fig. 12b and Supplementary Table 22).

Curating previously identified CAD prioritization gene sets

To assess the ability of V2G2P to prioritize disease-associated genes, we surveyed several CAD studies that used more than just GWAS and genomic positioning data to prioritize CAD loci and genes. Below is a summary of how each study created their gene set and how we accessed this data.

For the PoPS data reported in ref. 12, PoPS values were computed for all protein-coding genes within 500 kb of all GWAS signals and the gene with the highest PoPS value in each locus was prioritized, resulting in 221 genes. Data were obtained from supplementary table 25 of that study.

For the eQTL and splicing quantitative loci (sQTL) co-localization study in ref. 96, bulk RNA-seq data were collected from human coronary artery tissue samples from explanted transplant tissue, or collected from rejected transplant donors (138 individuals, from left anterior

descending coronary artery, right coronary artery and left circumflex artery). eQTL co-localization was performed to find eQTL-associated genes (eGenes) or to find sQTLs. The eQTL list was from supplementary table 12 column 'vdh_CAD_PPH4' with posterior probability > 0.8 (methods 'PPH4 > 0.8 to support evidence of a shared causal variant'). The splice variant list was from supplementary table 22 and subset for variants with posterior probability > 0.8 (column 'vdh_CAD_PPH4'). We then identified sGenes linked to the co-localized sQTLs by finding matching genes in supplementary table 20 (columns 'gene_id' and 'spliceid').

For the transcriptome-wide association study (TWAS)⁹⁷, which associated genotype and expression data across 15 tissues (7 from STARNET and 8 from GTEx), we used supplementary table 4 for significant TWAS genes (114 genes).

OpenTarget L2G⁹⁸ used a supervised machine-learning model to learn the weights of multiple evidence sources (distance, molecular QTL co-localization, chromatin interaction and variant pathogenicity) based on a gold standard of previously identified causal genes. The authors applied this model to a previous CAD GWAS dataset¹⁰. Prioritized genes had an L2G model score > 0.5 (table downloaded from <https://genetics.opentargets.org/Study/GCST005194/associations>).

The EC-specific eQTL co-localization study²⁹ used HAECs isolated from aortic trimmings from deceased heart donors and cultured with or without IL-1 β (53 individuals, bulk RNA-seq). ECs obtained from 157 donors were cultured with or without oxPL treatment (microarray). The authors performed eQTL mapping using Matrix eQTL and used the R package 'coloc' for co-localization. We obtained their data from table S5.

In another study¹⁰, variants were prioritized using probabilistic annotation integrator based on several features such as LD information, *P* value distribution, coding genes and H3K4me1 sites. Data were obtained from table 2 and online table 20 (labelled XX).

In an EC cell CRISPR screen for six phenotypes³⁰, the authors used a CRISPR screening approach to identify CAD risk variant-containing regulatory elements in 83 CAD GWAS loci that altered FACS-sortable signals for any of six pre-selected phenotypes in ECs (E-selectin, ICAM1, VCAM1, nitric oxide, reactive oxygen species and intracellular calcium). The identified 26 loci where perturbation of a variant-containing element affected one or more of these phenotypes (prioritizing a single gene in 21 of these loci). Data were obtained from figure 3a and supplementary table 4 of that study.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

Raw and processed data for Perturb-seq, ATAC-seq, H3K27ac ChIP-seq and RNA-seq in teloHAECs were deposited into the NCBI's GEO database under accession number GSE210523. This superseries is composed of the following subseries: GSE210489 (ATAC-seq), GSE210491 (ChIP-seq), GSE210522 (bulk RNA-seq of cytokine-treated parental lines and single guide CRISPRi knockdowns), GSE232400 (bulk RNA-seq of cells under flow, and *MAP3K3* double knockdowns), GSE212396 (pilot scRNA-seq studies) and GSE210681 (comprehensive Perturb-seq). FASTQ files for Perturb-seq and *TLNDR1* enhancer CRISPRi FlowFISH in teloHAECs are also available at the Impact of Genomic Variation on Function (IGVF) portal: Perturb-seq: <https://data.igvf.org/analysis-sets/IGVFD57943IIWZ/>, and CRISPRi FlowFISH: <https://data.igvf.org/analysis-sets/IGVFD59053MHCS/>. The following datasets were also used: CAD lead GWAS variants^{10,12} and phenome-wide association scan data¹². GWAS summary statistics for other traits and finemapping analysis were from H. Finucane and J. Ulirsch's analysis of UK Biobank data (<https://www.finucanelab.org/data>), and coding

and splice site annotations were from the RefGene database (from the UCSC Genome Browser dated 24 June 2017)⁶⁸. LD data from the 1000 Genome European ancestry database were accessed using "plink --ld-window-kb 1000 --ld-window 99999 --ld-window-r2 0.9". TF binding site information was from HOCOMOCO v.11 human full scan motifs (https://hocomoco11.autosome.ru/downloads_v11). Gene sets were from the MSigDB⁶⁷. We also used scRNA-seq data from explanted human right coronary artery ECs⁶⁹, endothelial *cis*-regulatory elements derived from snATAC-seq⁷², ENCODE datasets ENCSR000EVW (GATA2 ChIP-seq on HUVECs) and ENCSR000EOB (DNase-seq and DGF on HMVEC-dLy-Neo). We used UniProt for protein structure models for KRIT1 (O00522), CCM2 (Q9BSQ5), PDCD10 (Q9BUL8) and TLNDR1 (Q9HIK6). Previous prioritization calls for genes in CAD GWAS loci were from refs. 10,12,29,30,96–98 (details in the Methods).

Code availability

The V2G2P approach snakemake pipeline is available at GitHub (<https://github.com/EngreitzLab/V2G> or <https://doi.org/10.5281/zenodo.10357646>)⁹⁹, for V2G and https://github.com/EngreitzLab/cNMF_pipeline or <https://doi.org/10.5281/zenodo.10357454>¹⁰⁰, for G2P and V2G2P enrichment). For data analysis, data were processed using all packages required for running cNMF (v.1.2)²⁵. We additionally used R-3.6.1 and R-3.6.3 (ref. 101), edgeR 3.28.0 (refs. 58,62,64,65), seqLogo 1.52.0, MAST 1.12.0 (ref. 50), clusterProfiler 3.14.0 (refs. 66,102), org.Hs.eg.db 3.10.0, Seurat 3.0.2 (ref. 103), SeuratObject 4.0.0 (refs. 103–105), stats4 v3.6.1 and SingleCellExperiment 1.8.0 (ref. 106), for downstream analysis in R. The supporting packages in R used for data processing and figure generation were ggplot2 3.3.5 (ref. 107), ggpubr 0.4.0, ggrepel 0.9.1, gplots 3.1.1, gridExtra 2.3, scales 1.1.1, cowplot 1.1.1, dplyr 1.0.7, tidyr 1.1.3, textshape 1.7.1, reshape2 1.4.4, stringi 1.7.5 (ref. 108), conflicted 1.0.4, data.table 1.14.0, purrr 0.3.4, readxl 1.3.1, writexl 1.5.0, ramify 0.3.3, optparse 1.6.6, and all dependencies. Data were further processed using pysuspcenders 0.2.6 (ref. 109), pysam 0.19.1 (ref. 110), python 2.7.15, and LDSC (v.1.0.1)¹¹¹. Other software packages used were kallisto 0.48.0 (ref. 56), limma 3.42.2 (ref. 57), Bowtie (v.1.3.0) and Bowtie2 (v.2.4.2)¹¹², Macs2 2.2.7 (ref. 113), CellRanger 7.0.0 (ref. 114), FIMO (https://meme-suite.org/meme/meme_5.3.2/tools/fimo), Morpheus (<https://software.broadinstitute.org/morpheus>), IGV¹¹⁵, Open Targets^{98,116}, plink (v.1.90b6.21)¹¹⁷, MAGMA⁴, S-LDSC^{28,70}, PoPS⁵, FloJo (v.10.8.1), FCSalyzer 0.9.22alpha <https://sourceforge.net/projects/fcsalyzer/>, LocusZoom <http://locuszoom.org/>¹¹⁸, AlphaFold2.3 Multimer (v.3)⁸¹, UCSF ChimeraX (v.1.61), AlphaPickle⁸², Matplotlib (v.3.7.0), Seaborn (<https://seaborn.pydata.org/>) and Matlab R2018a (MathWorks). Phalloidin analysis was done using Fiji/ImageJ 2.9.0/1.53t with the LPX plugin. CRISPRi primer design used CRISPRDesigner (<https://github.com/EngreitzLab/CRISPRDesigner>). For data collection, the Opera Phenix imager (used for phalloidin stain analysis of teloHAECs) was run using Harmony 4.9.2137.273, Acapella 5.0.1.124082 and Oda 4.9.2137.273. The Zeiss confocal microscope LSM 880 (used for zebrafish imaging) was run using ZEN 2.3 SP1 software. The Sony MA900FP cell sorter (used for FlowFISH) was run using Cell Sorter Software v.3.2. Other instruments were used with the manufacturer-supplied intrinsic software, including EVOS microscopes, Applied Biosystems QuantStudio 5 (for RT-qPCR) and the ECIS Z-Theta instrument (Applied BioPhysics).

53. Fulco, C. P., Munschauer, M., Anyoha, R. & Munson, G. Systematic mapping of functional enhancer-promoter connections with CRISPR interference. *Science* **354**, 769–773 (2016).
54. Thakore, P. I. et al. Highly specific epigenome editing by CRISPR-Cas9 repressors for silencing of distal regulatory elements. *Nat. Methods* **12**, 1143–1149 (2015).
55. Gilbert, L. A. et al. Genome-scale CRISPR-mediated control of gene repression and activation. *Cell* **159**, 647–661 (2014).
56. Bray, N. L., Pimentel, H., Melsted, P. & Pachter, L. Near-optimal probabilistic RNA-seq quantification. *Nat. Biotechnol.* **34**, 525–527 (2016).
57. Law, C. W., Chen, Y., Shi, W. & Smyth, G. K. voom: precision weights unlock linear model analysis tools for RNA-seq read counts. *Genome Biol.* **15**, R29 (2014).

58. Chen, Y., Lun, A. T. L. & Smyth, G. K. From reads to genes to pathways: differential expression analysis of RNA-seq experiments using Rsubread and the edgeR quasi-likelihood pipeline. *F1000Research* **5**, 1438 (2016).
59. Huang, H. et al. Fine-mapping inflammatory bowel disease loci to single-variant resolution. *Nature* **547**, 173–178 (2017).
60. Marshall, J. L. et al. HyPR-seq: single-cell quantification of chosen RNAs via hybridization and sequencing of DNA probes. *Proc. Natl Acad. Sci. USA* **117**, 33404–33413 (2020).
61. Hart, T. & Moffat, J. BAGEL: a computational framework for identifying essential genes from pooled library screens. *BMC Bioinformatics* **17**, 164 (2016).
62. Soneson, C. & Robinson, M. D. Bias, robustness and scalability in single-cell differential expression analysis. *Nat. Methods* **15**, 255–261 (2018).
63. Nygaard, V., Rødland, E. A. & Hovig, E. Methods that remove batch effects while retaining group differences may lead to exaggerated confidence in downstream analyses. *Biostatistics* **17**, 29–39 (2016).
64. Robinson, M. D., McCarthy, D. J. & Smyth, G. K. edgeR: a Bioconductor package for differential expression analysis of digital gene expression data. *Bioinformatics* **26**, 139–140 (2010).
65. McCarthy, D. J., Chen, Y. & Smyth, G. K. Differential expression analysis of multifactor RNA-seq experiments with respect to biological variation. *Nucleic Acids Res.* **40**, 4288–4297 (2012).
66. Wu, T. et al. clusterProfiler 4.0: a universal enrichment tool for interpreting omics data. *Innovation* **2**, 100141 (2021).
67. Subramanian, A. et al. Gene set enrichment analysis: a knowledge-based approach for interpreting genome-wide expression profiles. *Proc. Natl Acad. Sci. USA* **102**, 15545–15550 (2005).
68. Karolchik, D., Hinrichs, A. S. & James Kent, W. The UCSC genome browser. *Curr. Protoc. Hum. Genet.* **71**, 18.6.1–18.6.33 (2011).
69. Wirka, R. C. et al. Atheroprotective roles of smooth muscle cell phenotypic modulation and the TCF21 disease gene as revealed by single-cell analysis. *Nat. Med.* **25**, 1280–1289 (2019).
70. Finucane, H. K. et al. Heritability enrichment of specifically expressed genes identifies disease-relevant tissues and cell types. *Nat. Genet.* **50**, 621–629 (2018).
71. Dey, K. K. et al. Contribution of enhancer-driven and master-regulator genes to autoimmune disease revealed using functionally informed SNP-to-gene linking strategies. *Cell Genomics* **2**, 100145 (2022).
72. Zhang, K. et al. A single-cell atlas of chromatin accessibility in the human genome. *Cell* **184**, 5985–6001.e19 (2021).
73. Huijoe, M. L. A., Gazal, S., Hormozdiari, F., van de Geijn, B. & Price, A. L. Disease heritability enrichment of regulatory elements is concentrated in elements with ancient sequence age and conserved function across species. *Am. J. Hum. Genet.* **104**, 611–624 (2019).
74. Padarti, A. & Zhang, J. Recent advances in cerebral cavernous malformation research. *Vessel Plus* **2**, 21 (2018).
75. Wei, S. et al. Cerebral cavernous malformation proteins in barrier maintenance and regulation. *Int. J. Mol. Sci.* **21**, 675 (2020).
76. Fischer, A., Zalvide, J., Faurobert, E., Albiges-Rizo, C. & Tournier-Lasserre, E. Cerebral cavernous malformations: from CCM genes to endothelial cell homeostasis. *Trends Mol. Med.* **19**, 302–308 (2013).
77. Cullere, X., Plovie, E., Bennett, P. M., MacRae, C. A. & Mayadas, T. N. The cerebral cavernous malformation proteins CCM2L and CCM2 prevent the activation of the MAP kinase MEK3. *Proc. Natl Acad. Sci. USA* **112**, 14284–14289 (2015).
78. Kleaveland, B. et al. Regulation of cardiovascular development and integrity by the heart of glass-cerebral cavernous malformation protein pathway. *Nat. Med.* **15**, 169–176 (2009).
79. Engreitz, J. M. et al. Local regulation of gene expression by lncRNA promoters, transcription and splicing. *Nature* **539**, 452–455 (2016).
80. Atri, D. S. et al. CRISPR–Cas9 genome editing of primary human vascular cells in vitro. *Curr Protoc.* **1**, e291 (2021).
81. Jumper, J. et al. Highly accurate protein structure prediction with AlphaFold. *Nature* **596**, 583–589 (2021).
82. mattarnoldbio. mattarnoldbio/alphapickle: Release v1.4.0. *Zenodo* <https://doi.org/10.5281/zenodo.5708709> (2021).
83. Yang, X. et al. A public genome-scale lentiviral expression library of human ORFs. *Nat. Methods* **8**, 659–661 (2011).
84. Bray, M.-A. et al. Cell Painting, a high-content image-based assay for morphological profiling using multiplexed fluorescent dyes. *Nat. Protoc.* **11**, 1757–1774 (2016).
85. Higaki, T. Quantitative evaluation of cytoskeletal organizations by microscopic image analysis. *Plant Morphol.* **29**, 15–21 (2017).
86. Kroll, F. et al. A simple and effective FO knockout method for rapid screening of behaviour and other complex phenotypes. *eLife* **10**, e59683 (2021).
87. Lu, F., Leach, L. L. & Gross, J. M. A CRISPR–Cas9-mediated FO screen to identify pro-regenerative genes in the zebrafish retinal pigment epithelium. *Sci. Rep.* **13**, 3142 (2023).
88. Moulton, J. D. & Yan, Y.-L. Using morpholinos to control gene expression. *Curr. Protoc. Mol. Biol.* **Chapter 26**, Unit 26.8 (2008).
89. Hoepfner, L. H. et al. Revealing the role of phospholipase Cβ3 in the regulation of VEGF-induced vascular permeability. *Blood* **120**, 2167–2173 (2012).
90. Wang, Y. et al. Dissecting VEGF-induced acute versus chronic vascular hyperpermeability: essential roles of dimethylarginine dimethylaminohydrolase-1. *iScience* **24**, 103189 (2021).
91. Zebrafish embryo medium. *Cold Spring Harb. Protoc.* **2011**, db.rec12478 (2011).
92. Machikhin, A. S., Volkov, M. V., Burlakov, A. B., Khokhlov, D. D. & Potemkin, A. V. Blood vessel imaging at pre-larval stages of zebrafish embryonic development. *Diagnostics* **10**, 886 (2020).
93. Thisse, C. & Thisse, B. High-resolution in situ hybridization to whole-mount zebrafish embryos. *Nat. Protoc.* **3**, 59–69 (2008).
94. Sudlow, C. et al. UK Biobank: an open access resource for identifying the causes of a wide range of complex diseases of middle and old age. *PLoS Med.* **12**, e1001779 (2015).
95. Roadmap Epigenomics Consortium. Integrative analysis of 111 reference human epigenomes. *Nature* **518**, 317–330 (2015).
96. Hodonsky, C. J. et al. Multi-ancestry genetic analysis of gene regulation in coronary arteries prioritizes disease risk loci. *Cell Genomics* **4**, 100465 (2024).
97. Li, L. et al. Transcriptome-wide association study of coronary artery disease identifies novel susceptibility genes. *Basic Res. Cardiol.* **117**, 6 (2022).
98. Mountjoy, E. et al. An open approach to systematically prioritize causal variants and genes at all published human GWAS trait-associated loci. *Nat. Genet.* **53**, 1527–1533 (2021).
99. Ma, X. R. & Engreitz J. M. EngreitzLab/V2G: V2Gv1.0.0. *Zenodo* <https://doi.org/10.5281/zenodo.10357646> (2023).
100. Kang, H., Guo, K. & Engreitz J. M. EngreitzLab/cNMF_pipeline: v1.0. *Zenodo* <https://doi.org/10.5281/zenodo.10357454> (2023).
101. The R Development Core Team. *R: A Language and Environment for Statistical Computing* (R Foundation for Statistical Computing, 2022).
102. Yu, G., Wang, L.-G., Han, Y. & He, Q.-Y. clusterProfiler: an R package for comparing biological themes among gene clusters. *OMICS* **16**, 284–287 (2012).
103. Stuart, T. et al. Comprehensive integration of single-cell data. *Cell* **177**, 1888–1902.e21 (2019).
104. Satija, R., Farrell, J. A., Gennert, D., Schier, A. F. & Regev, A. Spatial reconstruction of single-cell gene expression data. *Nat. Biotechnol.* **33**, 495–502 (2015).
105. Macosko, E. Z. et al. Highly parallel genome-wide expression profiling of individual cells using nanoliter droplets. *Cell* **161**, 1202–1214 (2015).
106. Amezquita, R. A. et al. Orchestrating single-cell analysis with Bioconductor. *Nat. Methods* **17**, 137–145 (2020).
107. Wickham, H. *ggplot2* (Springer-Verlag, 2016).
108. Gagolewski, M. stringi: fast and portable character string processing in R. *J. Stat. Softw.* **103**, 1–59 (2022).
109. Holt, J., Huang, S., McMillan, L. & Wang, W. Read annotation pipeline for high-throughput sequencing data. In *Proc. International Conference on Bioinformatics, Computational Biology and Biomedical Informatics* 605–612 (Association for Computing Machinery, 2013).
110. Li, H. et al. The Sequence Alignment/Map format and SAMtools. *Bioinformatics* **25**, 2078–2099 (2009).
111. Bulik-Sullivan, B. K. et al. LD score regression distinguishes confounding from polygenicity in genome-wide association studies. *Nat. Genet.* **47**, 291–295 (2015).
112. Langmead, B., Wilks, C., Antonescu, V. & Charles, R. Scaling read aligners to hundreds of threads on general-purpose processors. *Bioinformatics* **35**, 421–432 (2019).
113. Gaspar, J. M. Improved peak-calling with MACS2. Preprint at *bioRxiv* <https://doi.org/10.1101/496521> (2018).
114. Zheng, G. X. Y. et al. Massively parallel digital transcriptional profiling of single cells. *Nat. Commun.* **8**, 14049 (2017).
115. Robinson, J. T. et al. Integrative genomics viewer. *Nat. Biotechnol.* **29**, 24–26 (2011).
116. Ghousaini, M. et al. Open Targets Genetics: systematic identification of trait-associated genes using large-scale genetics and functional genomics. *Nucleic Acids Res.* **49**, D1311–D1320 (2021).
117. Purcell, S. et al. PLINK: a tool set for whole-genome association and population-based linkage analyses. *Am. J. Hum. Genet.* **81**, 559–575 (2007).
118. Boughton, A. P. et al. LocusZoom.js: interactive and embeddable visualization of genetic association study results. *Bioinformatics* **37**, 3017–3018 (2021).
119. Tsherniak, A. et al. Defining a cancer dependency map. *Cell* **170**, 564–576.e16 (2017).
120. Alcidi, E. A. & Tsukiyama, T. ATP-dependent chromatin remodeling shapes the long noncoding RNA landscape. *Genes Dev.* **28**, 2348–2360 (2014).
121. SenBanerjee, S. et al. KLF2 is a novel transcriptional regulator of endothelial proinflammatory activation. *J. Exp. Med.* **199**, 1305–1315 (2004).
122. Coma, S. et al. GATA2 and LMO2 control angiogenesis and lymphangiogenesis via direct transcriptional regulation of neuropilin-2. *Angiogenesis* **16**, 939–952 (2013).
123. Yu, M. et al. Computational estimates of annular diameter reveal genetic determinants of mitral valve function and disease. *JCI Insight* **7**, e146580 (2022).
124. Hogan, B. M., Bussmann, J., Wolburg, H. & Schulte-Merker, S. ccm1 cell autonomously regulates endothelial cellular morphogenesis and vascular tubulogenesis in zebrafish. *Hum. Mol. Genet.* **17**, 2424–2432 (2008).
125. Neuman, N. A. et al. The four-and-a-half LIM domain protein 2 regulates vascular smooth muscle phenotype and vascular tone. *J. Biol. Chem.* **284**, 13202–13212 (2009).
126. Wang, W. et al. Essential role of Smad3 in angiotensin II-induced vascular fibrosis. *Circ. Res.* **98**, 1032–1039 (2006).
127. Tsai, S. et al. TGF-β through Smad3 signaling stimulates vascular smooth muscle cell proliferation and neointimal formation. *Am. J. Physiol. Heart Circ. Physiol.* **297**, H540–H549 (2009).
128. Crispino, J. D. & Weiss, M. J. Erythro-megakaryocytic transcription factors associated with hereditary anemia. *Blood* **123**, 3080–3088 (2014).
129. Gruber, T. A. & Downing, J. R. The biology of pediatric acute megakaryoblastic leukemia. *Blood* **126**, 943–949 (2015).
130. Hauser, W. et al. Megakaryocyte hyperplasia and enhanced agonist-induced platelet activation in vasodilator-stimulated phosphoprotein knockout mice. *Proc. Natl Acad. Sci. USA* **96**, 8120–8125 (1999).
131. Pleines, I. et al. Mutations in tropomyosin 4 underlie a rare form of human macrothrombocytopenia. *J. Clin. Invest.* **127**, 814–829 (2017).
132. Meinders, M. et al. Sp1/Sp3 transcription factors regulate hallmarks of megakaryocyte maturation and platelet formation and function. *Blood* **125**, 1957–1967 (2015).

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Author contributions G.R.S. developed and implemented the systematic Perturb-seq method. G.R.S., B.C., E.S.L., R.M.G. and J.M.E. designed Perturb-seq experiments. H.K., G.R.S., X.R.M., T.Z., S.K.V., A.B., H.K.F. and J.M.E. developed and implemented analysis methods for Perturb-seq data. G.R.S., O.S.-G., V.S.L.-K. and S.K.V. conducted and analysed cell imaging experiments. G.R.S., S.F., V.S.L.-K., A.E.B. and R.Z. conducted additional assays in ECs. M.S.T. performed AlphaFold2 modelling. S.H. contributed to interpreting AlphaFold2 modelling data. G.R.S. and S.F. performed the co-IP experiments. G.R.S., S.F., D.T.B. and T.H.N. conducted bulk RNA-seq,

ATAC-seq and ChIP-seq experiments. G.R.S. and K.G. analysed bulk RNA-seq, ATAC-seq and ChIP-seq data. P.G., S.F. and G.M. created plasmids. N.C. and H.K.F. contributed to the PoPS analysis. K.A. provided GWAS data. R.S.A. and D.M. performed the zebrafish experiments. R.M.G. and J.M.E. supervised the work. All authors contributed to writing the manuscript.

Competing interests J.M.E. is a shareholder of Illumina and 10X Genomics, has received materials from 10X Genomics unrelated to this work, is an equity holder in and consultant for Martingale Labs, and has received guest speaker honoraria from GSK. M.S.T. holds equity and has received consulting fees from ROME Therapeutics, which is not related to this work. G.R.S., R.M.G., J.M.E., H.K. and X.R.M. are inventors on a provisional patent related to this work. All other authors declare no competing interests.

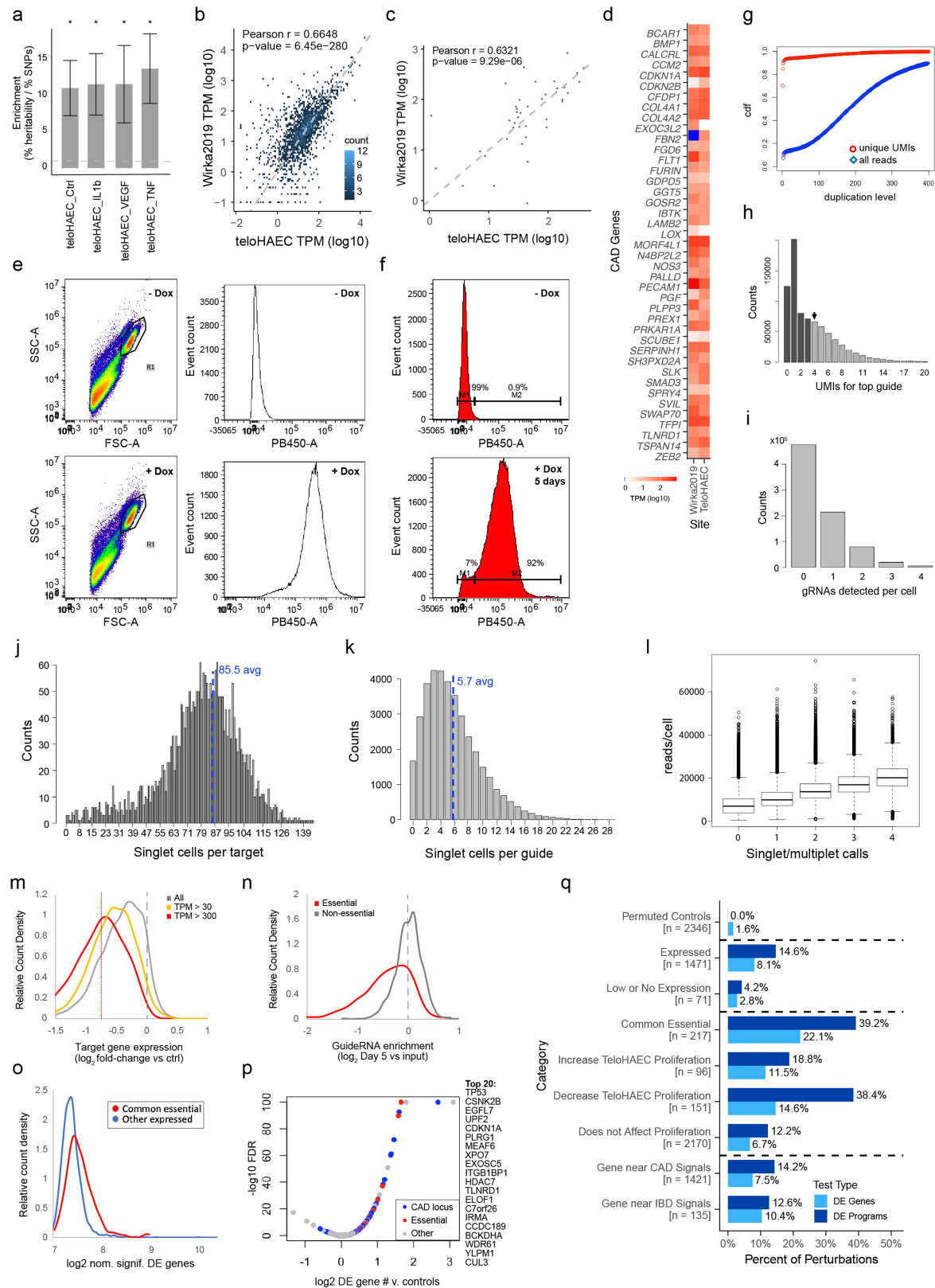
Additional information

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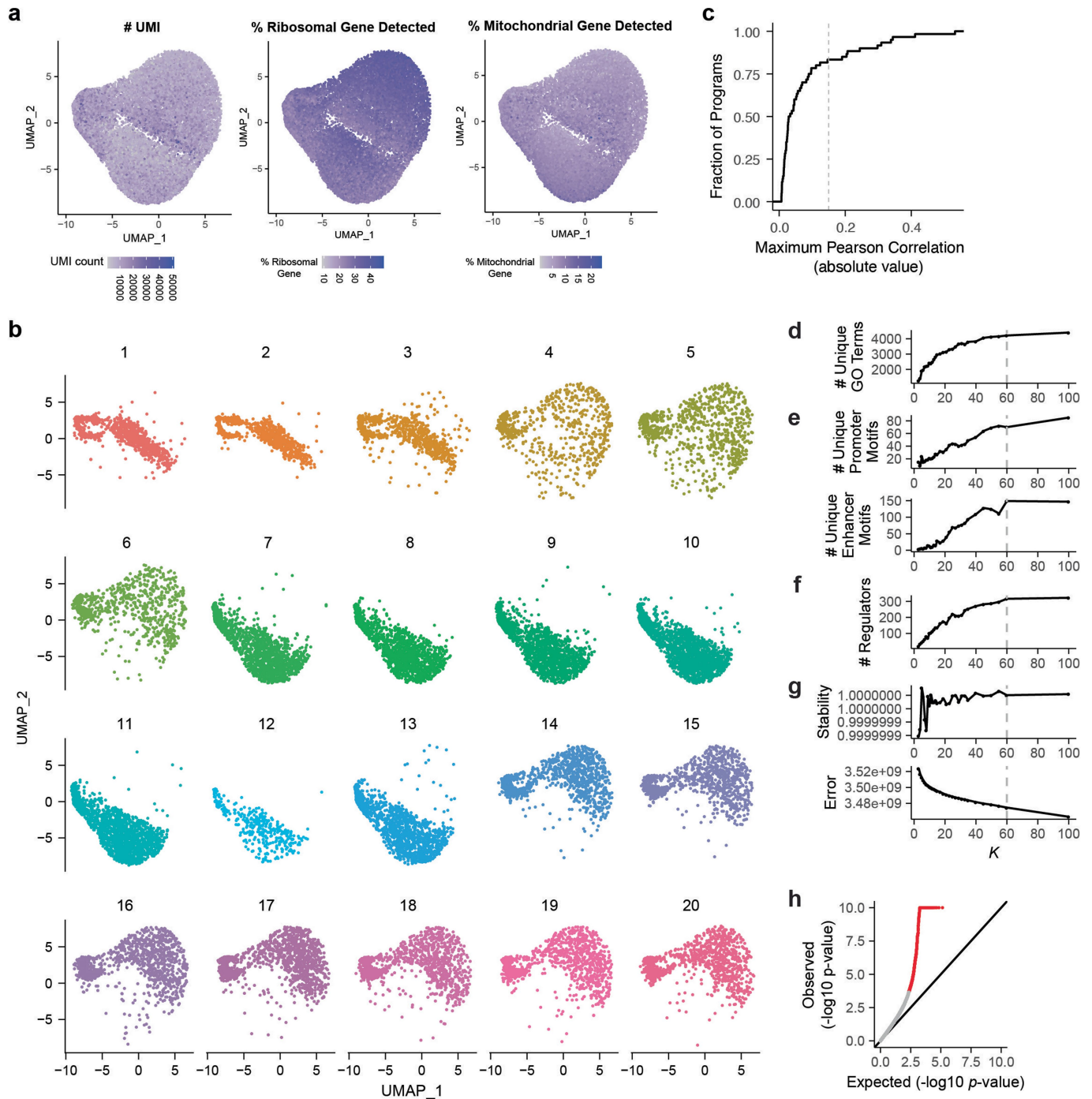


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Extended Data Fig. 1 | Establishing the TeloHAEC CRISPRi model and Perturb-seq details.

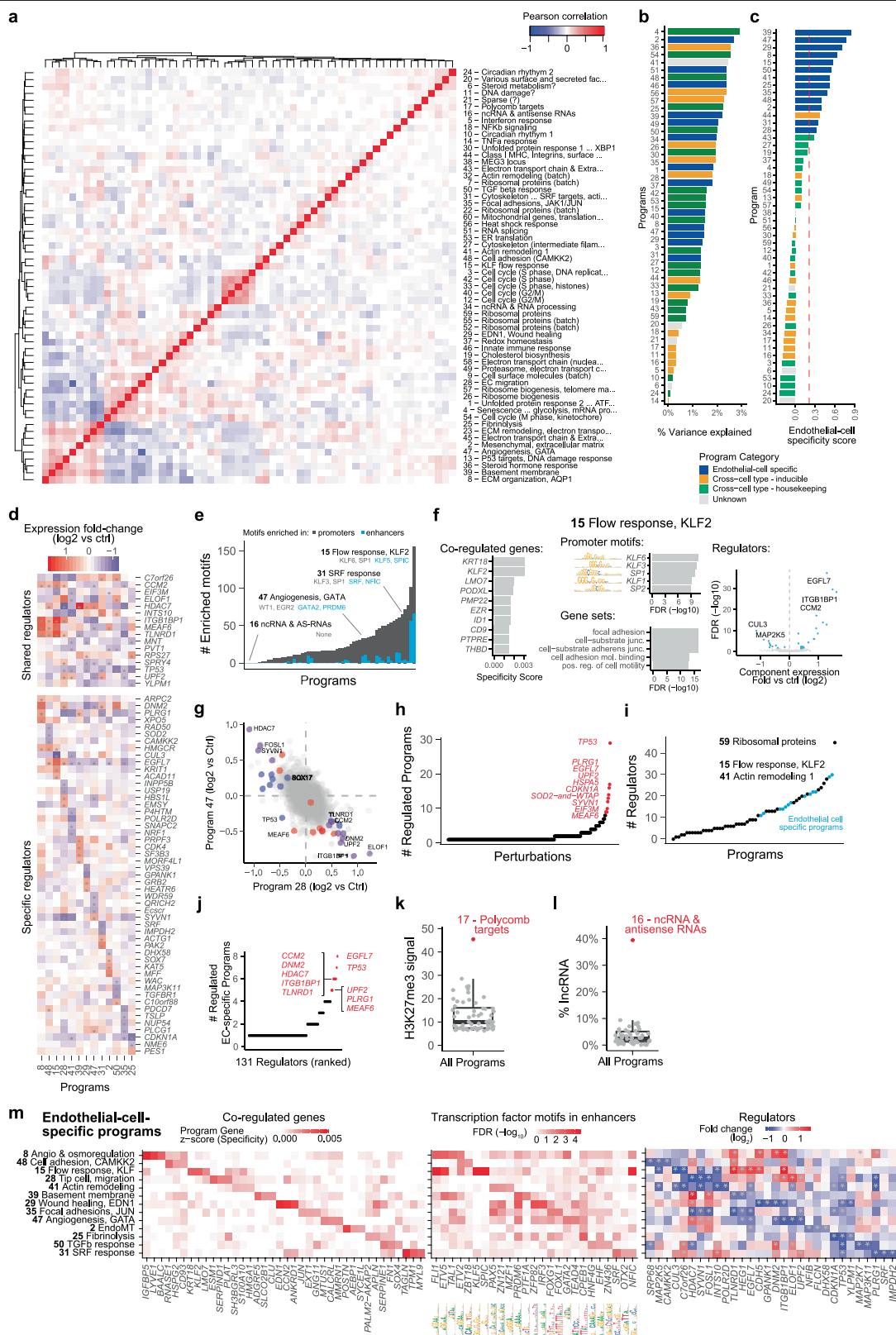
a, Enrichment of CAD heritability in TeloHAEC enhancers, from Stratified Linkage Disequilibrium Score Regression analysis (S-LDSC, see Methods), where enrichment is the percentage of heritability explained by variants in enhancers (%heritability), divided by the percentage of variants in enhancers (%SNPs). Enhancers in TeloHAEC (treated under the indicated conditions) were identified from ATAC-seq and H3K27ac ChIP-seq data ($n = 6$ for control ATAC, 3 for IL-1 β , TNF α or VEGF ATAC, 4 for control ChIP, and 2 for IL-1 β , TNF α or VEGF ChIP) by the Activity-by-Contact model. Error bars: standard error around the enrichment estimate, calculated by S-LDSC using jackknife (which resamples the data used for calculating heritability enrichment). P-values were calculated using the S-LDSC method²⁸, and FDR by the Benjamini-Hochberg method. *: FDR < 0.05, with specific FDR values of: Ctrl; 0.037, IL-1 β ; 0.015, TNF α ; 0.020 and VEGF; 0.041. Full S-LDSC results can be found in Supplementary Table 27. **b**, Scatter density plot of human right coronary artery endothelial cell single cell RNA-seq pseudobulk gene expression (from⁶⁹) versus TeloHAEC pseudobulk gene expression, for genes perturbed in this study. 2,107 of the 2,285 perturbed genes are expressed at TPM > 1 in healthy or diseased RCAECs. Rand p-values from two sided Pearson correlation test. **c**, As in **b**, but for the 41 V2G2P genes. **d**, Heatmap of gene expression (log₁₀ TPM) of the 41 V2G2P genes in diseased RCAECs and in TeloHAEC. All but one gene, *FBN2*, is expressed at > 1 TPM in RCAECs. **e**, FACS showing dox inducibility of KRAB-dCas9-IRES-BFP in TeloHAEC, after sorting but before the screen. Left panels: gating for viable individual cells. Right panels: counts of gated cells by fluorescence intensity in the BFP/PB450 channel. **f**, BFP channel counts of cells grown in parallel and concurrently with cells for the Perturb-seq screen. After expansion to 120 M cells, transduction, selection and 5-day doxycycline treatment, 92% of cells remain BFP positive. **g**, Cumulative distribution fraction for duplication levels of unique CBC-UMI-Guide combinations in deeply-sequenced dialout libraries ("unique UMIs", red) or all guide reads (blue) versus duplication level. Requiring four duplicates (dotted line) eliminates 90% of CBC-UMI-guide combinations (likely PCR chimaeras), while retaining >85% of total guide reads. **h**, UMIs for top guide per CBC. Arrow: the chosen 4 UMI threshold. **i**, Counts of singlets (1 gRNA, black bar), doublets (2) and higher multimers, as well as cells with no guide called (0), at the chosen thresholds of 4 UMIs for the top guide and 4 or more fold fewer for the next most frequent guide. **j**, Histogram of counts of singlet cells per target. Dotted line: average. **k**, As in **j**, but for singlet cells per guide. **l**, Read UMI counts for all transcripts per cell by singlet/multiplet status. The median UMI count for

doublets was 37% more than singlets. Assuming that droplets with two cells will have double the number of reads as singlets, this suggests 37% of doublets are due to two cells (9.3% of cells with guides) while the remainder (15.7% of cells with guides) are due to two guides in one cell, very close to the expectation from the infection MOI of 15%. $n = 352686, 214449, 79744, 19195$ and 5345 cells with 0, 1, 2, 3, or 4 guides, respectively. Boxplot centre line, median; box limits, upper and lower quartiles; whiskers, 1.5x interquartile range; points, outliers. **m**, Distribution of knockdown efficiency across target genes (log₂ expression in cells containing guideRNAs targeting the gene versus in cells containing negative control guideRNAs). Grey line: all targeted genes. Yellow and red lines: genes expressed at >30 and >300 TPM, respectively. Red dotted vertical line: 40% knockdown (average for 300 + TPM target genes). **n**, Distribution of fitness effects across all guideRNAs (log₂ ratio of guide frequency in singlet cells from the Perturb-seq experiment after 5 days of CRISPRi induction compared to guide frequency in the original guideRNA library). Guides targeting common essential genes (red) were depleted more frequently than guideRNAs targeting other genes. **o**, Relative count frequencies for the number of nominally significant differentially expressed (DE) genes per perturbed target (log₂ of genes with raw $p < 0.01$, and fold change >1.15 from EdgeR DE analysis), for the indicated subclasses of targets. **p**, Volcano plot showing log₂ (# DE genes for target)/(avg. # DE genes for non-expressed controls) versus -log₁₀ FDR (capped at 100). Significance was assessed by two sided binomial test versus DE gene counts for the 48 perturbed non-expressed control genes. Right: symbols for target genes with the strongest effects. 10.7% of all targets had a significant effect on transcription (FDR < .05 increased DE gene count), including 31.9% of common essential genes, and 9.0% of other genes. **q**, Percent of perturbations that have a significant transcriptional effect in Perturb-seq, as defined by either (i) "DE Genes", as per **p**, or (ii) "DE Programs": perturbations that lead to significant changes in program expression by MAST package⁵⁰ with 10X lane correction (FDR < 0.05), by each indicated class: Permuted Controls (statistical tests performed on randomly drawn cells with negative control or safe-targeting guides), Expressed (>=1 TPM in TeloHAEC bulk RNA-seq), Low or No Expression (<1 TPM), Common essential (as identified in DepMap¹¹⁹), TeloHAEC Proliferation (showing fitness effects, as per **n**, of +/-15%, FDR < 0.05), Gene near CAD GWAS signals (expressed genes nearby any CAD GWAS signal, see Methods: Defining variants in CAD GWAS signals), Gene near IBD signals (perturbed expressed genes nearby 10 selected IBD GWAS signals, with no genes overlapping those for CAD signals).



Extended Data Fig. 2 | QC metrics for single cells, and selection of number of components for cNMF. **a**, UMAPs showing number of UMIs per cell (left), percent ribosomal genes detected per cell (middle), percent mitochondrial genes detected per cell (right). **b**, UMAPs showing cells from each of the twenty 10X lanes. The differences in clustering along the UMAP_2 axis indicates a technical batch effect between 10X lanes. **c**, Cumulative distribution function (CDF) plot of the maximum absolute value of Pearson correlation between cNMF component expression in cells and batch. Dotted line: the $R \geq 0.15$ threshold used to call programs associated with batch. **d**, Gene set enrichment analysis for GO terms among co-regulated genes, as a function of the number of

components in the cNMF model (K). y-axis: the number of unique GO terms enriched across all programs for a given K . **e**, Number of unique motifs enriched among the promoters (top) or enhancers (bottom) of co-regulated genes across all components, as a function of K . **f**, Number of unique perturbations that have significant effect ($FDR < 0.05$) on one or more programs, as a function of K . **g**, Model-based evaluation of the choice of K . Stability of the components over 100 NMF runs (top) and element-wise square of error (bottom, see Methods). **h**, Quantile-quantile plot for effects of perturbations on program expression. X-axis: expected uniform distribution. Y-axis: $-\log_{10} p$ -value computed from MAST package⁵⁰. Red: p -value < 0.05 .

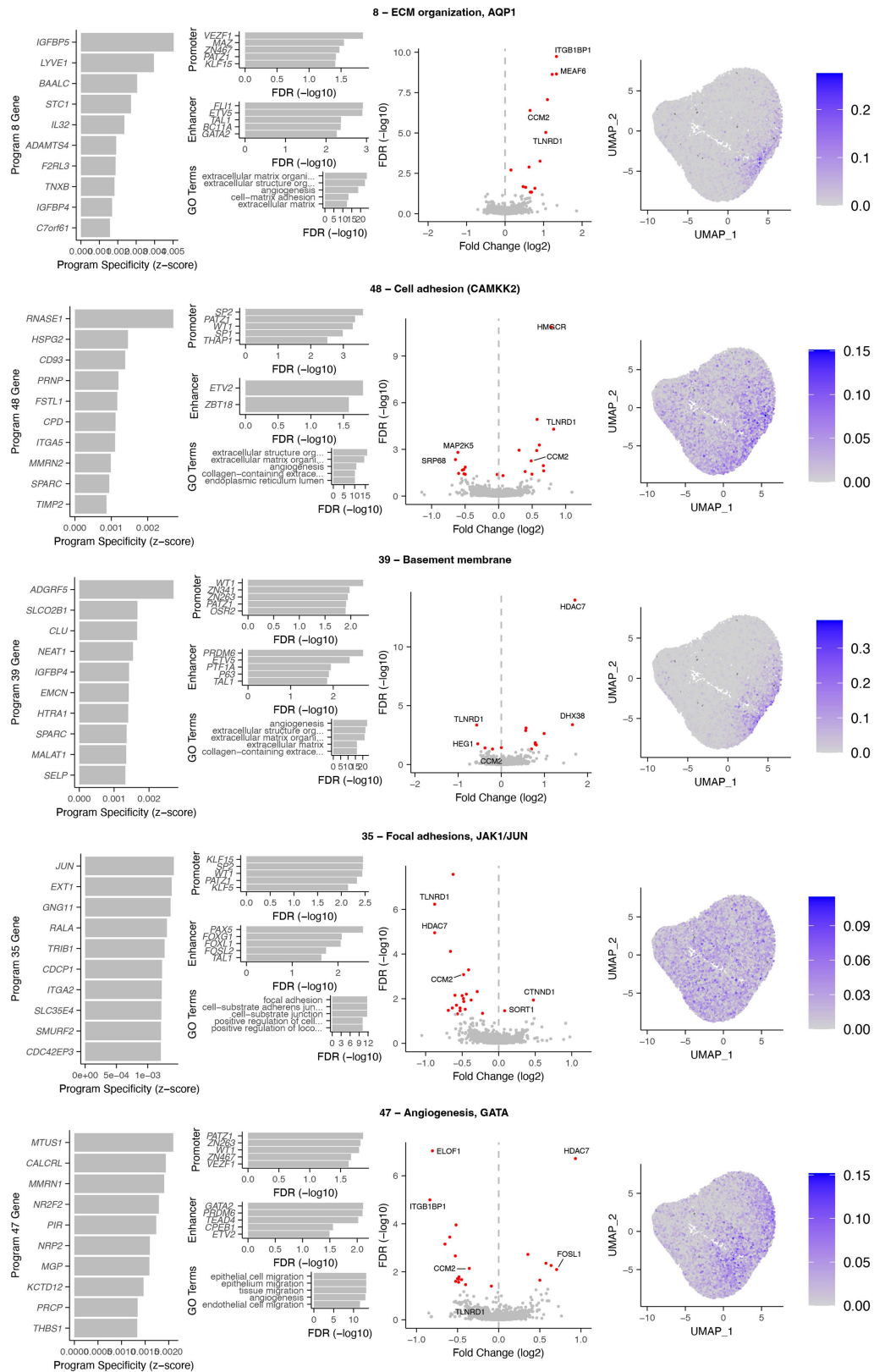


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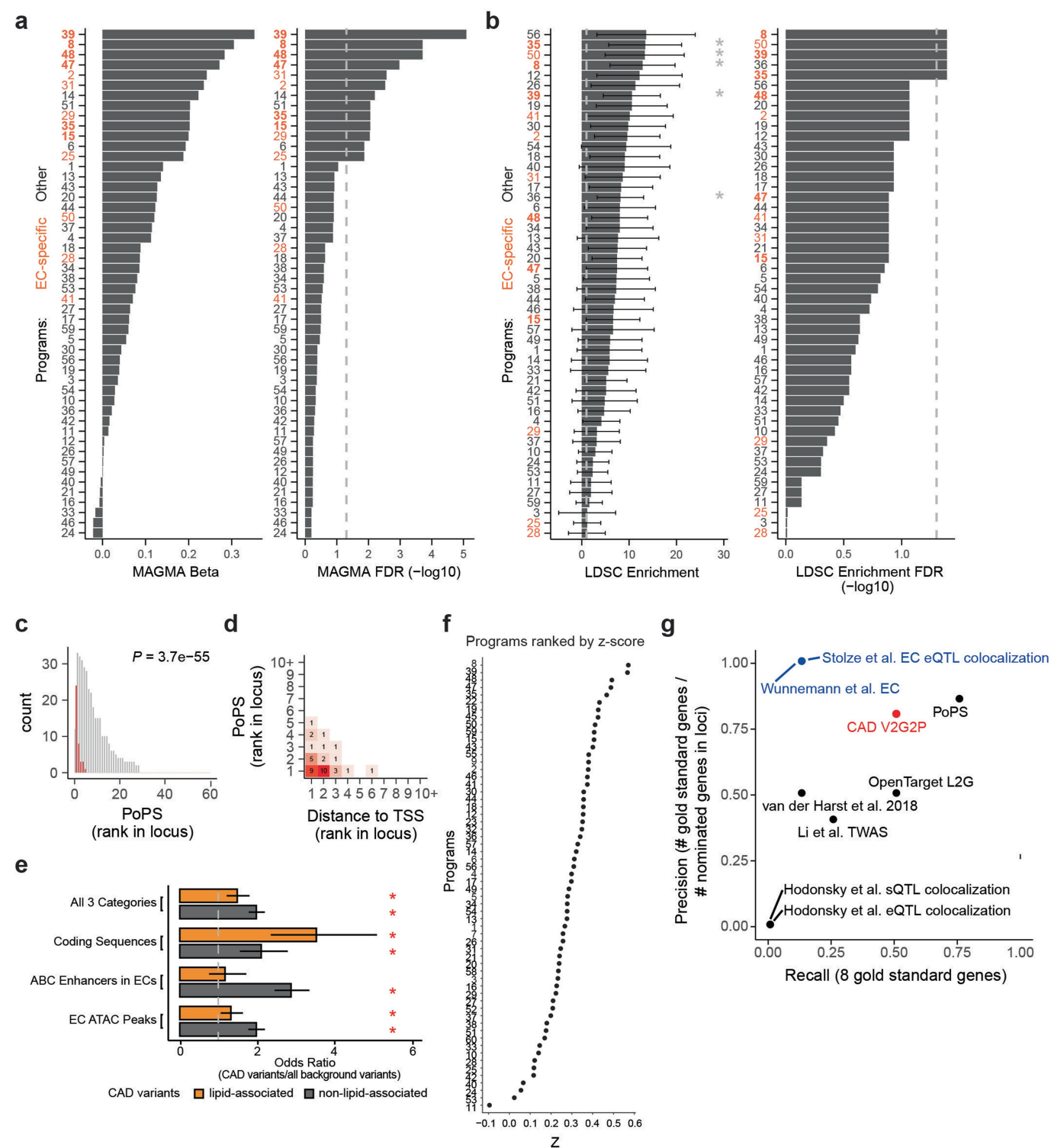
Extended Data Fig. 3 | Catalogue of gene programs. **a**, Correlation heatmap of cNMF components. Colour: Pearson's correlation of $\log_2$ fold-change in component expression across all perturbed genes. **b**, 50 programs ordered by variance explained (see Methods). **c**, 50 programs ordered by endothelial-cell specificity score – that is, the degree to which the co-regulated genes in the program are specifically expressed in endothelial cells versus in other cell types from FANTOM5 CAGE data (see Methods). Red line: z-score corresponding to top 10% of genes most specifically expressed in endothelial cells. **d**, Effects of selected regulators on the 13 endothelial-cell-specific programs. Heatmap: $\log_2$ fold-change in component expression in perturbation versus control. Top: 16 regulators shared between multiple endothelial cell-specific programs. Bottom: the 4 significant regulators (experiment-wide FDR < 0.05) per program with the most specific effects on that program relative to other endothelial-cell-specific programs. **e**, Programs ordered by number of enriched transcription factor motifs (see Methods). Grey: promoters. Blue: enhancers. Some programs only have enrichment for motifs in promoters. Some programs showed enrichment of distinct motifs in enhancers versus promoters, such as Program 47 (Angiogenesis, GATA2), with promoter enrichment in WT1 and EGR2 motifs, and enhancer enrichment in GATA2 and PRDM6 motifs. Among the programs with few or no enriched transcription factor motifs, we identified other likely proximal regulatory mechanisms: Program 17 expressed genes whose promoters were marked by H3K27me3 in endothelial cells (see also panel **k**), and the most significant regulator of this program was *SUZ12*, a component of the complex (PRC2) that writes this histone modification; and Program 16 pointed to a potential RNA surveillance program, since 40% of its program genes were noncoding RNAs (panel **l**), and its regulators included a component of the RNA exosome (*EXOSC5*) and the chromatin remodeller *INO80E*, which has previously been shown to regulate a subset of noncoding transcripts in yeast¹²⁰ (see also Supplementary Table 12). **f**, Annotations for an example program: 15. Left: top 10 program co-regulated genes. Middle, top: motifs enriched in promoters of the 300 program co-regulated genes. Middle, bottom: Gene Ontology terms enriched in the 300 program co-regulated genes.

Right: volcano plot of the effects of regulators on cNMF component 15 genes. Program 15 (Flow response, KLF2) appeared to correspond to a canonical endothelial cell response to laminar shear stress defined by the known flow-responsive transcription factor *KLF2*: the program was highly enriched for KLF motifs in promoters; included known flow-responsive genes such as *KRT18/19*, *NOS3*, and *KLF2* itself; and was significantly reduced by perturbations to *MAP2K5* (MEK5), a kinase known to activate the signaling pathway upstream of *KLF2*^{35,121}. **g**, Log₂ fold change in expression of programs 28 versus 47 for each perturbed gene relative to controls. Program 28 (Tip cell, migration) includes co-regulated genes that mark tip cell specification during sprouting angiogenesis (*ESM1*, *RHOC*, *PLAUR*), and Program 47 (Angiogenesis, GATA) includes co-regulated genes that are enriched in GATA2 & TAL1 motifs and that include *NRP2*, a co-receptor for VEGF-A, previously shown to act downstream of GATA2¹²²). Blue, red, and purple mark genes that are regulators of Program 28, Program 47, or both programs, respectively. Note that regulators that affect both programs do so in opposite directions. **h**, Perturbations ordered by the number of regulated programs. Red: top 10 perturbed genes. **i**, Programs ordered by the number of regulators. Blue: endothelial-cell-specific programs. The top 3 programs, by number of regulators, are labeled. **j**, 131 perturbed genes that are regulators of at least one endothelial-cell-specific program, ordered by the number of such programs that they regulate. Top 10 regulators are labelled, and included genes known to have important functions in ECs such as *EGFL7* and *ITGB1BP1/ICAP1*^{26,27}. **k**, Average H3K27me3 ChIP-seq signal in co-regulated gene promoters. The top program is Program 17 (Polycomb targets). See legend to **e** for more details. N = 50 programs. Boxplot centre line, median; box limits, upper and lower quartiles; whiskers, 1.5x interquartile range; points, all data points. **l**, Percent of noncoding RNA genes in program co-regulated genes. The top program is Program 16 (ncRNA & antisense RNAs). See legend to **e** for more details. N = 50 programs. Boxplot centre line, median; box limits, upper and lower quartiles; whiskers, 1.5x interquartile range; points, all data points. **m**, Same as Fig. 1d, but with transcription factor motif sequence logo. The enriched motifs have distinct sequence logos.



Extended Data Fig. 4 | Annotations for CAD-associated programs: 8, 35, 39, 47, 48. Left panels, top 10 program co-regulated genes. Program Specificity z-scores are the cNMF marker gene coefficients, indicating how specific this gene is to this program, relative to other programs (see Methods). Middle left panels, top: top 5 motifs enriched in the promoters or enhancers of the program co-regulated genes; bottom: top 5 GO terms enriched in program co-regulated

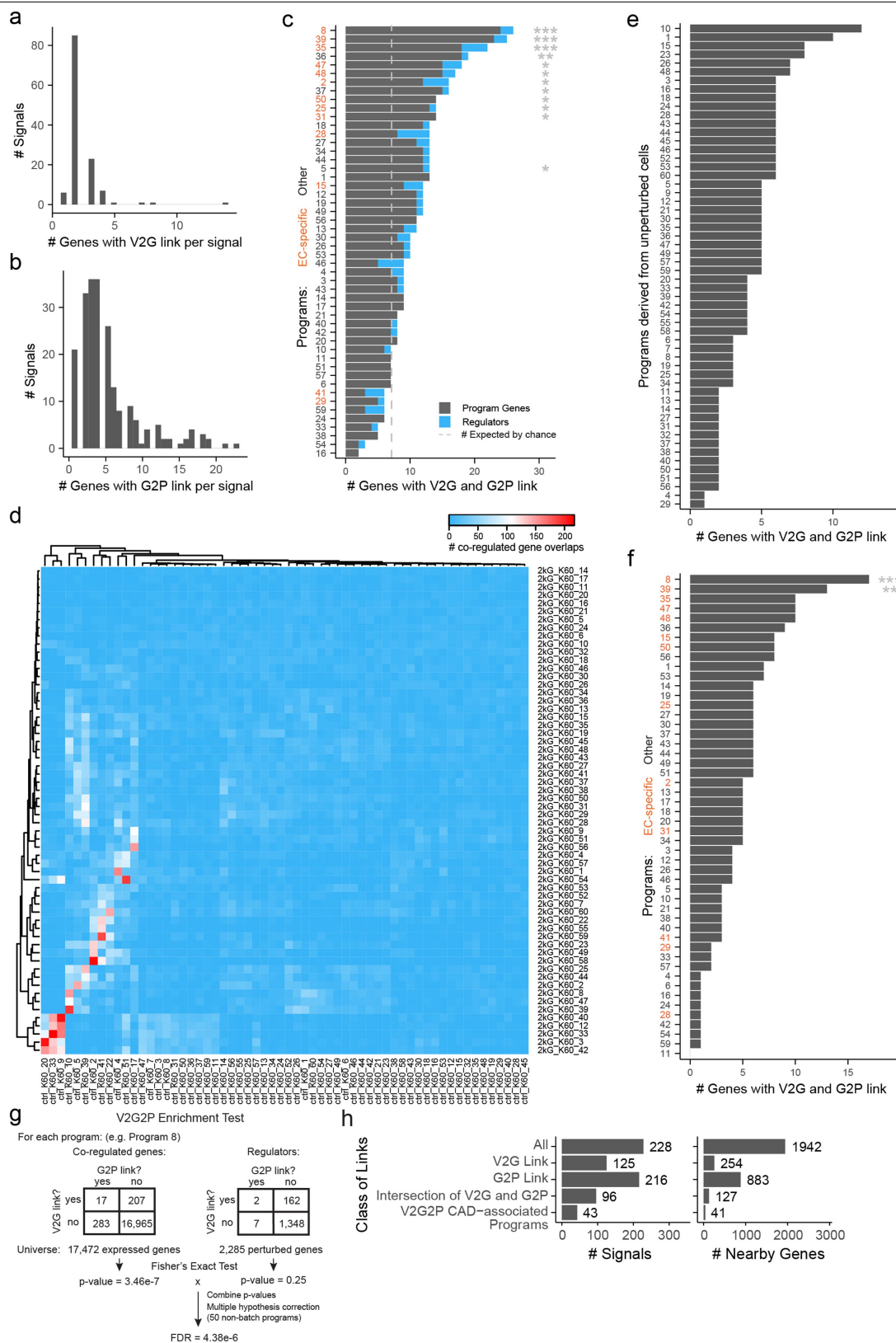
genes. Middle right panels, regulators of the program. Volcano plot shows effects of all perturbed genes on program expression. Red: FDR < 0.05. Labelled: top two significant regulators in each direction, plus *CCM2* and *TLNDR1*. Right panels, UMAP of program expression in a subset of cells (24,000, randomly selected).



Extended Data Fig. 5 | See next page for caption.

Extended Data Fig. 5 | Prioritization of CAD-associated programs and candidate CAD genes. **a**, Using MAGMA to prioritize gene programs enriched for CAD heritability (linking variants to program genes and 50 kb of flanking sequence, see Methods). Barplots show beta regression coefficient (left) and $-\log_{10}$ FDR (Benjamini-Hochberg adjusted enrichment p -value, right). Programs are ordered separately by beta or FDR value. Dotted line: FDR = 0.05. **b**, Using S-LDSC to prioritize gene programs enriched for CAD heritability (linking variants in endothelial cell chromatin accessible regions to genes within 50 Kb, see Methods). Barplots show enrichment (left) and $-\log_{10}$ FDR (Benjamini-Hochberg adjusted enrichment p -value, right). N = 300 (co-regulated program genes ranked by z-score coefficient, for each program). Error bars: standard error around the enrichment estimate, calculated by S-LDSC using jackknife (which resamples the data used for calculating heritability enrichment). P-values were calculated using the S-LDSC method²⁸, and FDR by the Benjamini-Hochberg method. *: FDR < 0.05. Dotted lines: 1 fold enrichment (left), or FDR 0.05 (right). **c**, CAD-associated V2G2P genes are ranked highly by an independent gene prioritization method, the Polygenic priority score (PoPS). For each of the 43 CAD GWAS signals including a CAD-associated V2G2P gene, we ranked nearby genes based on their PoPS scores. Red: 39 CAD-associated V2G2P genes (two genes, *EXOC3L2* and *PECAMI*, were not assigned scores by PoPS). Grey: all other nearby genes. p -value: two-sided Mann-Whitney U-test. **d**, Contingency table of PoPS and distance-to-TSS ranks for the 39 CAD-associated V2G2P genes. (two CAD-associated V2G2P genes were not assigned scores by PoPS). **e**, Odds ratios of variants in lipid-associated (N = 1,181) or non-lipid-associated (N = 3,313) CAD GWAS signals in (i) ATAC peaks in endothelial

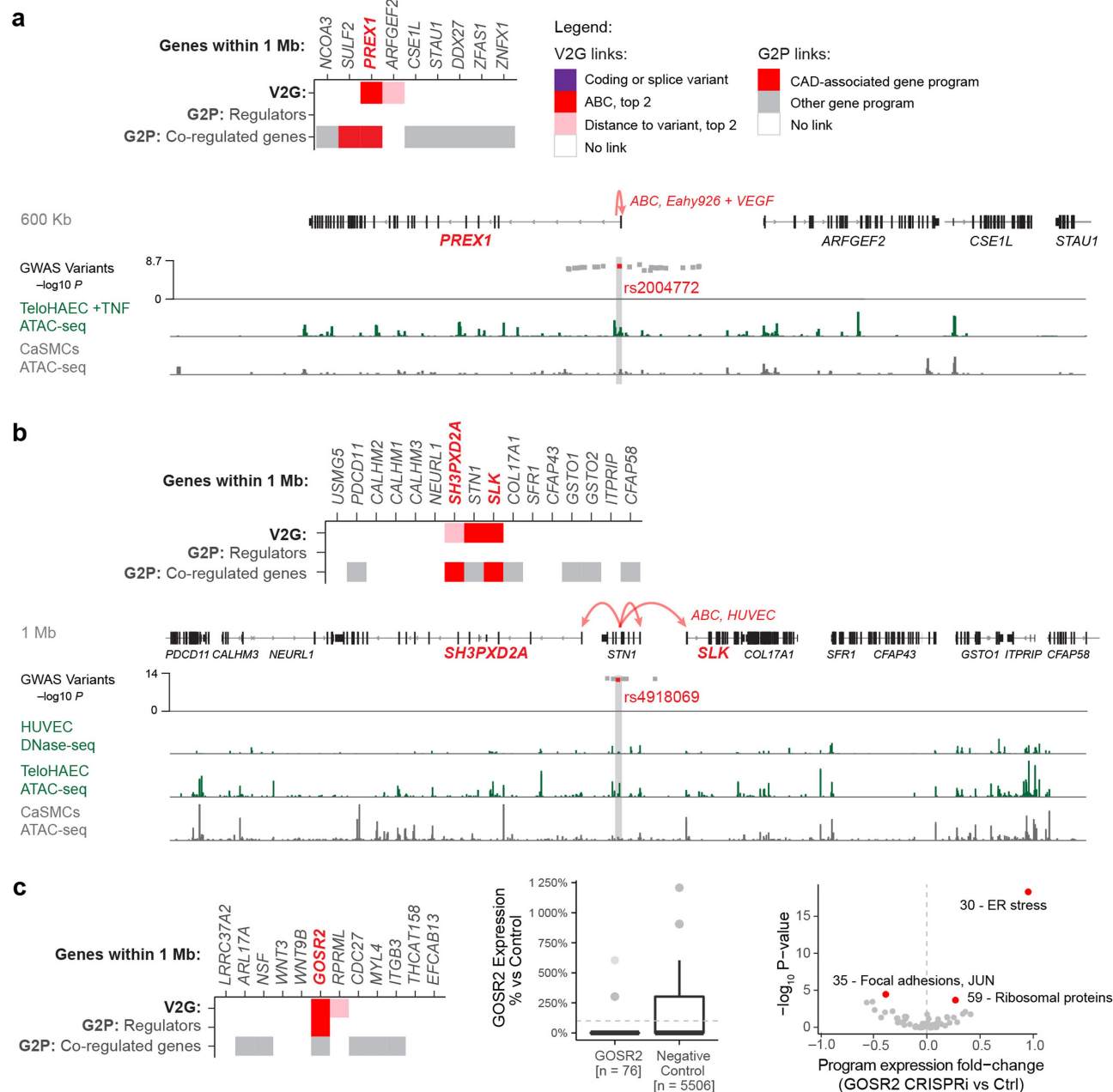
cells (N = 373,630 unique non-overlapping non-promoter features from 11 epigenomic datasets in ECs, see Methods), (ii) ABC enhancers in endothelial cells (N = 47,112 unique non-overlapping non-promoter features from 11 epigenomic datasets in ECs), (iii) coding sequences (N = 189,232 unique non-overlapping non-promoter features), or (iv) all three categories combined (N = 519,046 unique non-overlapping non-promoter features), compared to background variants (all SNPs from 1000 Genomes, excluding lipid-associated or non-lipid associated CAD GWAS variants, N = 9,955,2088 or N = 9,953,076, respectively; see Methods). Odds ratios were calculated as $((\text{CAD variants within the indicated genomic features})/(\text{all background variants within these features})) / ((\text{CAD variants outside of these features})/(\text{all background variants outside of these features}))$, and significance assessed by application of a two-sided Fisher's exact test to the contingency table of these data, with columns = CAD variants versus background variants and rows = inside features versus outside features. Error bars: 95% confidence interval. *: FDR < 0.05. Specific FDR values, from top to bottom, were 1.1e-4, 3.3e-33, 1.5e-8, 3.2e-6, 0.39, 6.0e-32, 0.011, 7.5e-31. Dotted line: odds ratio of 1. **f**, sc-linker prioritization for 60 EC Perturb-seq gene programs, ranked by z-score. The ranking of programs was similar to V2G2P analysis, but none of the programs reached significance. **g**, Precision/Recall (PR) plot for V2G2P and seven prior approaches to prioritize CAD locus genes. Recall: the fraction of the eight "gold standard" genes (with strong prior evidence for endothelial cell-specific roles in CAD) detected by each method. Precision: [number of "gold standard" genes called]/[number of genes called within these gold standard loci]. Red: V2G2P. Blue: other studies that prioritized CAD GWAS genes in endothelial cells.



Extended Data Fig. 6 | See next page for caption.

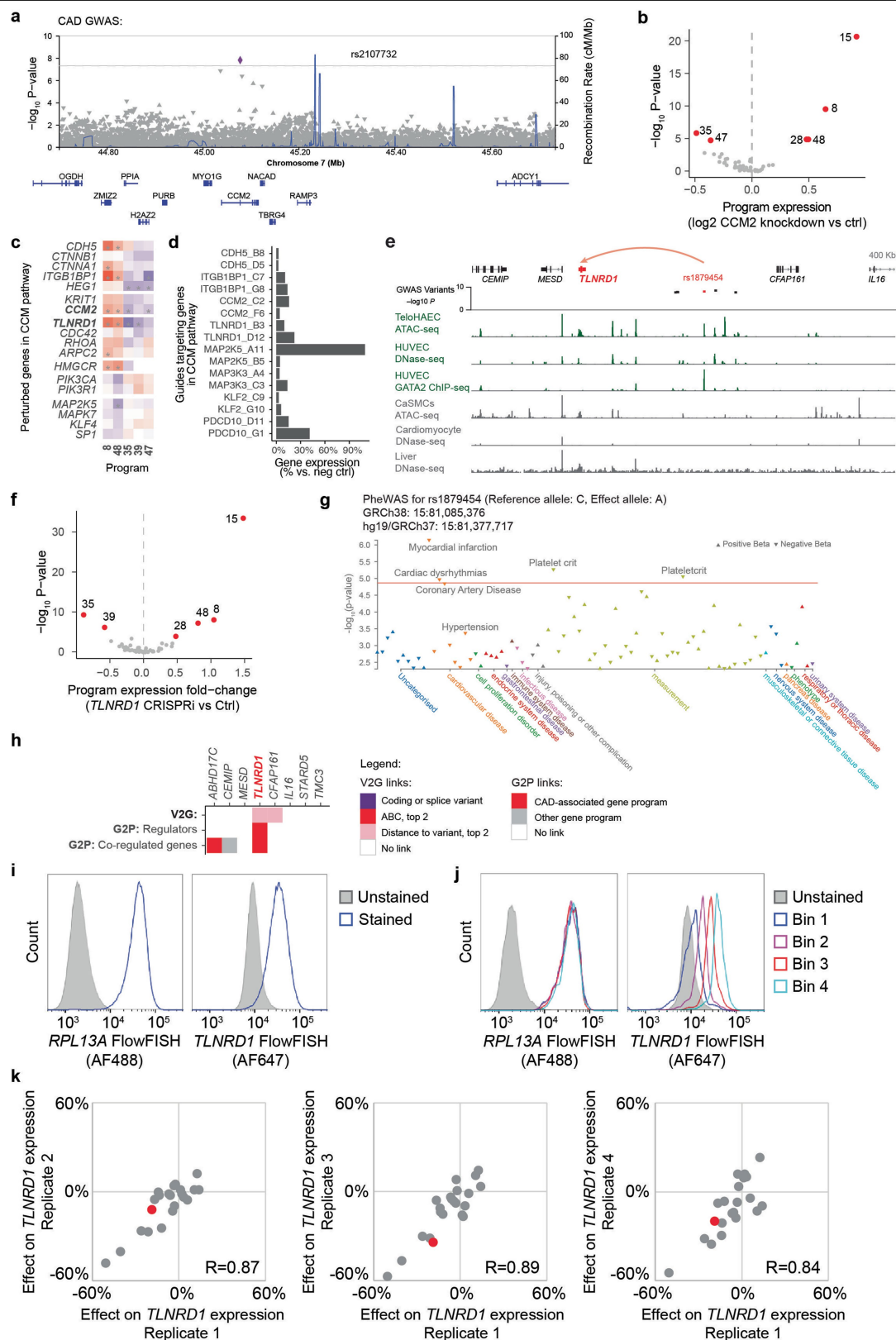
Extended Data Fig. 6 | Details for V2G2P analysis. **a**, Number of genes with V2G links, per non-lipid CAD GWAS signal. **b**, Number of genes with G2P links, per non-lipid CAD GWAS signal. **c**, The cell-type specificity of V2G links appeared to be important for identifying endothelial-cell-specific programs. Here, we repeated the V2G2P analysis (as outlined in Fig. 2), but linked variants to genes using cell-type-agnostic criteria (including ABC scores from any cell type and not just endothelial cells). The 50 programs are ordered (y-axis) by the number of program genes linked to CAD variants (x-axis). Grey dashed line: the number of genes linked to CAD variants that would be expected by chance. Orange labels: endothelial-cell-specific programs. Two significant non-endothelial-cell-specific programs were identified. Fisher exact test with FDR correction, as outlined in panel **g**. (* $FDR < 0.05$, ** $FDR < 0.005$, *** $FDR < 5e-4$). **d**, Number of overlapping co-regulated genes between control programs (ctrl, x-axis) and full library (2 kG, y-axis) programs. **e**, Using the full Perturb-seq dataset

appeared to be important for identifying the five CAD-associated programs. Programs discovered through cNMF analysis of only the “unperturbed” cells carrying negative control guideRNAs. In this version of the analysis, none of the programs are enriched for genes with V2G link. Fisher exact test with FDR correction, see procedure in panel **g**. (All programs have $FDR > 0.05$.) **f**, Enrichment of genes with V2G links, from the full library but only using co-regulated genes (not regulators, ***: $FDR < 0.0005$, **: $FDR < 0.005$). **g**, Steps of V2G2P enrichment test. Numbers shown as examples are from program 8. **h**, V2G2P analysis prioritizes a small subset of genes and GWAS signals compared to either V2G or G2P information alone. Barplots: counts for signals (left) or nearby genes (right), total (“All”) or those that have: a V2G link, a G2P link, both a V2G link and G2P link to any program, or both a V2G link and a G2P link to a significantly enriched V2G2P program.



Extended Data Fig. 7 | Variant-to-gene to program links refine causal gene predictions. **a**, V2G2P evidence at the *20p13.1* CAD GWAS locus. Top, heatmap lists genes within 1 Mb of the CAD GWAS signal in genomic order, and shows variant-to-gene (V2G) and gene-to-pathway (G2P) evidence, with the prioritized CAD-associated V2G2P gene(s) labelled in red bold font. Legend details: “ABC, top 2”: a noncoding variant overlaps a chromatin accessible peak in endothelial cells, and the ABC score is at least the second highest of all genes near the GWAS signal. “Distance to variant, top 2”: a noncoding variant overlaps a chromatin accessible peak in endothelial cells, and the gene is one of the two closest genes to the variant. Bottom, zoom-in on genes near the CAD GWAS signal, where rs2004772 is predicted by ABC to regulate *PREX1* in the Eahy926 endothelial cell line treated with VEGF. Red dot: prioritized variant in predicted enhancer. Grey dots: other variants within $R^2 < 0.9$ of the lead variant in the locus. Signal tracks below show ATAC-seq or DNase-seq for endothelial and coronary artery

smooth muscle cells (CaSMCs, another cell type relevant to CAD). **b**, As per **a**, showing V2G2P evidence at the *10p24.33* CAD GWAS signal, where three genes had V2G links (to an enhancer containing rs4918069) and two had gene to CAD-associated program links. HUVEC: human umbilical vein endothelial cells. **c**, V2G2P evidence at the *17q21.3* CAD GWAS locus, where we have previously linked rs17608766 to *GOSR2*²³. Heatmap, as in panel **a**. Middle: box plot of *GOSR2* reads per cell, normalized to control cell average. n: number of cells. Dotted line: control average (100%). Boxplot centre line, median; box limits, upper and lower quartiles; whiskers, 1.5x interquartile range; points, outliers. Counts for outliers, from top to bottom: *GOSR2*; 1 & 17, Control; 4 & 15. Right: volcano plot shows effect of *GOSR2* knockdown in Perturb-seq on the expression of the 50 non-batch gene programs. Red: FDR < 0.05, from two-sided statistical tests on program expression between perturbed versus control cells by the MAST package⁵⁰.

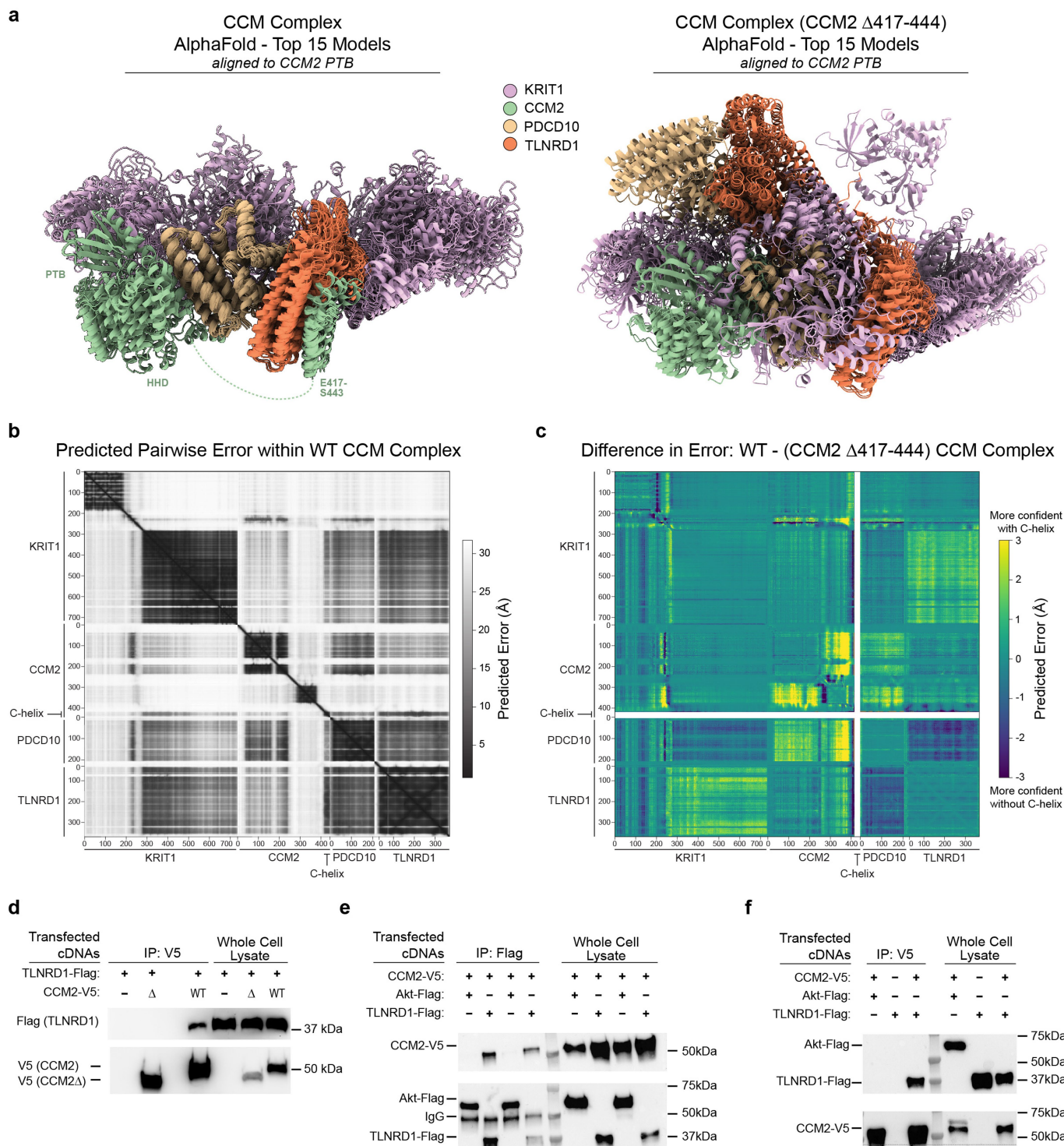


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Extended Data Fig. 8 | Regulatory connections amongst perturbed genes and the CCM pathway, and variant-to-gene links to *TLNRD1*. **a**, Locus zoom plot (<http://locuszoom.org/>) for CAD GWAS in a 1-Mb region around *CCM2*. P-values are from the joint association analysis in ref. 12. **b**, Volcano plot showing effect of *CCM2* knockdown in Perturb-seq on the expression of the 50 programs. Red: $FDR < 0.05$. Significance was assessed by two-sided statistical test on program expression between perturbed versus control cells by the MAST package⁵⁰. **c**, Effects of selected perturbed genes on CAD-associated programs (same as Fig. 3b, with significant effects marked with a * ($FDR < 0.05$)). Colour scale: $\log_2$ fold-change on program expression in Perturb-seq. Bold text: CAD-associated V2G2P genes. **d**, Knockdown efficiency for genes in the CCM pathway, in bulk RNA-seq (Fig. 3c). x-axis: gene expression for each target gene in cells receiving target guides, versus cells with control guides. y-axis: guide IDs for CCM pathway genes. **e**, 15q25.1 CAD risk locus, where rs1879454 is predicted to regulate *TLNRD1* (red arc). GWAS variants: $-\log_{10}$ GWAS P -value¹² for variants with $LD R^2 > 0.9$ with the lead SNP. Green signal tracks: epigenomic data from ECs. Grey signal tracks: data from other cell types. HUVEC: human umbilical vein ECs. CaSMCs: coronary artery smooth muscle cells. **f**, Volcano plot showing the effect of *TLNRD1* knockdown in Perturb-seq on the expression of the 50 programs. Red: $FDR < 0.05$. Statistical test as per **b**. **g**, Phenome-wide association study (PheWAS) for rs1879454, from Open Targets^{98,116}. P-values are the best GWAS p-values for association of this variant with each trait across all GWAS sources used by Open Targets (February 25, 2022 release), including UKBB. Orange line: p-value required for significance across all traits. No lipid

measure met the p-value threshold for inclusion in the plot, of 0.005. Note: the p-value for CAD is higher than that observed in ref. 12 because Open Targets does not currently contain summary statistics for the latest CAD GWAS. There were no measures of circulating lipids or blood pressure associated with this GWAS signal in a PheWAS analysis in ref. 12. **h**, Variant-to-gene-to-program evidence at the 15q25.1 CAD GWAS locus. Heatmap shows variant-to-gene (V2G) and gene-to-pathway (G2P) evidence for all genes within 1 Mb of the CAD GWAS signal, in genomic order, with the CAD-associated V2G2P gene labelled in red bold font. Legend details: “ABC, top 2”: a noncoding variant overlaps a predicted enhancer linked to this gene in endothelial cells, and ABC score is at least the second highest of all genes in the locus. “Distance to variant, top 2”: a noncoding variant overlaps a chromatin accessible peak near this gene in endothelial cells, and the gene is one of the two closest genes to the peak. “CAD-associated gene programs” are the five V2G2P programs: 8, 35, 39, 47, 48. **i**, Histograms of FlowFISH signal (arbitrary units of fluorescence) for *RPL13A* (left) and *TLNRD1* (right) in unstained versus stained teloHAEC expressing the gRNA pool against promoter and potential enhancers. The complete FlowFISH data can be found in Supplementary Table 28. **j**, As in **d**, but showing the *RPL13A* (left) and *TLNRD1* (right) signal after sorting of cells into four bins based on expression of *TLNRD1*. Results are typical of cells across the four independent samples. **k**, Scatter plots showing the strong correlation between effects on *TLNRD1* gene expression for all E-G pairs measured in each of four independent CRISPRi-FlowFISH screens. The enhancer containing rs1879454 is coloured in red, all others in grey. R: Pearson correlation coefficient.



Extended Data Fig. 9 | See next page for caption.

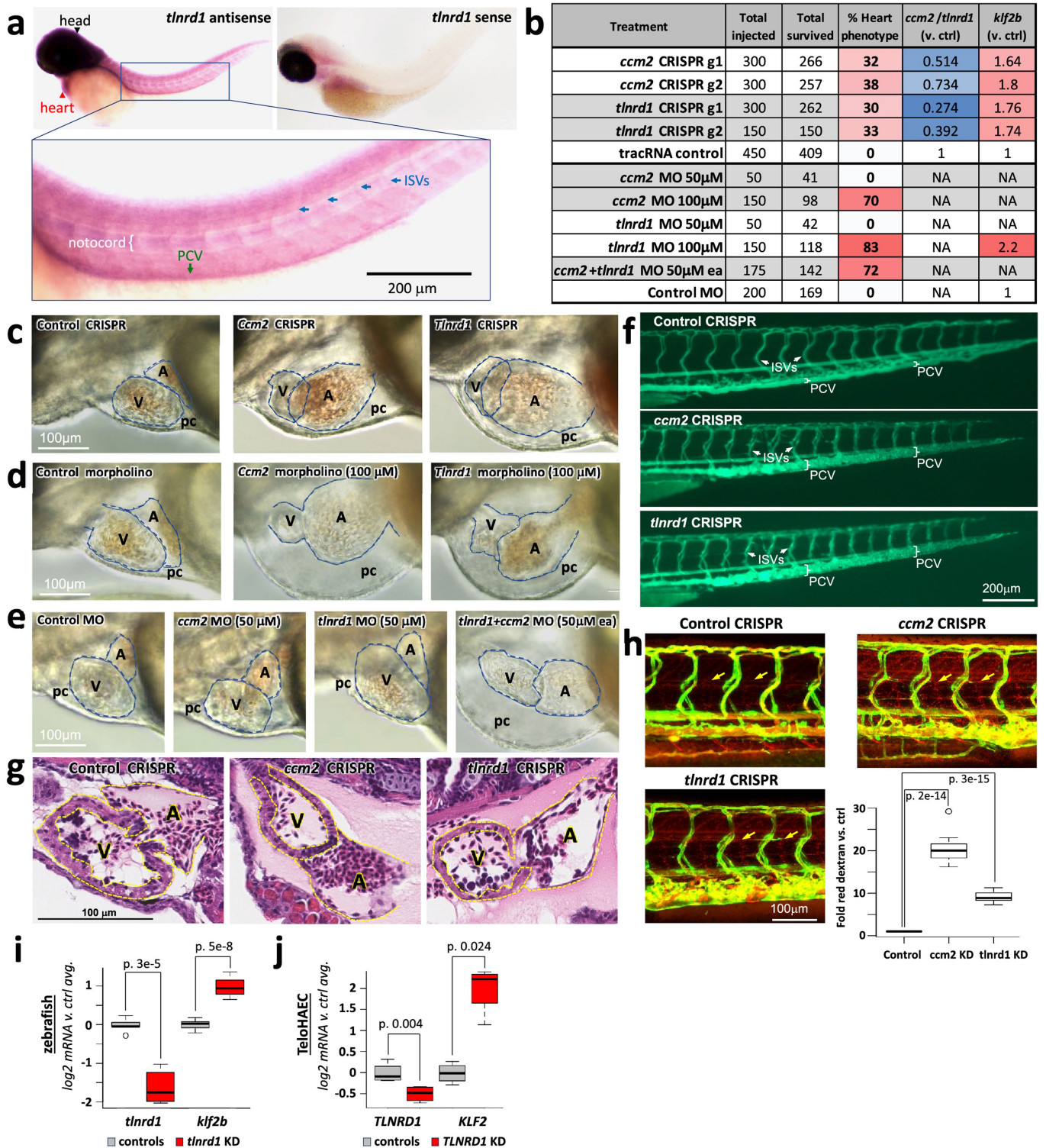
Extended Data Fig. 9 | TLNRD1 interaction with the CCM complex depends on the CCM2 C-terminal helix. **a.** The 15 top-ranking AlphaFold2 models of the CCM complex are shown, as predicted with and without the presence of CCM2 residues 417–444, and aligned to the CCM2 PTB domain (residues 55–237). The complex is consistently predicted in a high-confidence arrangement (left panel), with most variability in positions of the HHD domain of CCM2 and flexible regions of Krit1. Interactions are predicted between all members of the complex, including published interactions between CCM2-PDCD10. In contrast, multiple conformations are predicted in the absence of the CCM2 C-terminal helix (right panel). **b.** Predicted Alignment Error (PAE, Å) of all pairwise residue combinations in the WT CCM complex (extracted and plotted using AlphaPickler⁸²); lower error indicates higher confidence. **c.** Differences in PAE (Å) between the full CCM complex and the CCM complex lacking the CCM2 C-terminal helix; larger numbers represent higher confidence in the presence of the helix. The thick white lines correspond to deleted helix residues, which are omitted from comparison. Predictions within individual domains and proteins are largely unaffected, but within CCM2, the HHD and subsequent loops are predicted with reduced confidence upon helix deletion (yellow). Between domains and proteins, the largest differences are reduced confidence interactions between CCM2 and PDCD10 and between Krit1 and TLNRD1. Summing over the entire matrices, the increase in predicted error with deletion of the helix is 0.7 Å (22.9 versus 23.6 Å, $p = 10^{-56}$, two-tailed t-test). **d.** FLAG-tagged TLNRD1 and/or V5-tagged CCM2 full length (“WT”) or C-terminal

truncation (“Δ”) were expressed in HEK293T cells, as indicated. Extracts were co-immunoprecipitated with rabbit anti-V5 and blotted with mouse anti-Flag (top) or mouse anti-V5 (bottom). For gel source data, see Supplementary Fig. 1b. Similar results were seen in two separate experiments. **e.** HEK293 cells were transfected with V5-tagged CCM2 and either Flag-tagged TLNRD1 or Flag-tagged Akt (negative control). Cell lysates were either immunoprecipitated with mouse anti-Flag antibody-bound beads (IP Flag), or loaded directly on the gel (Input). The membranes were first probed with rabbit anti-V5 to detect CCM2 in the Flag precipitant and confirm the transfection of CCM2-V5. The membranes were then re-blotted with rabbit anti-TLNRD1 to evaluate the efficiency of Flag immunoprecipitation and validate the transfection of Akt-Flag and TLNRD1-Flag. Each pair of lanes came from independent biological replicates. For gel source data, see Supplementary Figure 1c. Similar results were seen in two separate experiments. **f.** HEK293 cells were transfected with V5-tagged CCM2 and Flag-tagged TLNRD1, or, as negative controls, either with CCM2-V5 and Akt-Flag or only TLNRD1-Flag. Cell lysates were either immunoprecipitated with anti-V5 beads (IP V5), or loaded directly on the gel (Input). The membranes were first blotted for Flag to detect TLNRD1 in the V5 precipitant and validate the transfection of Akt-Flag and TLNRD1-Flag. The membranes were re-blotted for V5 to evaluate the efficiency of V5 immunoprecipitation and confirm the transfection of CCM2-V5. For gel source data, see Supplementary Fig. 1d. Similar results were seen in two separate experiments.

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Extended Data Fig. 10 | Effects of *CCM2* and *TLNRD1* knockdown relative to *MAP3K3* knockdown and laminar flow. a–d, CRISPRi TeloHAEC with control non-targeting guides or with guides to *CCM2* or *TLNRD1* (two guides apiece) were treated with 2 µg/ml doxycycline for 3 days and nucleofected with Cas9 particles containing control non-targeting guides or with 3 guides targeting exon 3 of *MAP3K3* (Synthego). Cells were grown for 48 hours in media with 2 µg/ml doxycycline, and RNA harvested for bulk RNA-seq. The *CCM2* guides reduced target gene expression, on average, by 3.4–3.6 fold ($p < 2 \times 10^{-9}$), while *TLNRD1* guides reduced target gene expression by 9.4–9.9 fold ($p < 2 \times 10^{-43}$), consistent with the effects of these guides in our other bulk RNAseq data (Fig. 3c). *MAP3K3* transcript levels were not significantly reduced, but genome-mappable reads for the targeted exon were greatly reduced, and most of the remaining reads showed multiple mismatches, indicating efficient introduction of Cas9-targeted deletions. $N = 2$ per condition (from one experiment, 1 RNAseq library for each of 2 CRISPRi guides per target). Correlation coefficient (R), and p-values given in each panel are from a two-sided Pearson correlation test. **a**, The difference between the effect of *CCM2* knockdown in cells with *MAP3K3* knockdown and the effect of *CCM2* knockdown in control cells ($\log_2[\text{CCM2kd_with_MAP3K3kd}/\text{Control_with_MAP3K3kd}]/[\text{CCM2kd}/\text{Control}]$, Y-axis) was plotted against the effect of *CCM2* knockdown in control cells ($\log_2[\text{CCM2kd}/\text{Control}]$, X-axis), plotting all genes regulated at $p < 5 \times 10^{-4}$ in either contrast). Labelled genes are the top give up- or down-regulated genes by $\log_2$ fold change, on each axis. Diagonal line: slope -1 reference. **b**, As in **a**, but for the difference between the effect of *TLNRD1* knockdown in cells with *MAP3K3* knockdown and the effect of *TLNRD1* knockdown in control cells (Y-axis), versus the effect of *TLNRD1* knockdown in control cells (X-axis). The negative correlations in **a**, **b** indicate that *MAP3K3* perturbation partially reverses the transcriptomic effects of *CCM2* or *TLNRD1* knock down, consistent with a role of MEKK3/*MAP3K3* signalling in regulating transcription downstream of both *CCM2* & *TLNRD1*. **c**, As in **a**, but the difference between the effect of *MAP3K3* knockdown in cells with *CCM2* knockdown and the effect of *MAP3K3* knockdown in control cells (Y-axis) versus the effect of *MAP3K3* knockdown alone (X axis). **d**, As in **a**, but for the difference between the effect of *MAP3K3* knockdown in *TLNRD1* knockdown cells and the effect of *MAP3K3* knockdown in control cells (Y-axis) versus the effect of *MAP3K3* knockdown in control cells (X axis). The negative correlations in both **c** and **d** indicate that perturbation of *CCM2* or *TLNRD1* partially reverses the transcriptional effects of *MAP3K3* knockdown, consistent with the expectation

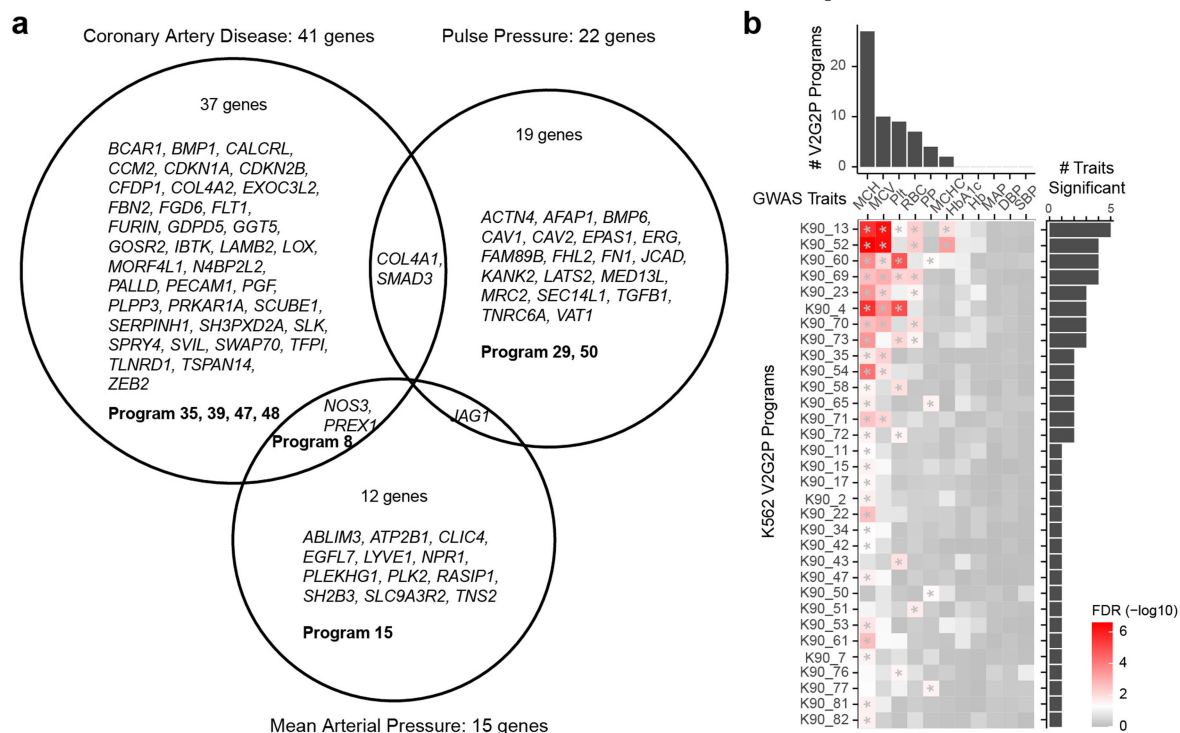
that decreased expression of upstream inhibitors can compensate for decreased expression of MEKK3. **e–h**, CRISPRi TeloHAEC with control non-targeting guides or with guides to *CCM2* or *TLNRD1* (2 guides apiece) were grown in static culture or subjected to flow in an Ibidi flow chamber for 48 h. In each case, cells were treated with 2 µg/ml doxycycline to induce the CRISPRi machinery for 5 days (3 days prior & 2 days after introduction of laminar flow). After phase contrast imaging, RNAseq libraries were prepared and sequenced to a depth of 10–12 million reads. $N = 2$ per condition (one experiment, with 1 RNAseq library for each of 2 CRISPRi guides per target). R and p-values, and labeled genes, as per **a–d**. **e**, The effects of *CCM2* knockdown in static culture (Y-axis) compared to the effect of flow in control cells (X-axis, showing all genes regulated at $p < 5 \times 10^{-4}$ in either contrast). Diagonal line: slope = +1 reference. **f**, As in **e**, but with Y-axis = effects of *TLNRD1* knockdown in static culture. The positive correlations indicate that *TLNRD1* or *CCM2* knockdown in static culture is similar to the effect of flow. **g**, The difference between the effect of flow in cells with *CCM2* knockdown and the effect of flow in control cells ($\log_2[\text{Flow_CCM2kd}/\text{Static_CCM2kd}]/[\text{Flow_Ctrl}/\text{Static_Ctrl}]$, Y-axis) versus the effect of flow in control cells ($\log_2[\text{Flow_Ctrl}/\text{Static_Ctrl}]$, X-axis). Diagonal line: slope = -1 reference. **h**, As in **g**, but showing the difference between the effect of flow in cells with *TLNRD1* knock down and the effect of flow in control cells (Y-axis) versus the effect of flow in control cells (X-axis). The negative correlations in **g**, **h** indicate that *CCM2* or *TLNRD1* knockdown cells have a weaker transcriptional response to flow than control cells, consistent with these cells already having a partial flow-like transcriptional phenotype in static culture (**e**, **f**). **i**, Representative images of CRISPRi teloHAEC with the indicated guides, under laminar flow. $N = 2$ per condition from one experiment (two distinct CRISPRi guides to *CCM2*, *TLNRD1* or controls), and with four images per guide. **j**, The normal alignment to flow in control teloHAEC (measured as the angle, relative to flow, of the long axis of each cell) is significantly abrogated in both *CCM2* & *TLNRD1* KD cells (increased average angle relative to flow). Average values for all cells in each of four images for each of two guides per target were calculated (35 to 103 cells per image). Significance was assessed by a two sided T-test on these average values. $N = 8$. Boxplot features, as per Extended Data Fig. 11. Note that alignment to flow is not completely blocked in *CCM2* or *TLNRD1* KD cells, since the average angle relative to flow does not reach the 45% value expected if orientation were entirely random. **k**, As per **j**, but measuring the ratio of the long versus short axis lengths for each cell (“length/width”) from the *fit ellipse* function in Fiji.



Extended Data Fig. 11 | See next page for caption.

Extended Data Fig. 11 | Additional analysis of *ccm2* and *tlnd1* CRISPR and morpholino knockdown zebrafish embryos. Zebrafish is a model system which has been extensively used to study CCM gene functions, and where *ccm2* has been shown to have characteristic effects in heart and vascular development^{34,35,43,44,78,124}. **a**, In situ analysis of *tlnd1* mRNA expression. *tlnd1* mRNA, detected by the anti-sense in situ probe, is expressed in the head (black arrowhead) and heart (red arrowhead, top left), and in the notochord (white bracket), posterior cardinal vein (PCV, green arrow) and intersegmental vessels (ISVs, blue arrows, bottom panel). This staining pattern is not seen with the negative control sense probe (top right). N = 10 from one experiment. This staining pattern is consistent with *TLNRD1* being most highly expressed in human endothelial cells (Tabula Sapiens atlas of gene expression, <https://tabula-sapiens-portal.ds.czbiohub.org/>). **b**, Quantitation of *ccm2* and *tlnd1* CRISPR/morpholino heart phenotypes, knockdown efficiency and effects on *klf2b* expression. The table summarizes the number of embryos injected with each guide, or with tracrRNA control (for CRISPR experiments), or with each experimental or control morpholinos at the indicated concentration(s), the number that survived, and the percent that showed a heart phenotype characterized by enlarged atrium, pericardial edema and slow blood flow in tail veins. “*ccm2/tlnd1* (v. ctrl)” summarizes qRT-PCR analysis of knockdown efficacy (*ccm2* or *tlnd1* levels in embryos with CRISPR guides to each of these target genes, versus control embryos). “*klf2b* (v. ctrl)” summarizes qRT-PCR quantification of *klf2b* in CRISPR or morpholino knockdown animals versus controls. “NA”: effects on *ccm2* or *tlnd1* were not measured for the morpholino studies, because morpholinos generally function by inhibiting translation; and *klf2b* levels were not tested for the indicated morpholino treatments. See also Supplementary Table 20. **c**, Light microscope images of CRISPR embryos. Representative images of Zebrafish 3dpf embryos injected with control, *ccm2* or *tlnd1* gRNA and Cas9 protein. A: atrium. V: ventricle. pc: pericardial space. For N and experimental replicates see **b** and Supplementary Table 20. **d**, Similar heart phenotypes in *ccm2* and *tlnd1* morpholino embryos. Knockdown of either *tlnd1* or *ccm2* with 100 μ M anti-*tlnd1* or anti-*ccm2* morpholino caused similar heart defects as seen by CRISPR knockdown, with no heart defects seen using 100 μ M control morpholino. For N and experimental replicates see **b** and Supplementary Table 20. **e**, Synergistic effect of *ccm2* and *tlnd1* morpholinos. As in **d**, but showing the synergistic phenotype of 50 μ M *tlnd1* & 50 μ M *ccm2*

morpholinos, which, individually, showed no phenotype, but together showed the heart phenotype in 72% of embryos. For N and experimental replicates see **b** and Supplementary Table 20. **f**, Vascular phenotype in *ccm2* and *tlnd1* morpholino embryos. Representative microangiogram images showing FITC dextran green (2000 kDa) injected in the vasculature in control, *ccm2* and *tlnd1* gRNA injected 3 dpf larvae. Brackets mark the thickness of the posterior cardinal vein (PCV). Arrows indicate the intersegmental vessels (ISVs). Experiments were repeated 3 times. N = 6 for control, and 5 for *ccm2* or *tlnd1* gRNAs, from one experiment. **g**, Ventricle wall thinning in *ccm2* and *tlnd1* morpholino embryos. Hematoxylin & eosin (H&E) stained sections of 3dpf embryos. A: atrium. V: ventricle. The space between dotted lines in the ventricle indicates ventricular wall thinning in *ccm2* or *tlnd1* CRISPR embryos. Cells within each chamber are blood cells. N = 3 for each treatment, from one experiment. **h**, More permeable vasculature in *ccm2* and *tlnd1* morpholino embryos. Representative images from vascular permeability analysis in control, *ccm2* and *tlnd1* gRNA injected zebrafish at 3 dpf. Red colour indicates Texas red dextran 70 KD, which was injected into the vasculature before imaging. Green: green fluorescence protein expression in the vasculature (*Tg:Flu GFP*). Both *ccm2* and *tlnd1* gRNA injected embryos displayed higher levels of red dye in the interspace between the vessels (arrows). Bottom right: quantitation of permeability (ratio of red dextran in interspace vs. controls. n = 10 for control, 13 for *ccm2* & 13 for *tlnd1*). Significance was assessed by two-sided T-test. Boxplot centre line, median; box limits, upper and lower quartiles; whiskers, 1.5x interquartile range; points, outliers. For the complete data, see Supplementary Table 31. **i**, *tlnd1* knockdown unregulated *klf2b* in zebrafish. qRT-PCR for knockdown of *tlnd1* & induction of *klf2b* in zebrafish embryos treated with CRISPR guides to *tlnd1*, or with control tracrRNA. Signal was normalized to Actin, and then to the average for controls. n = 9 for *klf2b* (6 for guide AF, 3 for guide AN.2). n = 5 for *tlnd1* (2 for guide AF, 3 for guide AN.2). Quantitation and boxplot features as in **h**. For the complete data, see Supplementary Table 30. **j**, *TLNRD1* knockdown upregulates *KLF2* in TeloHAEC. qRT-PCR for knockdown of *TLNRD1* and induction of *KLF2* in TeloHAEC with Cas9-guide nucleofection knock down of *TLNRD1* (or non-targeting guides, “Control”). Signal was normalized to *GAPDH*, and then to the average for controls. n = 4 separate samples. Quantitation and boxplot features as in **h**. For the complete data, see Supplementary Table 30.



Extended Data Fig. 12 | Application of V2G2P to other traits and other cell models. a, Venn diagram of V2G2P genes for coronary artery disease (CAD), pulse pressure (PP), and mean arterial pressure (MAP) GWAS traits in teloHAEC (using the same ABC-maps and Perturb-seq data, but disease variants for each trait). For MAP, we prioritized program 8 (ECM organization, AQP1, $FDR = 0.0135$) and program 15 (KLF2, flow response, $FDR = 0.0289$). For PP we prioritized program 50 (TGF β response, $FDR = 0.0046$) and program 29 (EDN1, wound healing, $FDR = 0.0316$). Several genes in the PP programs are known to regulate vascular tone and stiffness, including *FHL2*, *SMAD3*, and *TGFB1*^{125–127}. **b**, K562 V2G2P programs for mean corpuscular haemoglobin (MCH), mean corpuscular volume (MCV), platelet count (Plt), red blood cell count (RBC), pulse pressure (PP), mean corpuscular haemoglobin concentration (MCHC), average blood glucose level (HbA1c), haemoglobin count (Hb), mean arterial

pressure (MAP), diastolic blood pressure (DBP), systolic blood pressure (SBP). Overall, 32 programs were prioritized for 6 GWAS traits, ranging from 27 programs associated with MCH to 2 programs for MCHC. In general, traits that were not relevant to K562 erythroleukaemia cells had no K562 programs significantly associated with them (e.g. MAP, DBP & SBP). Programs associated with each trait contained genes related to that trait. For instance, the most significantly-enriched mean corpuscular haemoglobin program was K562 Program 13, which included many haemoglobin genes as well as the known regulators *GFI1B*¹²⁸ and *CBFA2T3*¹²⁹, while variants associated with platelet count showed most significant enrichment in K562 Program 4, which included genes known to be involved in megakaryocyte differentiation and platelet count such as *VASP*¹³⁰ and *TPM4*¹³¹, and which showed high enrichment of motifs for the known megakaryocyte regulators *SPI1*/*3*¹³².

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- ☐ ☒ The statistical test(s) used AND whether they are one- or two-sided
Only common tests should be described solely by name; describe more complex techniques in the Methods section.
- ☐ ☒ A description of all covariates tested
- ☐ ☒ A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons
- ☐ ☒ A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals)
- ☐ ☒ For null hypothesis testing, the test statistic (e.g. F , t , r) with confidence intervals, effect sizes, degrees of freedom and P value noted
Give P values as exact values whenever suitable.
- ☒ ☐ For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
- ☒ ☐ For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes
- ☐ ☒ Estimates of effect sizes (e.g. Cohen's d , Pearson's r), indicating how they were calculated

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Software and code

Policy information about [availability of computer code](#)

Data collection

The Opera Phenix imager (used for phalloidin stain analysis of TeloHAEC) was run using Harmony 4.9.2137.273, Acapella 5.0.1.124082 & Oda 4.9.2137.273. The Zeiss confocal microscope LSM 880 (used for zebrafish imaging) was run using ZEN 2.3 SP1 software. The Sony MA900FP cell sorter (used for FlowFISH) was run using "Cell Sorter Software" v3.2. Other instruments were used with the manufacturer-supplied intrinsic software, including EVOS microscopes, Applied Biosystems QuantStudio 5 (for qRT-PCR) and the ECIS Z-Theta instrument (Applied BioPhysics).

Data analysis

From the Code Availability section of the manuscript (numbers in parentheses are citation numbers from the manuscript): Data was processed using all packages required for running cNMF v1.2 (25). We additionally used R-3.6.1 and R-3.6.3 (98), edgeR 3.28.0 (57,61,63,64), seqLogo 1.52.0, MAST 1.12.0 (39), clusterProfiler 3.14.0 (65,99), org.Hs.eg.db 3.10.0, Seurat 3.0.2 (100), SeuratObject 4.0.0 (100–102), stats4 v3.6.1 and SingleCellExperiment 1.8.0 (103) for downstream analysis in R. The supporting packages in R used for data processing and figure generation were ggplot2 3.3.5 (104), ggpubr 0.4.0, ggrepel 0.9.1, gplots 3.1.1, gridExtra 2.3, scales 1.1.1, cowplot 1.1.1, dplyr 1.0.7, tidyr 1.1.3, textshap 1.7.1, reshape2 1.4.4, stringi 1.7.5 (105), conflicted 1.0.4, data.table 1.14.0, purrr 0.3.4, readxl 1.3.1, writexl 1.5.0, ramify 0.3.3, optparse 1.6.6, and all dependencies. Data was further processed using pysuspenders 0.2.6 (106), pysam 0.19.1 (107), python 2.7.15, and LDSC v1.0.1 (108). Other software packages used were kallisto 0.48.0(55), limma 3.42.2(56), Bowtie v 1.3.0 and Bowtie2 v2.4.2 (109), Macs2 2.2.7 (110), CellRanger 7.0.0 (111), FIMO (https://meme-suite.org/meme/meme_5.3.2/tools/fimo), plink v1.90b6.21 (112), MAGMA (2), S-LDSC (28,69), PoPS (3), FloJo v10.8.1, AlphaFold2.3 Multimer v3 (80), UCSF ChimeraX v1.61, AlphaPickle (81), Matplotlib v3.7.0, Seaborn (<https://seaborn.pydata.org/>) and MATLAB R2018a (MathWorks). Phalloidin analysis was done using Fiji/ImageJ 2.9.0/1.53t with the LPX plugin. CRISPRi primer design used CRISPRDesigner, <https://github.com/EngreitzLab/CRISPRDesigner>.

The Variant-to-Gene-to-Program (V2G2P) approach snakemake pipeline is available at: <https://github.com/EngreitzLab/V2G> (V2G) https://github.com/EngreitzLab/cNMF_pipeline/ (G2P and V2G2P enrichment).

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- A description of any restrictions on data availability
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From the Data Availability section of the manuscript (with reference numbers given in parentheses):

Raw and processed data for Perturb-seq, ATAC-seq, H3K27ac ChIP-seq, and RNA-seq in TeloHAEC were deposited in NCBI's Gene Expression Omnibus under accession number GSE210523. This superseries is composed of subseries: GSE210489 (ATAC-seq), GSE210491 (ChIP-seq), GSE210522 (bulk RNAseq of cytokine-treated parental lines & single guide CRISPRi knockdowns), GSE232400 (bulk RNAseq of cells under flow, and MAP3K3 double knockdowns), GSE212396 (pilot scRNA-seq studies) and GSE210681 (comprehensive Perturb-seq).

Other datasets used in these studies: CAD lead GWAS variants were derived from both Aragam et al.(12) and Harst et al.(10), PheWas data was from Aragam et al. (12), GWAS summary statistics for other traits, and finemapping analysis, were from Hilary Finucane and Jacob Ulirsch's analysis of UKBB data (<https://www.finucanelab.org/data>), coding and splice site annotations were from the RefGene database (from the UCSC Genome Browser dated 24 June 2017)(64), 1000 Genome European ancestry LD data was accessed using "plink --ld-window-kb 1000 --ld-window 99999 --ld-window-r2 0.9", TF binding site information was from HOCOMOCO v11 human full scan motifs (https://hocomoco11.autosome.ru/downloads_v11), gene sets were from the Molecular Signatures Database (MSigDB) (63), scRNAseq data from explanted human right coronary artery endothelial cells (65), endothelial cis-regulatory elements derived from snATAC-seq(68), ENCODE datasets ENCSR000EVW (GATA2 ChIP-seq on HUVEC) and ENCSR000EOB (DNase-seq and DGF on HMVEC-dLy-Neo), protein structure models for KRIT1 (UniProt O00522), CCM2 (UniProt Q9BSQ5, with and without deletion of residues 417-444), PDCD10 (UniProt Q9BUL8), and TLNRD1 (UniProt Q9H1K6), and previous prioritization calls for genes in CAD GWAS loci were from these studies: Aragam et al.(12), Hodonsky et al.(92), Li et al.(93), OpenTarget L2G (94), Stolze et al.(29), van der Harst and Verweij(10) and Wunnemann et al.(30) (with details in the Methods).

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Sample size

For Perturb-seq analysis, we collected on average 85.5 singlet cells per perturbed gene, based on pilot Perturb-seq studies described in the Methods under "Selection of singlet thresholds", and made available as part of the GEO superseries for this manuscript. We found that this sample size was sufficient to detect the effects of individual perturbations on programs (identifying "program regulators"). Sample size for FlowFISH (4 replicates, 15 to 37 guides per element, 117 negative control guides, etc.) were calculated based on our prior studies to enable the detection of effect sizes in the range of ~15-20% (Fulco, C. P., Nasser, J., Jones, T. R. & Munson, G. Activity-by-contact model of enhancer-promoter regulation from thousands of CRISPR perturbations. Nature Genetics 51, 1664–1669 (2019).). Sample sizes for endothelial cell and zebrafish phenotypes, and qRT-PCR, were based on our prior experience to allow 1) sufficient replicates for statistical testing (N>=4) and 2) power, specific to each assay, to detect effect sizes of ~25%.

Data exclusions

No data that passed uniformly-applied quality control filters was excluded from this study. For instance, we retained all cells called by CellRanger that were identified as singlets (1 cell, 1 guide) in the Perturb-seq analysis. Ten cNMF components that showed high correlation with batch (10x lane number) were excluded from further analysis (see Methods: Excluding components associated with batch effects).

Replication	The primary Perturb-seq screen was performed in a single large experiment, and individual gRNA validation studies were found to agree with the results of the original screen. For experiments studying the effects of individual gRNAs on endothelial cell phenotypes, we used two separate sgRNAs to each target (with each sgRNA validated for knock down efficacy by Perturb-seq and bulk RNAseq).
Randomization	Randomization was not necessary for these studies because they were performed on homogeneous cell lines (e.g. TeloHAEC) or non-outbred zebrafish lines.
Blinding	Blinding was not necessary for these studies because quantitation was performed using unbiased approaches (cNMF, established methods for FlowFISH, EdgeR for bulk RNAseq, qRT-PCR ddCT, etc.). Scoring of zebrafish phenotypes was performed by a single trained researcher, phenotype scoring was always performed in comparison to age-matched non-injected embryos and embryos injected with non-targeting controls, and the heart phenotype observed with ccm2 or tlnd1 KD embryos was not ambiguous or open to biased interpretation.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☐	☒ Antibodies
☐	☒ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☐	☒ Animals and other organisms
☒	☐ Human research participants
☒	☐ Clinical data
☒	☐ Dual use research of concern

Methods

n/a	Involved in the study
☐	☒ ChIP-seq
☐	☒ Flow cytometry
☒	☐ MRI-based neuroimaging

Antibodies

Antibodies used	Anti histone H3 lysine 27 acetyl (H3K27ac) antibody (Catalog number 39685 from Active Motif, Carlsbad, CA), mouse anti-Flag (F1804, Millipore Sigma), rabbit anti-V5 (13202, Cell Signaling Technology), rabbit anti-TLNRD1 (HPA071766, Sigma), mouse anti-V5 (ab27671, Abcam), anti-mouse HRP secondary (7076, Cell Signaling), and anti-rabbit HRP secondary (7074, Cell Signaling). Dilutions and applications are detailed in the Methods.
Validation	Citations for: anti H3K27ac (Grzybowski et al (2015) Mol Cell 58:886, Shah et al (2018) Mol Cell 72:162), rabbit anti-V5 (Hanke, T. et al. (1995) J Virol Methods 53, 149-56), mouse anti-FLAG (BioTechniques 1994, 16, 730-735), anti-TLNRD1 (Sigma provides no citations, but we confirmed the specificity by the band appearing only when FLAG-TLNRD1 was expressed in HEK293T human cells, running at the expected MW and in the same position as the FLAG tag signal), mouse anti-V5 (Soltanmohammadi N et al. Nat Commun 13:701 (2022) & 128 other citations at https://www.abcam.com/products/primary-antibodies/v5-tag-antibody-sv5-pk1-ab27671.html), anti-mouse HRP (PubMed ID: 36453426 & over 9000 other citations at https://www.cellsignal.com/products/secondary-antibodies/anti-mouse-igg-hrp-linked-antibody/7076), anti-rabbit HRP (PubMed ID: 36453425 & over 14,000 other citations at https://www.cellsignal.com/products/secondary-antibodies/anti-rabbit-igg-hrp-linked-antibody/7074). We also experimentally validated each antibody: we have extensively validated that anti H3K27ac marks active chromatin in human cell lines in our prior publications (e.g. Fulco, C. P., Nasser, J., Jones, T. R. & Munson, G. Activity-by-contact model of enhancer-promoter regulation from thousands of CRISPR perturbations. Nature Genetics 51, 1664-1669 (2019)), and we validated the anti-V5 and anti-FLAG antibodies by observing the correct size of tagged proteins and by the lack of signal when tagged protein was absent from the extracts.

Eukaryotic cell lines

Policy information about [cell lines](#)

Cell line source(s)	Eahy926, TeloHAEC and HEK 293T cell lines were purchased from ATCC: TeloHAEC (CRL-4052), Eahy926 (CRL-2922), HEKK293T (CRL-3216).
Authentication	Cell lines were authenticated by the provider. We further authenticated each line by analysis of microscopic morphology (e.g. TeloHAEC and Eahy926 displayed the characteristic EC cobblestone morphology, and TeloHAEC showed localization of VE-Cadherin to endothelial cell junctions), functionality (high transfectability and protein expression for HEK293T), mapping of ATAC-seq, ChIP-seq and RNAseq reads to human genome, and RNAseq profiles and responses (e.g., for Eahy926 & TeloHAEC, expression of EC-specific genes and observation of previously-observed responses to stimuli, like TNFalpha).
Mycoplasma contamination	Confirmed negative
Commonly misidentified lines (See ICLAC register)	None

Animals and other organisms

Policy information about [studies involving animals](#); [ARRIVE guidelines](#) recommended for reporting animal research

Laboratory animals	Zebrafish were used to assess cardiovascular phenotypes of <i>ccm2</i> or <i>tlr1d1</i> perturbations. Wild type AB, transgenic Tg(flk1:EGFP) (that express EGFP at the surface of blood vessels) and transgenic Tg(cmlc2:EGFP) (that express EGFP in heart muscle) zebrafish lines were used. FO embryos and larvae after morpholino or CRISPR injections, both male and female, were assessed in the range of 3 to 5 days post-fertilization.
Wild animals	No wild animals were used in these studies.
Field-collected samples	No field-collected samples were used in these studies.
Ethics oversight	All animal experiments were performed in accordance with relevant guidelines and regulations and with approval from the Mayo Clinic Institutional Animal Care and Use Committee.

Note that full information on the approval of the study protocol must also be provided in the manuscript.

ChIP-seq

Data deposition

- ☒ Confirm that both raw and final processed data have been deposited in a public database such as [GEO](#).
- ☒ Confirm that you have deposited or provided access to graph files (e.g. BED files) for the called peaks.

Data access links <i>May remain private before publication.</i>	Raw and processed data for Perturb-seq, ATAC-seq, H3K27ac ChIP-seq, and RNA-seq in TeloHAEC were deposited in NCBI's Gene Expression Omnibus under accession number GSE210523. This superseries is composed of subseries: GSE210489 (ATAC-seq), GSE210491 (ChIP-seq), GSE210522 (bulk RNAseq of cytokine-treated parental lines & single guide CRISPRi knockdowns), GSE232400 (bulk RNAseq of cells under flow, and MAP3K3 double knockdowns), GSE212396 (pilot scRNA-seq studies) and GSE210681 (comprehensive Perturb-seq). These GEO records were made publicly available on 11/14/2023.
Files in database submission	Raw .fastq files are provided for all sequencing runs and all samples. Additional supplementary files were provided based on the subseries. For GSE210489 (ATAC-seq) and GSE210491 (ChIP-seq): Excel format Macs2 peak calls (.xls extension) and TDF format read density traces, for rapid viewing in IGV (.tdf extension) were provided for each sample. For the bulk RNAseq subseries GSE210522 and GSE232400: A text file reporting TPM values for all genes, by symbol, across all samples. For the perturb-seq records, GSE212396 and GSE210681: for each sample, fastq files for both scRNAseq and for the dialout libraries used to assign guides to cells, CellRanger Filtered Feature output files (cells x genes counts matrix in .mtx format, and tsv files for cell barcodes and features/genes), and singlet calls files (calls.tsv) linking cell barcodes to guides. Additional tab separated files were provided for each experiment that linked genes to guides (_targets_and_guides.txt) and tabulating the effects of each perturbation on gene expression, from EdgeR (separate tab separated .txt files recording log2 fold changes & p.values). For the full scale perturb-seq screen (GSE210681) we also provided a matrix of genes x cells for all singlet cells (in RDS format), and cNMF output (the cells x components matrix, and genes x components matrices (for the raw cNMF scores and for specificity z scores), as tab separated text files), and MAST results (identifying perturbations with significant effects on programs, as tab separated text files).
Genome browser session (e.g. UCSC)	No longer applicable.

Methodology

Replicates	3 biological replicates for ATAC-seq, 2 biological replicates for H3K27ac ChIP-seq (double this for control conditions)
Sequencing depth	Between 10 and 30 million reads per sample.
Antibodies	Anti histone H3 lysine 27 acetyl (H3K27ac) antibody: Catalog number 39685 from Active Motif, Carlsbad, CA
Peak calling parameters	Reads were mapped to the genome using Bowtie2, and peaks called using MACS
Data quality	For ChIPseq libraries, between 56,000 and 82,000 peaks were found at FDR 0.1/p. 1e-3, with ~18,000 to 25,000 peaks being 5-fold over background. For ATAC-seq: 110,000-200,000 peaks were found at FDR 0.1/p. 1e-3 with ~60,000 to 110,000 being 5-fold enriched.
Software	Peaks were called using MACS2, (macs2 callpeak) with parameters: -p 1e-3 -g hs -f BAM --call-summits

Flow Cytometry

Plots

Confirm that:

- ☒ The axis labels state the marker and fluorochrome used (e.g. CD4-FITC).
- ☒ The axis scales are clearly visible. Include numbers along axes only for bottom left plot of group (a 'group' is an analysis of identical markers).
- ☒ All plots are contour plots with outliers or pseudocolor plots.
- ☒ A numerical value for number of cells or percentage (with statistics) is provided.

Methodology

Sample preparation

For scoring and sorting BFP+ CRISPRi cells after doxycycline treatment, cells were harvested and resuspended in EC medium at a density of $\sim 1\text{--}3 \times 10^6$ cells/ml. FlowFISH was performed according to our established methods (Fulco, C. P., Nasser, J., Jones, T. R. & Munson, G. Activity-by-contact model of enhancer–promoter regulation from thousands of CRISPR perturbations. *Nature Genetics* 51, 1664–1669 (2019). Briefly, we used the PrimeFlow RNA Assay Kit (Thermo Fisher; Catalog number: 88-18005) according to the manufacturer's instructions, using ThermoFisher probesets VA1-3010837-PF for TLNRD1 and VA4-13187-PF for RPL13A. The TLNRD1 stain was Alexa Fluor 647 (AF647, "Type 1"), while the RPL13A stain was Alexa Fluor 488 (AF488, "Type 4"). We diluted the stained and washed cells in PBS with 0.5% BSA to a concentration of 2×10^7 cells/ml and filtered using a 30 μm filter (CellTrics, Catalog number 04-004-2326).

Instrument

Diagnostic FACS for BFP positive cells was performed on a Beckman CytoFLEX S Flow Cytometer. The sort for BFP+ cells was performed on a MoFlo Astrios EQ Cell Sorter. For FlowFISH, we used a SONY MA900 sorter.

Software

FACS runs were performed using the manufacturer software package (e.g. CytExpert for diagnostic FACS). Basic analysis was performed using FloJo or FCSalyzer 0.9.22-alpha. FlowFISH analysis was performed using a Maximum Likelihood Estimator approach, according to our established methods (Fulco et al. *Nature Genetics* 51, 1664–1669 (2019). In brief, we counted gRNAs in each bin using Bowtie to map reads to a custom index, normalized gRNA counts in each bin by library size, then used a maximum-likelihood estimation approach to compute the effect size for each gRNA. We used the limited-memory Broyden–Fletcher–Goldfarb–Shanno algorithm (implemented in the R stats4 package) to estimate the most likely log-normal distribution that would have produced the observed guide counts, and the effect size for each gRNA is the mean of its log-normal fit divided by the average of the means from all negative-control gRNAs. As previously described, we scaled the effect size of each gRNA in a screen linearly, so that the strongest 20-guide window at the TSS of the target gene has an 85% effect, in order to account for non-specific probe binding in the RNA FISH assay (this is based on our observation that promoter CRISPRi typically shows 80–90% knockdown by qPCR). We averaged the effect sizes of each gRNA across replicates and computed the effect size of an element as the average of all gRNAs targeting that element. We assessed significance using a two-sided t-test comparing the mean effect size of all gRNAs in a candidate element to all negative-control guides. We computed the false-discovery rate (FDR) for elements using the Benjamini–Hochberg procedure and used an FDR threshold of 0.05 to call significant regulatory effects.

Cell population abundance

Cells were not a mix of cell types, and were all TeloHAEC. For BFP+ cell sorting, almost all cells had a stronger PB450-A signal than non-dox treated controls, and we sorted for cells with the top 15% PB450-A signal. For FlowFISH cells in the gRNA library-transduced pool with different CRISPRi perturbation guides were sorted into bins by percentage of normalized TLNRD1/RPL13A signal, as described below.

Gating strategy

For sorting BFP+ cells, live cells were gated by SSC-A and FSC-A. BFP-positive cells were identified by comparison to no Doxycycline control cells, and the top 15% of the BFP-positive peak (PB450-A) selected. The gating strategy is shown in Extended Data Fig. 1. FlowFISH parameters were essentially as per Fulco et al. *Nature Genetics* 51, 1664–1669 (2019), with the gating strategy illustrated in that publication. Briefly, to control for differences in staining efficiency for each cell, we normalized the fluorescence associated with the gene of interest to that of the control gene. Specifically, we used the color compensation tool to subtract a portion of each cell's AF647 signal based on the intensity of its AF488 signal such that the mean AF488 signal in the top and bottom 25% of cells based on AF647 was within 10%. If necessary, we then reduced the level of compensation until the fraction of cells with AF647 signal equal to 0 was no more than 5%. We set the gates for each bin on the compensated signal to capture quartiles of cells, with bins numbered according to the percentiles (i) 0–25% (ii) 25–50%, (iii) 50–75%, (iv) 75–100%.

- ☒ Tick this box to confirm that a figure exemplifying the gating strategy is provided in the Supplementary Information.